ANNEX 1

Horizon Europe (HORIZON)

Description of the action (DoA)

Part A

Part B

DESCRIPTION OF THE ACTION (PART A)

Associated with document Ref. Ares(2024)6673367 - 20/09/2024

COVER PAGE

Part A of the Description of the Action (DoA) must be completed directly on the Portal Grant Preparation screens.

PROJECT *Grant Preparation (General Information screen) — Enter the info.*	
Project number:	101178403
Project name:	PRocess Innovations for the Mineral industry focusing on ROasting and Calcination Kiln technologies and supported by novel pre- and postprocessing
Project acronym:	PRIM-ROCK
Call:	HORIZON-CL4-2024-TWIN-TRANSITION-01
Topic:	HORIZON-CL4-2024-TWIN-TRANSITION-01-41
Type of action:	HORIZON-RIA
Service:	HADEA/B/03
Project starting date:	first day of the month following the entry into force date
Project duration:	48 months

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PROJECT SUMMARY

Project summary

Grant Preparation (General Information screen) — Provide an overall description of your project (including context and overall objectives, planned activities and main achievements, and expected results and impacts (on target groups, change procedures, capacities, innovation etc)). This summary should give readers a clear idea of what your project is about.

Use the project summary from your proposal.

PRIM-ROCK addresses process efficiency through a multifaceted strategy addressing industrial processes from input to output, supported by an ambitious digital framework and transversal supporting activities. In particular, advanced techniques for the pre-processing of the raw material will be deployed to enhance its efficiency according to the requirements of the process, supplemented by simulations and decision support systems to increase the operational limits of flexibility in the feedstock domain. Additionally, the process intensification component of the project will design, develop, and validate innovative and efficient processes, as well as optimizing the existing ones. The aim is to develop a portfolio of technologies to a higher resource efficiency and lower level of GHG emissions of extractive industries. The project will focus on improving the resource efficiency of the calcination and roasting processes, that are commonly used in the mineral and cement industries.
To this end, process simulations in collaboration with AI data-driven models will be utilized and a digital twin for each process will be developed. Finally, waste reduction and re-utilization strategies will be investigated targeting to the real-time processes' adjustment capabilities to feedstock changes and tighter processing control solutions. All these components will be supplemented by transversal supporting activities, such as technoeconomic assessment, LCA, standardization etc. The PRIM-ROCK solutions will be demonstrated in 3 different ASPIRE sectors, namely Minerals (magnesite, laterite), Cement (limestone) and Non-ferrous metals (sphalerite, chalcopyrite).
The consortium is composed of 7 SMEs along with 6 RTDs, 3 universities and 5 large enterprises, working together to achieve over 30% resource efficiency enhancement in each use-case, 770ktCO2 averted, productivity increase in the range 9-16% and more that 1,360 new jobs by 2033.

LIST OF PARTICIPANTS

PARTICIPANTS

Grant Preparation (Beneficiaries screen) — Enter the info.

Number	Role	Short name	Legal name	Country	PIC
1	COO	NTUA	ETHNICON METSOVION POLYTECHNION	EL	999978142
2	BEN	CORE	CORE INNOVATION AND TECHNOLOGY OE	EL	917459078
3	BEN	EUT	FUNDACIO EURECAT	ES	928030235
4	BEN	ICCS	EREVNITIKO PANEPISTIMIAKO INSTITOUTO SYSTIMATON EPIKOINONION KAI YPOLOGISTON	EL	999654356
5	BEN	CIRCE	FUNDACION CIRCE CENTRO DE INVESTIGACION DE RECURSOS Y CONSUMOS ENERGETICOS	ES	999516907
6	BEN	ITA	INSTITUTO TECNOLOGICO DE ARAGON	ES	999509341
7	BEN	SCCH	SOFTWARE COMPETENCE CENTER HAGENBERG GMBH	AT	999663765
8	BEN	BUL	BRUNEL UNIVERSITY LONDON	UK	999749610
9	BEN	CEIN	INNCEINNMAT SL	ES	953847464

PARTICIPANTS

Grant Preparation (Beneficiaries screen) — Enter the info.

Number	Role	Short name	Legal name	Country	PIC

Associated with document Ref. Ares(2024)6673367 - 20/09/2024

10	BEN	IRIS	IRIS TECHNOLOGY SOLUTIONS, SOCIEDAD LIMITADA	ES	998993786
11	BEN	STAM	STAM SRL	IT	916965542
12	BEN	ADMR	ADVANCED MINERALS AND RECYCLING INDUSTRIAL SOLUTIONS IKE	EL	927557554
13	BEN	PRINS	PRINSUS I.K.E. TECHNOVLASTOS	EL	884099420
14	BEN	EUROC	EUROCORE CONSULTING	BE	901601712
15	BEN	MAGG	MAGGIOLI SPA	IT	996621457
16	BEN	CIBOS	CIBOS INNOVATION IDIOTIKI KEFALAIOUCHIKI ETAIREIA	EL	891970776
17	BEN	FORN	FORNACI CALCE GRIGOLIN SPA	IT	926631883
18	BEN	HEGOLD	HELLAS GOLD S.A.	EL	997219947
19	BEN	TERNAMAG	TERNA LEFKOLITHOI ANONYMI METALLEUTIKI EMPORIKI TECHNIKI VIOMICHANIKI ETAIREIA	EL	913286720
20	BEN	VDU	VYTAUTO DIDZIOJO UNIVERSITETAS	LT	999590627
21	BEN	UNE	ASOCIACION ESPANOLA DE NORMALIZACION	ES	984232326
22	BEN	KGHM	KGHM CUPRUM SPOLKA Z OGRANICZONA ODPOWIEDZIALNOSCIA - CENTRUM BADAWCZO- ROZWOJOWE	PL	999495470

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LIST OF WORK PACKAGES

Work packages

Grant Preparation (Work Packages screen) — Enter the info.

Work Package No	Work Package name	Lead Beneficiary	Effort (PersonMonths)	Start Month	End Month	Deliverables
WP1	Project specifications	14 - EUROC		1	12	D1.1 – Project specifications D1.2 – Expanded specifications
WP2	Simulation-driven design	5 - CIRCE		3	18	D2.1 – MFA report D2.2 – Raw material intelligence design D2.3 – IFRK models and design for calcination process D2.4 – IFRK models and design for roasting process D2.5 – Microwave rotary kiln models and design D2.6 – Plasma reactor models and design
WP3	Digitalisation framework development	15 - MAGG		6	18	D3.1 – PRIM-ROCK platform design principles D3.2 – Sensors and controls retrofitting report
WP4	Raw material intelligence system development	4 - ICCS		12	28	D4.1 – Feedstock intelligence system
WP5	Industry 4.0 framework deployment	2 - CORE		18	36	D5.1 – Control and automation system D5.2 – Process data models D5.3 – PRIM-ROCK IoT data platform D5.4 – Integrated HDT

WP6	Kilns and plasma reactor prototyping	1 - NTUA		12	28	D6.1 – IFRK lab scale prototype D6.2 – IFRK prototypes D6.3 – Microwave kiln prototype D6.4 – MIP reactor and heat exchanger prototypes

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Work packages

Grant Preparation (Work Packages screen) — Enter the info.

Work Package No	Work Package name	Lead Beneficiary	Effort (PersonMonths)	Start Month	End Month	Deliverables
						D6.5 – Feasibility assessment and impacts
WP7	Integrated prototypes and pilot deployment	3 - EUT		24	36	D7.1 – Validated systems
WP8	Pilot demonstrations	1 - NTUA		37	48	D8.1 – PRIM-ROCK demonstration
WP9	Adoption framework consolidation	12 - ADMR		29	48	D9.1 – TEA and efficiency gains D9.2 – Life Cycle Sustainability Assessment report D9.3 – Capacity building portfolio, health & safety D9.4 – Industrial uptake analysis D9.5 – Legislation and regulation
WP10	Early phase Dissemination Exploitation Communication	2 - CORE		1	18	D10.1 – Early PEDR [1st update] D10.2 – Standardization landscape, and strategy [1st update] D10.3 – Early PEDR [2nd update] D10.4 – Standardization landscape, and strategy [2nd update]
WP11	Mid-phase Dissemination Exploitation Communication	2 - CORE		19	36	D11.1 – Mid PEDR D11.2 – Standardization report I

WP12	Final phase Dissemination Exploitation Communication	2 - CORE		37	48	D12.1 – Final PEDR D12.2 – Standardization report II
WP13	Project Management	1 - NTUA		1	48	D13.1 – Project handbook D13.2 – Project management quality assessment Ver.1 D13.3 – Project management quality assessment Ver.2 D13.4 – Project management quality assessment Ver.3

Work package WP1 – Project specifications

Work Package Number	WP1	**Lead Beneficiary**	14 - EUROC
Work Package Name	Project specifications		
Start Month	1	**End Month**	12

Objectives

Objectives: (1) Collect, catalogue, and consolidate information describing the project's processes, methodologies and other technical elements in detail; (2) Refine KPIs, enhance the quantification of project's actions; (3) Define and clarify the critical process parameters and quality indicators; (4) Consolidate security and safety specifications.

Description

T1.1 Overall project technological specifications – (NTUA), M1-6, [All partners]
T1.1 focuses on the detailed technical definition of the overall PRIM-ROCK solution, outlining specifications for each key component, module, and functionalities. This includes Operating specifications: Elaborating the operating parameters for technologies and solutions to be implemented in the industrial end-users. Process and methodology specifications: Describing all additional resources, design tools, simulation software, managerial methods, strategies, and other resources committed by the consortium. Technology positioning: Reviewing the SoA in relevant processes, benchmarking, and updating the status quo by analysing new and past projects, initiatives and investigating market introductions of new products/technologies. Output: Technical specifications report (D1.1).

T1.2 End user requirements – (KGHM), M1-6, [All partners]
The aim of T1.2 is to establish a robust framework that facilitates a cooperative network between technology suppliers and researchers. The primary objective is to gain a thorough understanding of the end-user requirements and needs, as well as comprehend the industrial processes. Information collected during the proposal stage will be refined and enriched with the support of the demonstrators (process audits, previous developments, environmental audits, etc.), from existing data monitoring systems, from onsite measurements, if needed, and from estimations or models. The factors investigated will be dependent on the input provided by all partners and end user requirements analysis. Output: Detailed demo-case analysis (part of D1.1)

T1.3 Identification of critical parameters for monitoring and control – (IRIS), M1-6, [All partners]
T1.3 is dedicated to identifying the key performance and quality indicators, as well as the critical process parameters that influence the final product and must be considered by the developed solutions, such as operating temperature, kiln rotating speed, process time, mass loss, homogeneity index etc. A complete map of all important process variables will be obtained. This will serve as the guideline for the selection and design of the sensor. Additionally, a design of the monitoring Human-Machine Interface (HMI) and operator control workflow will be defined, and the interaction with the DSS and UI will be considered in the overall workflow. A monitoring configuration plan will be delivered to accompany the parameter listing. Output: Monitoring plan and parameter list (part of D1.1)

T1.4 Safety positioning– (EUROC), M1-12, [All partners]
To enhance the safety of the industrial environment and its employees, a comprehensive risk assessment will be conducted prior to the deployment of the PRIM-ROCK solution within the industrial use cases. The primary objective is to identify and mitigate hazards, thereby refining operational safety from the outset. Safety protocols relevant to the activities of the project will be compiled and delivered. The Consortium partners will analyse the procedures and assess if any additional considerations need to be included to ensure the safe operation of the PRIM- ROCK solution. Output: OHSE report (part of D1.2).

T1.5 Baseline establishment for assessments – (ADMR), M6-12, [All partners]
T1.5 aims to consolidate and define benchmarks for the transversal activities that underpin the Technology Adoption Framework (KER5). This encompasses a range of evaluations and activities, including: Technoeconomic assessment, Energy efficiency & intensity assessment, LCIA, Societal & environmental impacts assessment, Workplace effects (productivity, skills), business planning and sustainable investing (addressed in DEC), standardisation activities, data interoperability methodologies, clustering and safety protocols. Moreover, the goal will be to define benchmarks for the achievement of the process resource efficiency improvement metrics, building upon the work of T1.3. Output: Baseline indicators (part of D1.2)

Work package WP2 – Simulation-driven design

Work Package Number	WP2	**Lead Beneficiary**	5 - CIRCE
Work Package Name	Simulation-driven design		
Start Month	3	**End Month**	18
Objectives			
Objectives: (1) Analyze material flows for efficiency and waste reduction in mineral and cement industries; (2) Design a system for optimized raw material use; (3) Develop simulations of IFRKs for efficient calcination and roasting processes; (4) Design MRK for sustainable metal oxide reduction; (5) Create plasma reactor designs for effective H2S treatment and hydrogen recovery.			
Description			

T2.1 Material flow analysis – (CIRCE), M3-12, [NTUA, ITA, BUL, CEIN, EUT, End users]
This task is dedicated to developing a comprehensive methodology that enhances the circularity of material flows within the industrial processes, linking closely to the simulation-based design in T2.3 and T2.4. Starting with a thorough literature review, the methodology development entails creating tools for simulating material flows and integrating environmental considerations into the simulation algorithms. The aim is to identify the optimization parameters that improve both environmental and economic efficiencies. This methodology will undergo validation against real-world data, followed by the necessary adjustments to enhance its precision and adaptability. The final output will be a detailed report, serving as a valuable guide for implementing sustainable practices in the designed industrial process. Output: MFA report (D2.1).

T2.2 Raw material management process design – (ICCS), M3-15, [CIBOS]
In T2.2 ICCS will consolidate the preliminary designs for the RMI, tailoring these designs to support the specific endusers' processes (based on the output of T1.2) and addressing the unique characteristics of the raw materials, such as size distribution, composition and visual traits. This task includes the creation of an in-depth blueprint that maps the configuration of sensor networks, the selection of suitable sensor technologies (including XRF, hyperspectral imaging, LIBS), and the crafting of AI-driven algorithms for advanced data analysis and enhancement of the resource efficiency of the processes. Additionally, a comprehensive plan for the integration of these systems into existing ore processing workflows will be delivered, to ensure seamless functionality and improved outcomes in sorting, classification, and the determination of the optimal processing parameters. Output: Raw material intelligence system design (D2.2).

T2.3 IFRK simulations and design for the calcination process – (CIRCE), M3-16, [NTUA, End users]
The goal of T2.3 is the development of a multiscale simulation approach to accurately predict the performance of the IFRK for the calcination process, aiming to optimize efficiency, product quality, and energy consumption. NTUA will carry out the physicochemical simulations of the process, covering heat/mass transfer and chemical reactions, defining geometry/boundary conditions, chemical kinetics, simulation, and optimization. Simulations will include individual particle tracking and modelling of the surrounding gas phase as a whole, to simulate and examine the materials transformation process in detail. Software and modelling techniques, including the realizable k-ε model will be used. Factors such as particle motion and radiation heat transfer to accurately represent the calcination process will be examined. A comprehensive coupled DEM-CFD steady-state model will be developed by CIRCE to study in detail the flow dynamics, heat transfer and chemical reactions within the calcinatory, characterizing its operation in detail. The simulation results will be utilised to design and define the parameters of the IFRK to ensure optimal performance for the calcination process. Output: IFRK models and design for calcination process (D2.3).

T2.4 IFRK simulations and design for the roasting process – (ITA), M3-16, [BUL, End users]
The goal of T2.4 is the development of a multiscale simulation approach to accurately predict the performance of the IFRK for the roasting process. As presented in the Concept and Methodology section, this approach will be based on: (i) detailed chemical, physicochemical and thermo-fluid-dynamics simulations at different scales; and (ii) reduced order models. First, ITA will assess the kinetics of the hydrogen reduction reactions from molecular dynamics simulations (i.e., atomistic modelling). The simulations will rely on the reactive force field (ReaxFF) interatomic potentials, which allow to model large-scale systems where chemical reactions are occurring. Such simulations will allow for establishing a detailed reaction mechanism pathway with the different steps involved in the hydrogen reduction process of iron ores and for determining the kinetic parameters associated (i.e., activation energy, kinetic constant, order of reaction). The results will be compared with experimental data either comparing the kinetic parameters (i.e., activation energies, kinetic constants,

order of reaction) or thermodynamic parameters (i.e., heat of reactions), while they are available in the bibliography or from the results obtained experimentally in the project. As this task is highly computationally demanding, the scope of the task is limited to a few ore types (2-
3) to be selected in WP1. Second, ITA will implement a novel methodology to obtain the volumetric reaction rate at the particle level. To this end, the reaction kinetics obtained with ReaxFF will be considered at the gas-particle interface (surface reaction) to predict the overall volumetric reduction rate of the metal oxides and sulfides with H2 in relation with temperature and H2 partial pressure as well as the materials properties (particle size, specific surface, porosity) and process parameters (such as residence time). This methodology will be developed using open-source tools such as LIGGGHTS (Discrete Element Method, DEM), OpenFOAM (Finite Volume Method FVM – Computational Fluid Dynamics CFD) and Python. Once the workflow is fully implemented, it will be applied to study the roasting process at the particle scale under a wide range of scenarios (Design of Experiments, DoE) and the results will be postprocessed using reduced order techniques. The resulting ROMs could be later used as a source term of the full thermodynamics IFRK simulations (carried out by BUL). As this task is also highly computationally demanding, the scope of the task is limited to a few ore types (2-3) to be selected in WP1. BUL will develop the methodology for the thermo-fluiddynamics simulation of the roasting process at the reactor (IFRK) scale. The results will be validated by comparing with

experimental data from the lab-scale experiments. Once the tool is validated, then BUL will explore the operation of the IFRK under a wide range of operating conditions (Design of experiments), aiming at: (i) optimizing the reactor operation; and (ii) generating enough data to feed the ROMs in T6.2. Output: IFRK models and design for roasting process (D2.4).

T2.5 Microwave rotary kiln design – (CEIN), M3-18, [EUT]
In this task a microwave kiln for sulphide roasting using H2 will be designed by CEIN. Based on the behaviour of sulphides under microwave and thermal treatment, static microwave set ups will be developed to determine the primary process parameters in terms of power demand, produced gas and gasses additions following up the reaction, and solid byproducts. Investigations into the effects of electromagnetic field irradiation on sulphides under various conditions will be carried out using instrumented resonant applicators. These findings will guide the design of a continuous flow microwave kiln, equipped with a novel microwave applicator that produces a modular electromagnetic field, ensuring compatibility with the reaction requirements and the material conveying system. Microwave filters will be incorporated to assure open ends for the continuous flow of solids. A control system will also be developed to regulate the reactor's atmosphere, facilitating the injection of gases, including hydrogen. Installed microwave leakage detector integrated alarm, as well as H2 sensor alarm and portable multi-gas sensor and personnel sensor will be also used during operations. Finally, a study on the material conveyance will be conducted to verify adequate residence time and appropriate microwave irradiation exposure. Output: MRK design (D2.5).

T2.6 Plasma treatment of H2S simulations and design – (EUT), M6-18, [BUL, CEIN]
EUT will study several components and modifications in their equipment and modify them for PRIM ROCK's needs. EUT will consider devices before plasma, plasma reactions and after plasma. A further consideration will be security in the use of hydrogen sulphide before the plasma reactor, and hydrogen management after their generation/separation. We will perform several proofs of concept tests (PoC), supported by multiphysics simulations. BUL will participate by carrying out the materials selection, simulation and design of the Gas Cooling Heat Pipe (GCHP). Thermophysical and numerical modelling will be used to support the process. For the heat exchanger material, materials capable of withstanding high temperatures without compromising the structural integrity of the component will be selected, with water (although alternatives will be investigated) as working fluid and a suitable architecture will be defined (e.g., finless). The Heat Pipe design will aim at keeping the internal operation temperature of the heat pipe below 300°C through optimised removal of the heat that is absorbed from the H2S gas. Radiative cooling will also be inherently included in the design of the same heat pipe system. Output: MIP reactor models and design (D2.6).

Work package WP3 – Digitalisation framework development

Work Package Number	WP3	**Lead Beneficiary**	15 - MAGG
Work Package Name	Digitalisation framework development		
Start Month	6	**End Month**	18
Objectives			
Objectives: (1) Define the data model for the project; (2) Design the digital framework and data integration platform; (3) Design the advanced sensor network; (4) Design the process control system and interfaces.			
Description			

T3.1 Data reference architecture design – (IRIS), M6-18, [MAGG, ICCS, CIBOS, CORE, SCCH, STAM]
Based on the outputs of T1.1-3, the objective of this task is to develop the IT architecture and specify the technical requirements for the Scadalytics Industrial IoT data platform, involving: 1) data storage and processing of heterogeneous data types such as real-time and batch data, sensor data, operational data, etc., 2) integration mechanisms and definition of interfaces and data models to improve interoperability and coupling between all involved modules, 3) specifications of the BE and FE components, 4) gateways, connectors and responsibilities definition, as well as 5) cybersecurity measures. The data reference architecture will be coherent with the Scadalytics paradigm, as a middleware solution considering heterogeneity, scalability, reliability, as well as security requirements, and employing industrial standard technologies (Kafka, MQTT) for the middleware and robust tools for data storage (InfluxDB, Object storage-s3 compatible) for optimized time-series data and flexibility, while establishing monitoring and management capabilities. The input of domain experts (e.g., solution developers) will be requested. Output: PRIM-ROCK's platform design principles (D3.1).

T3.2 Advanced sensor network design – (IRIS), M8-18, [MAGG, NTUA, EUT, CEIN, ICCS, CIBOS, FORN, STAM, KGHM]
T3.2 involves the creation of a tailored monitoring and control plan for each pilot, aligning with the specifications, requirements, critical process parameters and safety regulations established in WP1. To ensure comprehensive sensing capabilities for the pilot demonstrations, IRIS will utilise a novel Multirange Spectroscopic Platform. It will be designed and engineered as a versatile test rig, which will integrate a selection of sensing techniques, including SWIR, NIR, VIS spectroscopy, Raman spectroscopy, Fluorescence and LIBs spectroscopy, according to each pilot's specific monitoring challenges. A validation process will be performed comparing the results obtained with the in- line sensors and those obtained with conventional analytical tools available at the premises of academic institutions and the industrial partner facilities. The collected information will serve as the basis for the development of the most suitable analytical platform configuration for each pilot. The selected sensing configurations (FOS, the combinations from the VISUM Fusion multispectral platform, commercial sensors) will be evaluated and defined for each use- case, considering critical parameters to be monitored and controlled, as well as the optimal way to integrate them. This will be enabled by the Scadalytics platform (see also Task 5.3), for gathering real-time heterogeneous data from multiple, simultaneous sources (interrogators, cameras, industrial equipment), and making it easily available according to IoT guidelines. From this datafusion, aggregated metrics will be calculated to provide meaningful insights about the process's status and to establish a baseline for subsequent efficiency assessments. Ultimately, the optimum spectral data fusion techniques will be defined using innovative hybrid ensembling. This approach will lead to a robust and cost effective in-line monitoring solution, equipped with a smart sensing device with automated data acquisition system and machine learning algorithms to enable real-time data analysis and predictive modelling. FORN and STAM will implement industrial IoT on site for direct collection of operational data, used in process validation. Output: Sensor network design (part of D3.2).

T3.3 Process control and interface design – (MAGG), M10-18, [IRIS, NTUA, EUT, CEIN, ICCS, CIBOS, STAM, KGHM]
This task provides a detailed definition of the functional architecture and specifications of the Process Control System (PCS) that will allow to monitor and manage the industrial processes foreseen in the pilots. Based on the monitoring and control plan designed in T3.2, and the corresponding sensing configurations, this task will design PCS functional architecture that will allow end-users to define the industrial processes and to specify the variables to be monitored. The PCS will include mechanisms to associate sensors to specific variables and to automatically feed the corresponding data into the identified variables. This aspect allows to have a direct calculation/measure of KPIs for the assessment of energy and resource efficiency in all stages of the industrial processes. The PCS will be complemented with horizontal features such as: 1) Data Logging and Historian: where to record and store process data over time for analysis, troubleshooting, and compliance purposes; 2) Alarm Management: to monitor and manage alarms generated by the system to alert operators of abnormal or hazardous conditions; 3) Integration of safety and security protocols relevant to the activities of the project; 4) Integration with Enterprise Systems for data sharing, reporting, and analysis. Finally, the task will define the user interface (UI) that provides a graphical representation of the processes and aggregates all the needed functionalities allowing operators to interact with the system, execute processes, monitor process variables, evaluate alternatives through simulation, and, finally, make adjustments and corrections if needed. Output: PCS design and architecture (part of D3.2).

Work package WP4 – Raw material intelligence system development

Work Package Number	WP4	**Lead Beneficiary**	4 - ICCS

Work Package Name	Raw material intelligence system development		
Start Month	12	**End Month**	28

Objectives

Objectives: (1) Develop monitoring and control systems for the raw material intelligence (RMI); (2) Deliver two integrated sorting prototypes (3) Develop and deliver an intelligent DSS for classification; (4) Deliver non-destructive testing configurations;

Description

T4.1 Development of monitoring and control systems for the RMI – (ICCS), M12-24 [CIBOS]
T4.1 focuses on the development and integration of advanced sensorisation and control systems needed for the data collection and control signals of the RMI system (classification, sorting, crushing). This task involves the integration of various sensors, such as XRF, hyperspectral imaging, and LIBS, in offline or online modes, based on the sensor network design of T3.2, to accurately detect and analyse the composition of ore streams. The integrated control systems will be responsible for dynamically adjusting the processing parameters, ensuring maximum efficiency, and throughput while maintaining high accuracy in separation. The systems will be developed and prototyped, with a subsequent validation phase in WP8. The outcome will be an integrated sensor and control solution, tailor-made for high-precision ore processing applications, ready to be implemented in the RMI system. Output: Advanced sensorisation and control system prototype (part of D4.1).

T4.2 Advanced and adaptive automated sorting prototyping – (ICCS), M12-24, [CIBOS]
Two twin AI-assisted, sorting prototypes for tuneable segregation of mineral ore streams will be developed and deployed in this task. The primary objective is to identify and evaluate key features of the ore's chemical structure, and to utilize them in the joint optimization process of maximizing the sorting performance, while aligning with the requirements of the whole valorisation process. An intelligent sorting mechanism for raw materials in the calcination process ($MgCO_3$ and $CaCO_3$) will be delivered, installed and validated at NTUA premises. The cognition of this system will be accomplished by integrating an ensemble of non-destructive, vision sensory devices, such as hyperspectral and microscopy imaging. A twin sorting mechanism for the roasting process raw material (FeNi, ZnS, CuS) will be designed, developed and installed in EUT premises, tailored to recognize and segregate minerals within ore streams. Running these twin intelligent systems in parallel will allow for a thorough and extensive exploratory analysis on the state-of-the-art deep learning network architectures and ore pre-treatment techniques. Characterization of samples in both setups will utilize deep learning models based on CNN architectures, complemented by a pneumatic subsystem capable of separating samples with a precision and accuracy greater than 95%. Output: Two automated sorting prototypes (part of D4.1).

T4.3 Intelligent Decision Support System for classification – (CIBOS), M16-28, [ICCS]
In T4.3 ICCS will develop a comprehensive intelligent Decision Support System for raw-material pre-processing optimisation (RMI-DSS), designed to assist in the determination process of the optimal sorting thresholds of mineral ore streams. The system will integrate advanced techniques and algorithms (e.g. tree-based, rule-based algorithm, etc.) to analyse the physical and chemical properties of the mined minerals. Considerations for the system's analysis will include the previously determined optimal, valorisation flow-specific crushing size, operational data from the intelligent sorting mechanisms for all raw-materials, and the overall requirements of the upcoming valorisation chain. This will facilitate an accurate estimation of the sorting thresholds for the raw materials present in the stream. The DSS's ability to dynamically and in real-time decide on sorting criteria is a testament to the adaptive capacity of the intelligent sorting mechanisms. Output: RMI-DSS (part of D4.1).

T4.4 Crushing size identification for ore streams – (ICCS), M18-28, [CIBOS]
The aim of this task is to characterize ore streams to determine the most efficient crushing size for subsequent processes. To this end, ICCS will leverage optical sensors with microscopy imaging sensors and hyperspectral cameras, capable of densely sampling the electromagnetic spectrum up to 5000nm. During this task, trials within ore processing chains will be conducted with a pre-selected set of ore crushing sizes using the lab-scale sorting system in ICCS premises. The sorted samples will be processed, off-line using analytical techniques, such as Raman spectroscopy, FTIR and X-ray to a) validate the system's ability to separate minerals and b) identify the optimal crushing size tailored to the specific valorisation scenario. Output: NDT real time techniques analysis (part of D4.1).

Work package WP5 – Industry 4.0 framework deployment

Work Package Number	WP5	**Lead Beneficiary**	2 - CORE
Work Package Name	Industry 4.0 framework deployment		
Start Month	18	**End Month**	36

Objectives

Objectives: (1) Develop and deliver the control system, (2) Develop data-driven models for the calcination and roasting processes; (3) Integrate and coordinate hybrid digital twins and the corresponding human-machine interface (4) Develop a 3D visualization of the processes.

Description

T5.1 Automation & control development – (MAGG), M18-30, [IRIS, KGHM]
In T5.1 MAGG will develop the Process Control System (PCS) tailored for the PRIM-ROCK proposed processline, building upon the functional architecture outlined in T3.3 and ensuring that the PCS is robust, accurate, and responsive to the specific needs of processes. The PCS will be adaptable to support the integration of the optimization algorithms developed in T5.2 and the UI delivered in T5.3. The goal is to provide a flexible PCS that facilitates the monitoring, management, and manual control of the processes, engineered with the capacity to handle high-fidelity data, laying the groundwork for eventual automated control and decision-making capabilities. The development of a control infrastructure that is compliant with safety and quality standards, anticipating the operational demands of the calcination and roasting processes will be prioritized. Output: PCS development (D5.1).

T5.2 Data driven model development – (SCCH), M18-36, [CORE]
In this task, the data-driven models for monitoring and optimisation for both the calcination and roasting processes will be developed. The approach will be based upon modern AI-based techniques and novel transfer learning and hybrid modelling to enhance performance and generalize model applicability. In the first phase of this task, we will develop the data pre-processing, data quality, and feature engineering/selection of the collected process/use-case data. Modern AIbased modelling techniques will be utilized to design predictive models, that will exploit to the extent possible expert knowledge and physics-based information. We will consider standard modelling techniques, such as linear/nonlinear regression models, as well as more advanced types of neural networks, such as recurrent neural networks and LSTM, variational autoencoders and Attention Networks to study the latent space. In the second phase of this task, we will advance these predictive models through the design of transfer learning techniques both with respect to the integration of alternative data sources as well as the model fine tuning. In the first case, we will utilize alternative data sources (e.g., data collected from the lab-scale prototype developed in T6.1 and simulation data from physics-based models developed in WP2) to perform domain generalization and feature alignment (i.e., to remove the effect of certain process details or material properties from the data). This will increase the size of relevant training data. In the second case, the trained model is further improved through hybrid combinations with alternative models (e.g., the physics-based simulation models of WP2) to further increase its performance and robustness. SCCH will focus on the calcination process, and CORE will focus on the roasting process (both via IFRK and MRK). Once integration starts in M30, the models will be adapted to make use of the real-world data from the test setups and validated to ensure that they accurately represent the real process. Output: Data-driven models for calcination and roasting processes (D5.2).

T5.3 Data orchestration middleware – (IRIS), M18-36, [MAGG, CORE, SCCH, STAM]
In T5.3, IRIS will adapt its Scadalytics Data Management Platform to incorporate the middleware software components and services, emphasizing flexibility, robustness and security. The goal is to orchestrate heterogeneous data sources including raw sensing data, data from SCADA/ERP, PLC, other industrial IoT or field devices and external socioeconomic data streams. Adopting flexibility principles through a microservices architecture will enable seamless data integration. To orchestrate these layers, the resources of each environment will be clustered to form distinct computation clusters. The management and orchestration of these clusters will be achieved via mixed container environments (e.g., Kubernetes & multi-cluster orchestration tools). Through this orchestrator, relevant Communication Middleware components will be deployed on appropriate environments. It will include frameworks and tools for secure data management & distribution, e.g., cloud-message-brokers (Kafka, MQTT, Orison Broker, etc.). It will also be responsible for the generation of the real-time-data needed by analytics, simulation and digital twin components that rely on streaming data inputs. Output: PRIM-ROCK IoT data platform (D5.3).

T5.4 Integrated process digital twinning (DSS & UI) – (CORE), M22-36, [SCCH]

In T5.4 a Decision Support System (DSS) with an integrated User Interface (UI) will be created to enhance the resource efficiency of the calcination and roasting processes by recommending the optimal set of process parameters and settings. The DSS will integrate diverse streams of heterogeneous information from various models (data-driven models, physicsbased models or hybrid models) to provide a comprehensive view of the thermal processes. This system will deliver realtime insights and alerts, empowering operators to make informed decisions quickly and accurately towards optimizing the process for maximum efficiency. Based on user-requirements and critical parameters defined in WP1, a prototype of the UI will be initially designed, which will be subsequently tested and refined based on user feedback. Once the prototype is finalized, the actual UI will be developed integrating the data models and the DSS. Finally, a user acceptance testing will be conducted to confirm the UI's usability, intuitiveness, and alignment with project specifications. Output: PRIM-ROCK DSS optimizer (part of D5.4).

T5.5 3D visualisation – (STAM), M22-36, [CORE, SCCH, MAGG]
Task 5.5 will deliver an interactive 3D viewer for process visualization leveraging advanced data interface technologies and multi agent immersive frameworks. This tool will combine "standard" 3D viewer that are widely adopted to view orthophotos, BIM, and 3D models, with a specialised adaptation layer. This layer will enable different users and devices to view and interact with the models both in simulation mode, as well as in the so-called "monitoring" mode – the latter updating the viewer in real-time according to the actual monitoring of the process. The primary objective of the task is twofold: to facilitate a virtual interaction with the processes through Human- Machine Interfaces (HMI) accessible to a diverse range of stakeholders and devices, and to provide sophisticated visualization capabilities that enhance process monitoring. The 3D models of the process will overlap with relevant data or HMI elements (gauge charts, data series, alerts etc.) to unleash the potential of remote support, data-driven decision making, as well as deep bottle-necks analysis for operators, planners and designers. Output: Integrated3D process visualization module (Part of D5.4).

Work package WP6 – Kilns and plasma reactor prototyping

Work Package Number	WP6	**Lead Beneficiary**	1 - NTUA
Work Package Name	Kilns and plasma reactor prototyping		
Start Month	12	**End Month**	28
Objectives			
Objectives: (1) Construct a lab-scale IFRK for data generation and tests, (2) Finalise the design of the IFRKs for maximum resource efficiency (ROMs, process simulations), (3) Construct two TRL6 IFRKs for the calcination and roasting processes, (4) Construct a microwave rotary kiln for metal sulphides reduction, (5) Construct a microwaveinduced plasma reactor for H2 recovery, (6) Construct a heat exchanger for gas cooling, (7) Assess the overall feasibility of the processes.			
Description			

T6.1 Lab-scale IFRK prototype integration – (NTUA), M12-17
In T6.1, a small, lab-scale IFRK prototype unit will be manufactured, aiming to generate data to validate the simulations. An additional goal will be to assess and optimise the ability of the IFRK to handle different granulometries, with focus on processing fine and ultrafine divided solids effectively and without losses. Then, a feasibility study to confirm that the design minimizes raw material losses and enhances overall process efficiency will be carried out, to support the scaleup process. Output: Lab-scale IFRK prototype (D6.1).

T6.2 Reduced Order Models – (ITA), M14-20, [CIRCE]
The goal of this task is to build ROMs, based on the thermophysical models developed in WP2, to feed the process simulations (ASPEN) performed in T6.3. ITA will develop ROMs of the roasting process, built on the physics-based thermophysical models of BUL (T2.4). To this end, ITA will use a ROM generation library such as Twinkle and the results of a DoE of virtual experiments of the roasting IFRK, provided by BUL. ITA will support BUL in the definition of the DoE and then BUL will run the resulting extensive set of simulations of the thermophysical models These physicsinformed ROMs will be converted into executables that will be used by NTUA in T6.3 for the process optimization. Likewise, CIRCE will develop the ROMs for the calcination process. Output: (part of D6.2).

T6.3 IFRKs design optimisation – (NTUA), M16-23
Based on the outputs of T6.2 process simulations will be performed (e.g. using ASPEN) to obtain the optimal

design configurations towards the enhancement of the resource efficiency. The simulation will focus on modelling the calcination and roasting processes within the IFRK, including heat transfer, mass transfer, chemical reactions, and fluid dynamics. Through comprehensive modeling and sensitivity analysis, critical process parameters will be identified (maximum temperature, temperature ramp up, staying time etc.) to enhance the overall efficiency.Optimization techniques will also be applied to increase resource efficiency and minimize energy consumption. The activities will be reported in D6.2 and will include a detailed report, summarizing key findings and recommendations for IFRK design optimization. Output: Optimized IFRK design report (part of D6.2).

T6.4 Prototyping of IFRKs – (NTUA), M22-28, [FORN, HEGOLD, TERNAMAG]
The task includes prototyping the IFRKs to validate the design and optimize performance ensuring the reliability of calcination and roasting processes. Prototyping will include design specifications, selecting appropriate materials, constructing of the prototype kiln, conducting comprehensive performance testing under simulated conditions, and verifying the design through optimization based on test results. The activities will be reported in D6.2 and will include documentation of the prototyping process, construction details, performance analyses, and conclusions considering the overall performance and performance optimisation. Output: IFRK prototypes (part of D6.2).

T6.5 Prototyping of microwave rotary kiln for roasting – (CEIN), M19-28, [NTUA]
In this task CEIN will undertake the development and construction of an initial of microwave kiln prototype designed for continuous flow. The final prototype will result from detailed engineering and construction, to be safely validated using feedstock and hydrogen within a controlled cabinet environment. The task will also involve developing a system to monitor thermal and microwave parameters, which will be incorporated into basic software for comprehensive process parameter analysis. Additionally, a detailed examination of the conveying system will be performed to ensure its integration into the microwave applicator and to validate the process effectiveness. A gas circuit system will be designed to accommodate reactive gases like hydrogen, including provisions for extraction and collection during operation. The compact and portable microwave roasting pilot unit will encompass the newly developed microwave kiln, power source, and critical subsystems including raw material handling, monitoring, control software, gas treatment, and additional ancillary components like the chiller-water circuit, dust filtering, and solids collection. Output: Microwave rotary kiln prototype (D6.3).

T6.6 Prototyping of hydrogen recovery plasma reactor – (EUT), M19-28
In this task EUT will perform the construction, assembly and test of the plasma reactor prototype for the separation of the hydrogen sulphide. The works are related to the modification of waveguides, and material supply in the form of gas or the form of powder. Security will be present to avoid errors and potentially harmful accidents. In the first stages, the hydrogen sulphide will be supplied artificially, until the integration planned in WP7 becomes possible. Another important issue is the recuperation of the produced hydrogen and the devices related to it. This task is related to works performed in internal projects at EUT for hydrogen production from water steam or the decomposition of plastics to recover methane or related ions. The capture of H2 needs technology not sufficiently developed and an accurate process to maximize the yield. Output: Plasma treatment reactor prototype (part of D6.4).

T6.7 Prototyping of heat exchanger – (BUL), M13-28
BUL will work with EUT and will develop a prototype that will be integrated at the outlet of the reactor for the quenching of the gasses. The GCHP will be manufactured based on the results for T2.6. One lab-scale unit will be constructed and will be integrated with the remaining technology developed under the project. The unit will include all the necessary sensors to monitor and fully characterise the behaviour of the HPCE during operation. This will allow for a validation of the models used in T2.6. The manufacturing techniques employed for the construction of the unit will also ensure that the system lifespan is substantially higher than currently used technology and that efficient cooling control is exerted over the gasses. Output: Heat exchanger prototype (part of D6.4).

T6.8 Process feasibility assessment – (ADMR), M13-28, [EUROC]
This task encompasses activities related to the technology adoption framework. These activities include the implementation of the Technoeconomic Assessment (TEA), conducting Life Cycle Sustainability Assessments and utilizing Multi-Criteria Decision Analysis (MCDA) to ensure overall sustainability. Furthermore, it involves the collection of data to evaluate societal and workplace effects, exploring community benefits, environmental conservation, and workplace dynamics. It also guides community engagement efforts and proposes training programs to address skill gaps. Additionally, an assessment of health and safety impacts will be conducted to analyze the project's influence on workplace health and safety standards. Finally, the energy model for the virtual scale up study will be designed and initiated. EUROC will also consider the OHS aspects during the prototyping phase. Output: Consolidated assessment report (D6.5).

Work package WP7 – Integrated prototypes and pilot deployment

Work Package Number	WP7	**Lead Beneficiary**	3 - EUT
Work Package Name	Integrated prototypes and pilot deployment		
Start Month	24	**End Month**	36
Objectives			
Objectives: (1) Integrate all PRIM-ROCK prototypes into the pilot units; (2) Deploy the digital technologies and refine; (3) Validate the PRIM-ROCK pilots			
Description			
T7.1 Pilot unit integration – (EUT), M24-30, [CEIN, NTUA, ICCS, CIBOS, BUL, MAGG, IRIS, EUROC] T7.1 is focused on the integration, delivery and installation of the four prototypes, coordinating the integration of the necessary components provided by the other partners to EUT and NTUA. CEIN will integrate the microwave roasting pilot based on portable and compacted production unit, that will be designed to be easy to install. A user- friendly software will be improved, as well as the connection to process the ores. A system to capture and store the liberated gases will be improved to purify and extract the gases for further treatment. ICCS&CIBOS will integrate the RMI, IRIS and MAGG will integrate the sensors and controls correspondingly. BUL will integrate the heat exchanger. EUROC will monitor and report safety considerations, updating existing protocols. Output: Pilot integration report (part of D7.1). T7.2 Greek pilot validation – (NTUA), M30-36, [CEIN, EUT, ICCS, CIBOS, BUL, MAGG, IRIS, CORE, SCCH, STAM, TERNAMAG, HEGOLD, FORN] The task includes conducting a pilot validation of the calcination process set up in Lavrion. The installation and commissioning of the pilot-scale plant will be conducted, along with all necessary auxiliary equipment to ensure compliance with safety standards and regulations. Calcination trials (over 70 trials) will be executed to evaluate the kiln's performance under set operating conditions, monitoring all process parameters for further analysis. Performance evaluation will address energy efficiency, product quality, operational reliability, followed by operational optimization. The activities will be reported in D7.1 and will include the reporting of all pilot validation activities, assessment of the plant's performance and recommendations for optimisation. Output: Testing and validation of the Greek pilot (part of D7.1). T7.3 Spanish pilot validation– (EUT), M30-36, [CEIN, NTUA, ICCS, CIBOS, BUL, MAGG, IRIS, CORE, SCCH, STAM, KGHM] T7.3 is focused on the validation of the Spanish pilot (roasting process), set up in EUT. As part of this task, the integrated pilot will process at least 5 batches of materials in the necessary number of experiments (estimated over 60, will be defined by the output of the design of experiments). The work will be focused on two aspects: (a) roasting reduction efficiency and (b) hydrogen recovery through MIP including the gas cooling. The performance evaluation will focus on assessing energy efficiency, product quality, and operational reliability, subsequently leading to operational optimization. Findings from this evaluation, including all pilot validation activities, an assessment of the plant's performance, and optimization recommendations, will be documented in report D7.2. Output: Testing and validation of the Spanish pilot (part of D7.1).			

Work package WP8 – Pilot demonstrations

Work Package Number	WP8	**Lead Beneficiary**	1 - NTUA
Work Package Name	Pilot demonstrations		
Start Month	37	**End Month**	48
Objectives			
Objectives: (1) Demonstrate the PRIM-ROCK solutions in 4 industrial processes; (2) Assess the technological performance of the demonstrations.			

Description
T8.1 Demonstration of industrial processes #1 – (FORN), M37-48, [All partners] T8.1 addresses the emulation of the lime production process in the Greek pilot unit, operated by NTUA. This involves demonstrating the IFRK tailored to the calcination process, the RMI system and the associated DSS interface and effectiveness. For the calcination process, the KPIs on productivity and resource efficiency will be validated. Measurements will be taken with regards to CO2 quality. The demonstrations will be carried out both with reference samples and with samples dispatched by FORN. Key to cover the PRIM-ROCK IUC#1 objectives is the demonstration through live presentation of the digital twin in action, illustrating its capability to accurately simulate and optimize the calcination process under various scenarios. This will include real-time visualizations of operational parameters like temperature and pressure distributions within the kiln, demonstrating how the model can anticipate and mitigate potential issues, such as feedstock non-compliance or operational inefficiencies. The digital process-line optimisation results will be validated with experimental data, initiating iterative training processes and fine-tuning. During the demonstrations, operational parameters (e.g., rotation speed, inclination, temperature, staying time, heating ramp up for the kiln) will be closely monitored and assessed with regards to impact on resource efficiency, energy intensity and emissions-related metrics. Safety aspects will be monitored and reported. Output: Industrial process #1 demo results (part of D8.1). T8.2 Demonstration of industrial processes #2 – (TERNAMAG), M37-48, [All partners] Task 8.2 will comprise activities linked to the demonstration of the Magnesite production line in the Greek pilot unit. Similarly to T8.1, first the RMI system will dynamically and optimally pre-process the magnesite before calcinated through the IFRK. All the operational and control parameters will be monitored at real-time. A significant highlight of the demonstration will be the colour detection technology employed in optical sorting, which significantly improves the quality and efficiency of magnesite ore separation while also hand picking will also be used for the selection of high purity, special material. For the calcination process, the KPIs on productivity and resource efficiency will be validated. Measurements will be taken with regards to CO2 quality. The digital twin functions will also be demonstrated and finetuned. The KPIs for validating the IUC #2 will serve the PRIM-ROCK IUC#2 objectives including the production of very clean CO2 for capture, that will be validated through the virtual toolkit, and increase of material recovery through sorting, filtering and the IFRK up to 30%. Safety aspects will be monitored and reported. Output: Industrial process #3 demo results (part of D8.1). T8.3 Demonstration of industrial processes #3 – (NTUA), M37-48, [All partners] In this task the demonstration of Ferronickel production will be carried out in the Spanish pilot unit. The RMI system will pre-process the nickeliferous laterite, to be subsequently roasted in the IFRK to reduce iron, nickel and cobalt oxides. The key highlight will be the replacement of carbon with hydrogen in the reduction process, significantly reducing environmental impact and energy usage. Measurements will be taken with regards to H2 quality. The digital framework will be demonstrated here as well, with emphasis given in the real-time monitoring of the process and the enhancement of the resource efficiency. KPIs will be obtained regarding the individual performance of each solution and the overall performance of the pilot-line. Safety aspects will be monitored and reported. Output: Industrial process #2 demo results (part of D8.1). T8.4 Demonstration of industrial processes #4– (KGHM), M37-48, [All partners] This task is focused on the demonstration activities in relation to the roasting of metal sulphides (ZnS and CuS) in both pilot units. The demonstration will comprehensively display the entire process-line including the pre-processing of the metal sulphides through the RMI system, followed by the roasting process in a H2 atmosphere utilising both the IFRK (Greek pilot unit) and the MRK (Spanish pilot unit) prototypes. The highlight of this demonstration will be the recovery of the H2 through the MIP reactor and the controlled gas cooling from the heat exchanger. Relevant operational parameters will be monitored, controlled and optimized through the digital functions. Analysis on the utilisation of the recovered H2 for e-Methanol production will be conducted. KPIs will be validated to provide valuable insights into how the PRIM-ROCK IUC#4 objectives serve the metal sulphide processing industry, making it more efficient, costeffective, and environmentally friendly. Output: Industrial process #4 results (part of D8.1). T8.5 Technological assessment of demonstrations – (NTUA), M42-48, [All partners] This task will pursue the technological assessment of the developed technological solutions and will provide quantitative results on the monitoring parameters and KPIs identified in previous tasks. Indicators such as resource efficiency, improved application deployment time, production time efficiency, quality of products, ratio of estimated energy gains, Carbon reduction as well as use-case specific KPIs will be addressed. In addition, a set of KPIs and correspondent targets values will be selected to measure the absolute sustainability beyond the SoA. Tests will include short term trials

of coupled plants, modifications, optimization, long term materials production campaign and load variations. The last period of the experimental work is a long-term trial, demonstrating continuous operation of the production units. The

produced materials and water will be collected and sent for analysis in laboratories in NTUA. Output: Demo assessment results (part of D8.1).

Work package WP9 – Adoption framework consolidation

Work Package Number	WP9	Lead Beneficiary	12 - ADMR
Work Package Name	Adoption framework consolidation		
Start Month	29	End Month	48

Objectives

Objectives: In this WP a set of assessments and studies will be conducted on the PRIM-ROCK solutions, involving: (1) Technical and economic feasibility; (2) Environmental, economic and social impacts; (3) Workplace effects and improvements; (4) Health and Safety; (5) Replication and uptake; (6) Policies and regulations.

Description

T9.1 Technoeconomic and energy assessments – (ADMR), M29-48, [All partners]
This task will evaluate the feasibility and economic viability of the PRIM-ROCK breakthrough innovations. This involves analysing costs, market potential, financial risks, and the overall return on investment. We will utilize a combination of engineering and financial modelling to forecast project expenditures and revenues, benchmarking against industry standards to ensure competitiveness and profitability. This task will also measure the energy efficiency and intensity of the project's processes. Energy consumption will be benchmarked against outputs to calculate energy intensity ratios. Strategies for minimizing energy use while maximizing output will be developed, considering the adoption of renewable energy sources and energy recovery systems to reduce the carbon footprint. Output: D9.1 TEA and energy assessments

T9.2 LCA, LCC, LCSA – (PRINS), M29-48, [All partners]
LCSA will assess the overall sustainability of the proposed innovations by systematically evaluating the environmental, economic and social impacts throughout its life cycle (cradle-to-gate). This includes raw material extraction, manufacturing, and end-of-life disposal. The goal is to identify and quantify the environmental, economic and social loadings and draw a comprehensive picture of the project's overall sustainability footprint and develop mitigation strategies. The task's objectives will be carried out in the following steps: (i) Life Cycle Assessment (LCA) covers the assessment of the environmental sustainability aspect of PRIM-ROCK's processes. It focuses on the entire value chain from the extraction of raw materials up to the final products and the EoL phase of waste flows by following the ISO14040 and ISO14044 recommendations. Specifically, the methodology involves 4 distinct analytical steps: Goal & Scope definition, Inventory analysis (data collection and evaluation), Impact Assessment and Interpretation of the results. Hot spots will be identified, and new solutions will be proposed towards optimization and strategies development (M29-48); (ii) Socio-economic sustainability assessment will be carried out through Social LCA (s-LCA) and Life Cycle Costing (LCC). S-LCA is used to assess the social impacts of the product based on UNEP's Life Cycle Initiative Guidelines. The social data will be collected in collaboration with project partners and various stakeholders. Different social indicators will be calculated and aggregated on scores and indexes to assess the social impact of the products/processes (M29-48). LCC analysis will reveal the economic feasibility and cost-effectiveness of the proposed processes, following the same methodology described in (i), accounting for the relative cost flows. Relevant financial data will be calculated while the appropriate economic indicators will be established to estimate and minimize the economic impacts and possible risks, pinpointing which processes should be further improved to reduce any possible economic load (M29-48); (iii) Life Cycle Sustainability Assessment (LCSA) will concatenate environmental, economic, and social indicators to perform the overall sustainability assessment of project's innovations utilising Multi-Criteria Decision Analysis (MCDA). For this purpose, a set of evaluation criteria (covering all three aspects of sustainability) will be developed in collaboration with project partners (M36-48). Output: Life Cycle Sustainability Assessment report (D9.2).

T9.3 Capacity building, workplace effect maximisation – (EUROC), M29-48, [All partners]
Aiming to enhance workplace dynamics, productivity, and skill utilization, a structured methodology will be followed including assessment of the impact of the proposed technologies in labour, identification of skill gaps and development of targeted training programs to address the specific needs within the workforce. Through capacity building enhancement within the organizations, workplace dynamics can be fostered, maximizing efficiency and effectiveness. Output: Capacity building portfolio (part of D9.3).

T9.4 Health and safety impacts assessment– (EUROC), M29-48, [All partners]

This task will evaluate the health and safety performance of the PRIM-ROCK technologies. This involves assessing the identified hazards, through the evaluation of likelihood and severity, and development of a mitigation plan to control the risks arising from the proposed technologies. A monitoring and surveillance program will be developed so as to evaluate the effectiveness of the control measures and assess the remaining health and safety risk. An emergency response plan will be drafted to address potential health and safety emergencies. Output: Standardisation report (part of D9.3).

T9.5 Virtual scale up and expansion study– (CORE), M29-48, [NTUA]
Based on the outputs of the final designs and digital twins, this task will carry out a scalability study with two components: (1) suggest configurations within the energy system's boundary conditions, based on the user's inputs that will set the objective function for each scenario, maximizing renewable integration as a constraint and (2) virtual linking of additional added value technologies, mainly CCUS and synthetic fuel production. A Reinforcement Learning (RL) algorithm will be employed to train the planning model. Planning a scalable and expandable energy system will benefit from RL, due to the algorithm's ability in making sequential decisions in complex and changing environments. This approach also provides the advantage of being able to consider long-term effects of decisions and adapts well to changing objective functions. Black-box models will be employed for the electrolyser, CCUS, IFRK and other complex components, meaning that the model will classify these elements as energy sources and sinks, while still considering weighed values assigned to their lifecycle (e.g., CAPEX, OPEX, production uptime, product purity or quality). This will require the collaboration of domain experts and will be validated with experimental data towards the end of the prototype validation process. Output: Industrial uptake analysis (D9.4).

T9.6 Legislative and regulatory monitoring– (VDU), M29-48, [All partners]
T9.6 will provide policy recommendations to the EU and EEA based TG4-5 to overcome the barriers identified and capitalize on favourable technological, infrastructural and institutional lock-ins. To expedite accelerated adoption of the PRIM-ROCK technology portfolio in the process industry, the policy recommendations will include command-andcontrol instruments including bans, targets and performance standards; financial instruments including both penalties and incentives; management and planning instruments; along with education, RDI and dissemination instruments. The set of recommended policy instruments will be crafted on grounds of the consortium's scientific and other output, observation techniques employed in industry and other events, stakeholder interviews (TG1-4), and a Qualitative Comparative Analysis (QCA) that allows for causal analysis of changes sought in complex settings such as the process industries. Output: (D9.5) Scientific articles and policy report on the recommendations for TG4

Work package WP10 – Early phase Dissemination Exploitation Communication

Work Package Number	WP10	**Lead Beneficiary**	2 - CORE
Work Package Name	Early phase Dissemination Exploitation Communication		
Start Month	1	**End Month**	18
Objectives			
Objectives: (1) Produce a strategic communication and dissemination plan and a PRIM-ROCK identity brand; (2) Raise awareness of PRIM-ROCK objectives and achievements; (3) Engage and maintain a strong stakeholder network through a series of events and build synergies to facilitate knowledge exchange and transfer with similar projects, as well as relevant initiatives; (4) Develop and implement a strategy to identify relevant existing and emerging standards, to inform the technical work, and to engage effectively with standardization bodies to influence standardization efforts aligned with project goals.			
Description			

T10.1: Dissemination and communication activities – (CORE), M1-18, [All partners]
The early phase will be dedicated to raising awareness among the target groups of the PRIM-ROCK's scope and objectives, outlined in §2.2 and to be specified in PEDR. The project's brand, including all the aesthetic elements (logo, templates for project deliverables, presentations, leaflets, website, etc) will be crafted and translated into all project languages to extend our reach across Europe. Media such as videos capturing the key development stages of the project will be produced and shared. Dissemination activities will target both the academic and industry professional communities with scientific publications and public presentations on the results of the R&D activities, aligned with the Open Science policy. Online events like webinars, congresses, etc. will hosted by partners or in conjunction with clusters (e.g., P4Planet, Made in Europe) and sectorial associations to further spread our results.
Output: Brand and visual identity of the project, project website, draft PEDR (part of D10.1).

T10.2 Exploitation and innovation management activities – (STAM), M01-18, [All partners]
In this early phase a detailed analysis of the PRIM-ROCK Key Exploitable Results (KERs) will be conducted, exploring their market potential, exploitability, and future impact to industry. For all KERs, an IP strategy that meets the principles "as open as possible, as closed as necessary", will be initiated. Based on a literature search and PESTLE analysis, barriers to innovation, i.e., deployment barriers from the supply side and adoption barriers from the demand side, and success factors will be analysed. Additionally, in this task the main stakeholders of the project will be enquired and profiled, also developing a plan to engage them according to the different needs and interests in the project. Output: Market analysis, exploitation part of the PEDR (part of D10.1).

T10.3 Standardization landscape analysis and strategy – (UNE), M1-18, [All partners]
Starting from the needs of all the partners and in collaboration with them, existing standards, as well as those under development, of relevance for the project, together with the standardization organizations and technical committees involved, will be identified and reported (D10.2, M6). This early document will be an asset for informing the development work to ensure compliance with relevant standards. In addition, a strategy for an effective contribution to standardization will also be prepared (D10.2, M18). This will include different options for a contribution and will identify the most relevant Standardization Technical Committees to be targeted for a relevant impact. The possibility and the extent to which a contribution to standardization can be made will be investigated through interaction with the relevant standardization technical committees (i.e. monitoring their information, attending meetings, establishing formal liaisons, organizing joint events, etc.), disseminating the project and getting feedback from their members. This will enable to further assess further the possibility and the extent to which a contribution to standardization can be made. Output: Standardization landscape, applicable standards and strategy (D10.2).

Work package WP11 – Mid-phase Dissemination Exploitation Communication

Work Package Number	WP11	**Lead Beneficiary**	2 - CORE
Work Package Name	Mid-phase Dissemination Exploitation Communication		
Start Month	19	**End Month**	36
Objectives			
Objectives: (1) Focus on further engagement of the targeted audiences with the project; (2) Disseminate evolving results through events and publications and generate additional interest in PRIM-ROCK; (3) Inform target markets about the technological breakthroughs and business benefits of PRIM-ROCK; (4) Define exploitation routes, analyse target market and competitive landscape; (5) Actively engage in and contribute to ongoing standardization efforts, influencing the development or modification of standards in line with the project's advancements and goals.			
Description			

T11.1: Dissemination and communication activities – (CORE), M19-36, [All partners]
The task will continue to pursue the goals and the activities described in T10.1. The middle phase will be dedicated to growing and consolidating awareness, informing stakeholders and groups about technological breakthroughs and business benefit. Output: 1 project video, 8 publications (part of D11.1).

T11.2 Exploitation and innovation management activities – (STAM), M19-36, [All partners]
In this Mid-phase, the identified barriers to innovation and success factors will be further and deeper analysed based on the feedback from external stakeholders, such as sister R&D projects). The exploitation routes of each KER will be also evaluated. Specifically, for KERs with a commercial orientation, the value proposition will be designed to analyze the unique benefits that PRIM-ROCK provides to its target customers and users. A comprehensive market analysis will be carried out to assess market size, outlook and dynamics, which will be complemented by a SWOT analysis. Additionally, the competitive landscape will be assessed through a Porter 5 Forces analysis. In parallel, EUROC will conduct a Sustainable Business Investment assessment in a comprehensive investment strategy ensuring the long-term viability and impact of PRIM-ROCK innovations. Data-driven analytics and feedback mechanisms will be used to adapt investment strategies in response to market trends, stakeholder feedback, and evolving sustainability standards. Output: PEDR (part of D11.1).

T11.3 Contribution to standardization I – (UNE), M19-36, [All partners]
Based on the preparatory work done in T10.3, collaboration with standardization organizations will be continued. Depending on the ongoing works of the technical committees, this may involve providing information or participating in ongoing works, or supporting the development of standards via comments to drafts or the direct participation in standardization working groups within the relevant technical committees. Depending on this external situation, the progress of the project results and the exploitation strategies of the partners, as well as gaps or barriers in existing standards or topics not yet addressed by standards, the best possible contributions to new standardization activities will be identified, evaluated, selected and launched. Output: Standardisation report I (D11.2).

Work package WP12 – Final phase Dissemination Exploitation Communication

Work Package Number	WP12	**Lead Beneficiary**	2 - CORE
Work Package Name	Final phase Dissemination Exploitation Communication		
Start Month	37	**End Month**	48

Objectives

Objectives: (1) Focus on maximizing target market and industry awareness on PRIM-ROCK's exploitable results (KERs); (2) Disseminate results through all available channels, (3) Leverage communication and dissemination efforts and support the project sustainability and its effective exploitation and market replication; (4) Design PRIM- ROCK Unique Selling Points (USPs); (5) Develop PRIM-ROCK Exploitation Roadmap and business model (underpinned by a compelling Business Case); (6) Execute and advance identified standardization activities from T11.3, focusing on the modification of existing standards or the development of new standards according to the project's developments.

Description

T12.1: Dissemination and communication activities – (CORE), M37-48, [All partners]
The goal of the final phase is to ensure the long-term impact of the project by establishing appropriate lines of communication to maximize influence on policy and decision makers within targeted communities (industry, research, academia). At least 18 publications will be made. During this phase, the final PEDR will be consolidated and the final KPIs will be achieved. Output: PEDR (part of D12.1).

T12.2 Exploitation and innovation management activities – (STAM), M37-48, [All partners]
Based on the competition analysis and the identified target customer needs (produced in the previous phase), the Unique Selling Points (USPs) of the PRIM-ROCK KERs will be designed. A compelling Business Case, and an exploitation roadmap to reach TRL9, will be developed for the PRIM-ROCK KERs aiming for commercial use. The roadmap will be accompanied by suitable business models (using the Business Model Canvas tool) built upon the foundations of sustainable business model arche-types. Output: PEDR (part of D12.1).

T12.3 Contribution to standardization II – (UNE), M37-48, [All partners]
Following the strategic decisions made in T11.3 regarding the standardization potential and the optimal routes for their development, the corresponding standardization procedures will be deployed. The main outcome of the task is to leverage PRIM-ROCK's contributions towards the development of new standards in areas pertinent to the project's objectives and results, such as processing procedures, best practices for implementation, criteria and indicators for decision. This may include providing information to technical committees, participating in ongoing works, submitting proposals for the modification of existing standards or promoting the elaboration of new standards (like technical specifications, TS). The development of fast-track standards like CEN-CENELEC Workshop Agreements (CWA) is usually the best-suited option in the frame of a Horizon Europe project. Output: Standardisation report II (D12.2).

Work package WP13 – Project Management

Work Package Name	Project Management		
Start Month	1	**End Month**	48

Objectives

Objectives: This WP is structured to (1) ensure that PRIM-ROCK achieves its objectives within the time frame and budget as outlined in the contract; (2) provide strategic, technical (including risk management) and ethical (and inclusive) coordination; (3) manage and monitor project resources (staff, budget, equipment, etc.); (4) carry out quality control and data management; (5) handle reporting to the European Commission (EC) and; (6) ensure consistent internal communication as well as with the EC.

Work Package Number	WP13	**Lead Beneficiary**	1 - NTUA

Description
T13.1 – Administrative and financial project management – (NTUA), M1-48, [All partners] This task includes the comprehensive legal, contractual, financial and administrative oversight of the project. Responsibilities include: a) compiling and overseeing technical and financial progress reports; b) performing final reviews and submitting all required documents, such as deliverables and progress reports; c) managing costs, controlling budget and accounting; d) arranging and documenting Project Management Team meetings; e) monitoring and controlling project progress in alignment with the GA; f) addressing legal and contractual issues, such as the maintenance of the GA, CA and amendments (if needed). NTUA will be the Project Coordinator (PC), acting as the main contact point between the EC Project Officer and the Consortium. The PC's additional duties involve informing all partners on the EC communications and requirements, gather all the needed information from the partners and respond to the EC. NTUA will lead the internal communications within the Consortium, ensuring all partner are informed. Regular meetings and a variety of communication channels (regular virtual/hybrid/physical meetings; periodic reports or others) will be utilised to ensure an optimal information flow among all partners. The Project management will be supported by setting up and using collaborative software tools: documentation and common data repositories and sharing tools. Output: Project handbook (M6), Management reports (part of D13.1). T13.2 Technical and scientific project management – (NTUA), M1-48, [All partners] In collaboration with the WP leaders, NTUA will monitor the technical and scientific activities of the project to ensure the fulfilment of both the overall project objectives and individual WP targets, and the effective coordination of interdisciplinary efforts across the work plan. Regular technical progress reports will be compiled to monitor advancement, allowing for necessary adjustments and alterations to the project's activities. Effective risk management will be prioritized, aimed at controlling the critical risks identified in Table 3.1e, and other risks that may emerge, to ensure a proactive approach and effective implementation of mitigation strategies. Ultimately, NTUA, supported by the experienced WP leaders, will ensure that technical work is executed in line with project objectives, proposing modifications and reallocating resources when needed. Output: (part of D13.1). T13.3 Ethics management and inclusiveness – (VDU), M1-48, [All partners] This task ensures the protection of privacy and confidentiality for individuals whose data is collected, as the project involves extensive data collection and storage practices. This includes gaining participants' informed consent, maintaining data anonymity and confidentiality, and securely storing and processing sensitive information. Additionally, it promotes an inclusive and gender-sensitive approach to PRIM-ROCK research, emphasizing equitable participation. Ethical considerations are embedded throughout the project lifecycle, from the design and execution of data collection to the analysis and dissemination of findings. The project team commits to strict standards to guarantee that individuals' rights and welfare are respected and protected. Output: (part of D13.1).

LIST OF DELIVERABLES

Deliverables

Grant Preparation (Deliverables screen) — Enter the info.

The labels used mean:

Public — fully open (automatically posted online)

Sensitive — limited under the conditions of the Grant Agreement

EU classified —RESTREINT-UE/EU-RESTRICTED, CONFIDENTIEL-UE/EU-CONFIDENTIAL, SECRET-UE/EU-SECRET under Decision 2015/444

Deliverable No	Deliverable Name	Work Package No	Lead Beneficiary	Type	Dissemination Level	Due Date (month)
D1.1	Project specifications	WP1	1 - NTUA	R — Document, report	SEN - Sensitive	6
D1.2	Expanded specifications	WP1	14 - EUROC	R — Document, report	SEN - Sensitive	12
D2.1	MFA report	WP2	5 - CIRCE	R — Document, report	SEN - Sensitive	12
D2.2	Raw material intelligence design	WP2	4 - ICCS	R — Document, report	SEN - Sensitive	15
D2.3	IFRK models and design for calcination process	WP2	5 - CIRCE	OTHER	SEN - Sensitive	16
D2.4	IFRK models and design for roasting process	WP2	6 - ITA	OTHER	SEN - Sensitive	16
D2.5	Microwave rotary kiln models and design	WP2	9 - CEIN	OTHER	SEN - Sensitive	18
D2.6	Plasma reactor models and design	WP2	3 - EUT	OTHER	SEN - Sensitive	18
D3.1	PRIM-ROCK platform design principles	WP3	10 - IRIS	OTHER	SEN - Sensitive	18
D3.2	Sensors and controls retrofitting report	WP3	15 - MAGG	R — Document, report	SEN - Sensitive	18

D4.1	Feedstock intelligence system	WP4	4 - ICCS	DEM — Demonstrator, pilot, prototype	SEN - Sensitive	28
D5.1	Control and automation system	WP5	15 - MAGG	OTHER	SEN - Sensitive	30
D5.2	Process data models	WP5	7 - SCCH	OTHER	SEN - Sensitive	36

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Deliverable No	Deliverable Name	Work Package No	Lead Beneficiary	Type	Dissemination Level	Due Date (month)
D5.3	PRIM-ROCK IoT data platform	WP5	10 - IRIS	OTHER	SEN - Sensitive	36
D5.4	Integrated HDT	WP5	2 - CORE	OTHER	SEN - Sensitive	36
D6.1	IFRK lab scale prototype	WP6	1 - NTUA	DEM — Demonstrator, pilot, prototype	SEN - Sensitive	18
D6.2	IFRK prototypes	WP6	1 - NTUA	DEM — Demonstrator, pilot, prototype	SEN - Sensitive	28
D6.3	Microwave kiln prototype	WP6	9 - CEIN	DEM — Demonstrator, pilot, prototype	SEN - Sensitive	28
D6.4	MIP reactor and heat exchanger prototypes	WP6	3 - EUT	DEM — Demonstrator, pilot, prototype	SEN - Sensitive	28

D6.5	Feasibility assessment and impacts	WP6	12 - ADMR	R — Document, report	SEN - Sensitive	28
D7.1	Validated systems	WP7	1 - NTUA	DEM — Demonstrator, pilot, prototype	SEN - Sensitive	36
D8.1	PRIM-ROCK demonstration	WP8	1 - NTUA	R — Document, report	SEN - Sensitive	48
D9.1	TEA and efficiency gains	WP9	12 - ADMR	R — Document, report	SEN - Sensitive	48
D9.2	Life Cycle Sustainability Assessment report	WP9	13 - PRINS	R — Document, report	SEN - Sensitive	48
D9.3	Capacity building portfolio, health & safety	WP9	14 - EUROC	R — Document, report	PU - Public	48
D9.4	Industrial uptake analysis	WP9	2 - CORE	R — Document, report	PU - Public	48

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Deliverable No	Deliverable Name	Work Package No	Lead Beneficiary	Type	Dissemination Level	Due Date (month)
D9.5	Legislation and regulation	WP9	20 - VDU	R — Document, report	PU - Public	48
D10.1	Early PEDR [1st update]	WP10	2 - CORE	R — Document, report	SEN - Sensitive	6
D10.2	Standardization landscape, and strategy [1st update]	WP10	21 - UNE	R — Document, report	PU - Public	6

D10.3	Early PEDR [2nd update]	WP10	2 - CORE	R — Document, report	SEN - Sensitive	18
D10.4	Standardization landscape, and strategy [2nd update]	WP10	21 - UNE	R — Document, report	PU - Public	18
D11.1	Mid PEDR	WP11	2 - CORE	R — Document, report	SEN - Sensitive	36
D11.2	Standardization report I	WP11	21 - UNE	R — Document, report	PU - Public	36
D12.1	Final PEDR	WP12	2 - CORE	R — Document, report	SEN - Sensitive	48
D12.2	Standardization report II	WP12	21 - UNE	R — Document, report	PU - Public	48
D13.1	Project handbook	WP13	1 - NTUA	R — Document, report	SEN - Sensitive	6
D13.2	Project management quality assessment Ver.1	WP13	1 - NTUA	R — Document, report	SEN - Sensitive	18
D13.3	Project management quality assessment Ver.2	WP13	1 - NTUA	R — Document, report	SEN - Sensitive	36
D13.4	Project management quality assessment Ver.3	WP13	1 - NTUA	R — Document, report	SEN - Sensitive	48

Deliverable D1.1 – Project specifications

Deliverable Number	D1.1	**Lead Beneficiary**	1 - NTUA
Deliverable Name	Project specifications		
Type	R — Document, report	**Dissemination Level**	SEN - Sensitive
Due Date (month)	6	**Work Package No**	WP1
Description			
Specifications including the outputs of T1.1-T1.3			

Deliverable D1.2 – Expanded specifications

Deliverable Number	D1.2	**Lead Beneficiary**	14 - EUROC
Deliverable Name	Expanded specifications		
Type	R — Document, report	**Dissemination Level**	SEN - Sensitive
Due Date (month)	12	**Work Package No**	WP1
Description			
Assessments baseline and safety specifications			

Deliverable D2.1 – MFA report

Deliverable Number	D2.1	**Lead Beneficiary**	5 - CIRCE
Deliverable Name	MFA report		
Type	R — Document, report	**Dissemination Level**	SEN - Sensitive
Due Date (month)	12	**Work Package No**	WP2
Description			
Material flow analysis for the end-user processes. Detailed report, serving as a valuable guide for implementing sustainable practices in the designed industrial process.			

Deliverable D2.2 – Raw material intelligence design

Deliverable Number	D2.2	**Lead Beneficiary**	4 - ICCS
Deliverable Name	Raw material intelligence design		
Type	R — Document, report	**Dissemination Level**	SEN - Sensitive
Due Date (month)	15	**Work Package No**	WP2
Description			

Design for the integrated RMI system. In-depth blueprint analysis report that maps the configuration of sensor networks, the selection of suitable sensor technologies (including XRF, hyperspectral imaging, LIBS), and the crafting of AI-driven algorithms for advanced data analysis and enhancement of the resource efficiency of the processes.

Deliverable D2.3 – IFRK models and design for calcination process

Deliverable Number	D2.3	**Lead Beneficiary**	5 - CIRCE
Deliverable Name	IFRK models and design for calcination process		
Type	OTHER	**Dissemination Level**	SEN - Sensitive
Due Date (month)	16	**Work Package No**	WP2
Description			
Physicochemical & thermophysical simulations and designs for the IFRKs for the calcination processes. DEM-CFD steady-state model for flow dynamics, heat transfer and chemical reactions within the calcinatory, characterizing its operation in detail.			

Deliverable D2.4 – IFRK models and design for roasting process

Deliverable Number	D2.4	**Lead Beneficiary**	6 - ITA
Deliverable Name	IFRK models and design for roasting process		
Type	OTHER	**Dissemination Level**	SEN - Sensitive
Due Date (month)	16	**Work Package No**	WP2
Description			
IFRK models and design for roasting process, analysing kinetics of the hydrogen reduction in relation with temperature and hydrogen partial pressure as well as the materials properties			

Deliverable D2.5 – Microwave rotary kiln models and design

Deliverable Number	D2.5	**Lead Beneficiary**	9 - CEIN
Deliverable Name	Microwave rotary kiln models and design		
Type	OTHER	**Dissemination Level**	SEN - Sensitive
Due Date (month)	18	**Work Package No**	WP2
Description			
Simulation results and design of the microwave rotary kiln for sulphide roasting, amalysing static microwave set ups and primary process parameters in terms of power demand, produced gas and gasses additions following up the reaction, and solid by-products. Description of control and safety systems material conveyance, identification of residence time and appropriate microwave irradiation exposure.			

Deliverable D2.6 – Plasma reactor models and design

Deliverable Number	D2.6	**Lead Beneficiary**	3 - EUT
Deliverable Name	Plasma reactor models and design		
Type	OTHER	**Dissemination Level**	SEN - Sensitive

Due Date (month)	18	**Work Package No**	WP2
Description			
Simulation results and design of the MIP reactor for H2 recovery. Results analysis of from proofs of concept tests, supported by multiphysics simulations. Simulation and design of the Gas Cooling Heat Pipe.			

Deliverable D3.1 – PRIM-ROCK platform design principles

Deliverable Number	D3.1	**Lead Beneficiary**	10 - IRIS
Deliverable Name	PRIM-ROCK platform design principles		
Type	OTHER	**Dissemination Level**	SEN - Sensitive
Due Date (month)	18	**Work Package No**	WP3
Description			
IoT reference architecture and integration guidelines. Design principles for IT architecture, technical requirements for the Scadalytics Industrial IoT data platform specifications.			

Deliverable D3.2 – Sensors and controls retrofitting report

Deliverable Number	D3.2	**Lead Beneficiary**	15 - MAGG
Deliverable Name	Sensors and controls retrofitting report		
Type	R — Document, report	**Dissemination Level**	SEN - Sensitive
Due Date (month)	18	**Work Package No**	WP3
Description			
Report on the monitoring and control network system, user interface (UI) definition. Result analysis comparing in-line sensors data versus conventional analytical tools available.			

Deliverable D4.1 – Feedstock intelligence system

Deliverable Number	D4.1	**Lead Beneficiary**	4 - ICCS
Deliverable Name	Feedstock intelligence system		
Type	DEM — Demonstrator, pilot, prototype	**Dissemination Level**	SEN - Sensitive
Due Date (month)	28	**Work Package No**	WP4
Description			
Integrated system comprised of T4.1-4 outputs. Integrated sensor and control solution, tailor-made for high-precision ore processing applications, ready to be implemented in the RMI system. Utilisation of hyperspectral cameras, capable of densely sampling the electromagnetic spectrum up to 5000nm. Results analysis RMI efficiency towards off-line analytical techniques			

Deliverable D5.1 – Control and automation system

Deliverable Number	D5.1	**Lead Beneficiary**	15 - MAGG
Deliverable Name	Control and automation system		

Type	OTHER	**Dissemination Level**	SEN - Sensitive
Due Date (month)	30	**Work Package No**	WP5
Description			
Process Control System and associated documentation analysing monitoring management, and control of the processes for eventual automated control and decision-making capabilities			

Deliverable D5.2 – Process data models

Deliverable Number	D5.2	**Lead Beneficiary**	7 - SCCH
Deliverable Name	Process data models		
Type	OTHER	**Dissemination Level**	SEN - Sensitive
Due Date (month)	36	**Work Package No**	WP5
Description			
Data-driven models' for calcination and roasting processes able to make use of the real-world data from the test setups and to ensure real process accuracy. Report analysing data structure, models ontology and AI-based techniques integrated.			

Deliverable D5.3 – PRIM-ROCK IoT data platform

Deliverable Number	D5.3	**Lead Beneficiary**	10 - IRIS
Deliverable Name	PRIM-ROCK IoT data platform		
Type	OTHER	**Dissemination Level**	SEN - Sensitive
Due Date (month)	36	**Work Package No**	WP5
Description			
Orchestration software that may serve various data sources, Report analysing Scadalytics Data Management Platform, microservices architecture, Communication Middleware components and real-time-data analytics, simulation and digital twin components			

Deliverable D5.4 – Integrated HDT

Deliverable Number	D5.4	**Lead Beneficiary**	2 - CORE
Deliverable Name	Integrated HDT		
Type	OTHER	**Dissemination Level**	SEN - Sensitive
Due Date (month)	36	**Work Package No**	WP5
Description			
Integrated HDT at process line level with 3D process visualization module. integrated digital twinning analysis performing real-time insights and alerts, optimizing the process for maximum efficiency. Description of virtual interaction with the processes through Human-Machine Interfaces for providing sophisticated visualization capabilities to enhance process monitoring.			

Deliverable D6.1 – IFRK lab scale prototype

Deliverable Number	D6.1	**Lead Beneficiary**	1 - NTUA
Deliverable Name	IFRK lab scale prototype		
Type	DEM — Demonstrator, pilot, prototype	**Dissemination Level**	SEN - Sensitive
Due Date (month)	18	**Work Package No**	WP6
Description			
Technical report of the lab scale IFRK. Results analysis for evaluating raw material losses and simulation models validation. TRL4 for operational data generation and model training.			

Deliverable D6.2 – IFRK prototypes

Deliverable Number	D6.2	**Lead Beneficiary**	1 - NTUA
Deliverable Name	IFRK prototypes		
Type	DEM — Demonstrator, pilot, prototype	**Dissemination Level**	SEN - Sensitive
Due Date (month)	28	**Work Package No**	WP6
Description			
From ROMs and process simulations to the IFRK prototyping. Documentation of the prototyping process, construction details, performance analyses, and conclusions considering the overall performance and performance optimisation. Technical specifications: 2m, 4-10kg/h, 6 heating zones up to 1200oC.			

Deliverable D6.3 – Microwave kiln prototype

Deliverable Number	D6.3	**Lead Beneficiary**	9 - CEIN
Deliverable Name	Microwave kiln prototype		
Type	DEM — Demonstrator, pilot, prototype	**Dissemination Level**	SEN - Sensitive
Due Date (month)	28	**Work Package No**	WP6
Description			
Report on the Microwave rotary kiln prototyping for sulphide roasting. Documentation for development and construction of an initial of microwave kiln prototype, detailed engineering for conveying system and overall piping for using safely hydrogen. Technical specifications: tube reactors of 5-12cm diameter, 915MHz, 5-30kg/h.			

Deliverable D6.4 – MIP reactor and heat exchanger prototypes

Deliverable Number	D6.4	**Lead Beneficiary**	3 - EUT
Deliverable Name	MIP reactor and heat exchanger prototypes		
Type	DEM — Demonstrator, pilot, prototype	**Dissemination Level**	SEN - Sensitive
Due Date (month)	28	**Work Package No**	WP6

Description
Report on the MIP reactor prototyping for H2 recovery and the heat exchanger, Detailed documentation of construction, assembly of the plasma reactor prototype for the separation of the hydrogen sulphide, modification of waveguides, and material supply in the form of gas or the form of powder. Technical specifications: (projected 70-100% H2S concentration, at flow rates of 0.5-3l/sec, preheating temperature)

Deliverable D6.5 – Feasibility assessment and impacts

Deliverable Number	D6.5	**Lead Beneficiary**	12 - ADMR
Deliverable Name	Feasibility assessment and impacts		
Type	R — Document, report	**Dissemination Level**	SEN - Sensitive
Due Date (month)	28	**Work Package No**	WP6
Description			
Consolidated pre-assessment report on the feasibility of PRIM- ROCK solutions related to the technology adoption framework. OHS aspects for prototyping phase			

Deliverable D7.1 – Validated systems

Deliverable Number	D7.1	**Lead Beneficiary**	1 - NTUA
Deliverable Name	Validated systems		
Type	DEM — Demonstrator, pilot, prototype	**Dissemination Level**	SEN - Sensitive
Due Date (month)	36	**Work Package No**	WP7
Description			
Technology validation of the two pilot units, delivery and installation of the four prototypes to EUT and NTUA. Report analysis for prototypes integration, sensors, controls and IoT commissioning, Results from calcination trials (over 70 trials), performance evaluation will address energy efficiency, product quality, operational reliability, followed by operational optimization.			

Deliverable D8.1 – PRIM-ROCK demonstration

Deliverable Number	D8.1	**Lead Beneficiary**	1 - NTUA
Deliverable Name	PRIM-ROCK demonstration		
Type	R — Document, report	**Dissemination Level**	SEN - Sensitive
Due Date (month)	48	**Work Package No**	WP8
Description			
Report on the demonstration of the pilot units and the technological assessments. Integrated pilot process utilisation for at least 5 batches of materials in the necessary number of experiments (estimated over 60, will be defined by the output of the design of experiments). Documentation on productivity and resource efficiency validation, CO2 quality, H2 recovery and IoT technological assessment.			

Deliverable D9.1 – TEA and efficiency gains

Deliverable Number	D9.1	**Lead Beneficiary**	12 - ADMR
Deliverable Name	TEA and efficiency gains		
Type	R — Document, report	**Dissemination Level**	SEN - Sensitive
Due Date (month)	48	**Work Package No**	WP9
Description			
Technoeconomic assessment of the processes and energy gains. 35% reduced process energy footprint for Hydrogen reduction of pure oxide and sulphide minerals			

Deliverable D9.2 – Life Cycle Sustainability Assessment report

Deliverable Number	D9.2	**Lead Beneficiary**	13 - PRINS
Deliverable Name	Life Cycle Sustainability Assessment report		
Type	R — Document, report	**Dissemination Level**	SEN - Sensitive
Due Date (month)	48	**Work Package No**	WP9
Description			
Environmental (LCA), social (s-LCA), economic (LCC) and overall sustainability (LCSA) Life Cycle assessment report. Documentation on environmental, social and cost assessment. LCA decrease global warm potential GWP100 by >80% (excluding H2 production)			

Deliverable D9.3 – Capacity building portfolio, health & safety

Deliverable Number	D9.3	**Lead Beneficiary**	14 - EUROC
Deliverable Name	Capacity building portfolio, health & safety		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	48	**Work Package No**	WP9
Description			
Report on the workplace dynamics, hazards and risks. Documentation for capacity building portfolio in relation to the impact of the proposed technologies in labour, identification of skill gaps, effectiveness evaluation of the control measures for health and safety risk.			

Deliverable D9.4 – Industrial uptake analysis

Deliverable Number	D9.4	**Lead Beneficiary**	2 - CORE
Deliverable Name	Industrial uptake analysis		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	48	**Work Package No**	WP9
Description			
AI-based study of the scale up potential. Documentation on results from possible configurations within the energy system's boundary conditions and virtual linking of additional added value technologies			

Deliverable D9.5 – Legislation and regulation

Deliverable Number	D9.5	**Lead Beneficiary**	20 - VDU
Deliverable Name	Legislation and regulation		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	48	**Work Package No**	WP9
Description			
Policy report on the legislative and regulatory aspects. Documentation on the outputs of observation techniques applied involving policy makers and financiers			

Deliverable D10.1 – Early PEDR [1st update]

Deliverable Number	D10.1	**Lead Beneficiary**	2 - CORE
Deliverable Name	Early PEDR [1st update]		
Type	R — Document, report	**Dissemination Level**	SEN - Sensitive
Due Date (month)	6	**Work Package No**	WP10
Description			
Initial and updated reports on the DEC strategy and results [1st update]. Documentation on Brand and visual identity of the project, project website, dissemination, communication and exploitation activities to be performed as a means to promote and exploit the project results. Identification of key project stakeholders, communication tools and channels.			

Deliverable D10.2 – Standardization landscape, and strategy [1st update]

Deliverable Number	D10.2	**Lead Beneficiary**	21 - UNE
Deliverable Name	Standardization landscape, and strategy [1st update]		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	6	**Work Package No**	WP10
Description			
Standardization landscape and applicable standards (M6). Reporting on existing standards, as well as those under development, of relevance for the project, Strategy identification for an effective contribution to standardization.			

Deliverable D10.3 – Early PEDR [2nd update]

Deliverable Number	D10.3	**Lead Beneficiary**	2 - CORE
Deliverable Name	Early PEDR [2nd update]		
Type	R — Document, report	**Dissemination Level**	SEN - Sensitive
Due Date (month)	18	**Work Package No**	WP10

Description
Initial and updated reports on the DEC strategy and results, [2nd update]

Deliverable D10.4 – Standardization landscape, and strategy [2nd update]

Deliverable Number	D10.4	**Lead Beneficiary**	21 - UNE
Deliverable Name	Standardization landscape, and strategy [2nd update]		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	18	**Work Package No**	WP10
Description			
Standardization landscape and applicable standards, strategy and activity (M18)			

Deliverable D11.1 – Mid PEDR

Deliverable Number	D11.1	**Lead Beneficiary**	2 - CORE
Deliverable Name	Mid PEDR		
Type	R — Document, report	**Dissemination Level**	SEN - Sensitive
Due Date (month)	36	**Work Package No**	WP11
Description			
Update of D10.1			

Deliverable D11.2 – Standardization report I

Deliverable Number	D11.2	**Lead Beneficiary**	21 - UNE
Deliverable Name	Standardization report I		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	36	**Work Package No**	WP11
Description			
Intermediate report on the activities performed to interact with the standardization system and contribute to generate new or modified standards			

Deliverable D12.1 – Final PEDR

Deliverable Number	D12.1	**Lead Beneficiary**	2 - CORE
Deliverable Name	Final PEDR		
Type	R — Document, report	**Dissemination Level**	SEN - Sensitive
Due Date (month)	48	**Work Package No**	WP12

Description
Update of D11.1

Deliverable D12.2 – Standardization report II

Deliverable Number	D12.2	**Lead Beneficiary**	21 - UNE
Deliverable Name	Standardization report II		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	48	**Work Package No**	WP12
Description			
Final report on the activities performed to interact with the standardization system and contribute to generate new or modified standards.			

Deliverable D13.1 – Project handbook

Deliverable Number	D13.1	**Lead Beneficiary**	1 - NTUA
Deliverable Name	Project handbook		
Type	R — Document, report	**Dissemination Level**	SEN - Sensitive
Due Date (month)	6	**Work Package No**	WP13
Description			
Setting up and using collaborative software tools: documentation and common data repositories and sharing tools. Project procedure manual			

Deliverable D13.2 – Project management quality assessment Ver.1

Deliverable Number	D13.2	**Lead Beneficiary**	1 - NTUA
Deliverable Name	Project management quality assessment Ver.1		
Type	R — Document, report	**Dissemination Level**	SEN - Sensitive
Due Date (month)	18	**Work Package No**	WP13
Description			
Quality assessment of overall project implementation for P1			

Deliverable D13.3 – Project management quality assessment Ver.2

Deliverable Number	D13.3	**Lead Beneficiary**	1 - NTUA
Deliverable Name	Project management quality assessment Ver.2		
Type	R — Document, report	**Dissemination Level**	SEN - Sensitive

Due Date (month)	36	**Work Package No**	WP13

Description
Quality assessment of overall project implementation for P2

Deliverable D13.4 – Project management quality assessment Ver.3

Deliverable Number	D13.4	**Lead Beneficiary**	1 - NTUA
Deliverable Name	Project management quality assessment Ver.3		
Type	R — Document, report	**Dissemination Level**	SEN - Sensitive
Due Date (month)	48	**Work Package No**	WP13
Description			
Quality assessment of overall project implementation for P3			

LIST OF MILESTONES

Milestones

Grant Preparation (Milestones screen) — Enter the info.

Milestone No	Milestone Name	Work Package No	Lead Beneficiary	Means of Verification	Due Date (month)
1	Preliminary component designs	WP2	5 - CIRCE	Presentation of first version of KER designs at the M12 meeting. Preliminary design of 1. RMI system, 2. IFRKs for the calcination processes and for metal roasting 3. microwave rotary kiln for sulphide roasting and 4. Plasma reactor. Report presenting main components and flows for each KER	12
2	Preliminary digital architecture design	WP3	15 - MAGG	Draft data model documentation, preliminary IoT reference architecture, workflow interaction with the DSS and UI, integration mechanisms and definition of interfaces and data models to improve interoperability and coupling between all involved modules, gateways, connectors and responsibilities definition. Preliminary report presenting digitalisation framework.	12
3	RMI process line prototype completed	WP4	4 - ICCS	Working prototype in test operation. Prototype ready for testing campaign, monitoring and control systems and pneumatic subsystem capable of separating samples commissioned. Short report and documentation of RMI prototype commissioned.	24

4	Start of validations	WP7	3 - EUT	Completion of T7.1 and start of T7.2-3. Protypes fully commissioned and ready to utilised for validation campaign. Integrated pilots commissioned in EUT and NTUA premises. Report presenting pilot units integration and related documentation will be delivered.	30

Milestones

Grant Preparation (Milestones screen) — Enter the info.

Milestone No	Milestone Name	Work Package No	Lead Beneficiary	Means of Verification	Due Date (month)
5	First round of adoption framework assessments	WP9	12 - ADMR	Report presenting preliminary results related to (1) Technical and economic feasibility; (2) Environmental, economic and social impacts; (3) Workplace effects and improvements; (4) Health and Safety; (5) Replication and uptake; (6) Policies and regulations.	42

LIST OF CRITICAL RISKS

Critical risks & risk management strategy

Grant Preparation (Critical Risks screen) — Enter the info.

Risk number	Description	Work Package No(s)	Proposed Mitigation Measures
1	Restricted access to end-users' data due to confidentiality issues (WP1,9, L/S:2/1)	WP1	NDA, close collaboration with partners, use of cost proxy values to fill in data gaps
2	Excessive number of computational resources required for CFD simulations (WP2, L/S:2/2)	WP2	In this case, PRIM-ROCK can resort to simplified models of combustion and/or turbulence, coarse-grained meshes or even cloud computing.

3	Inaccurate or inadequate data collection (WP2,3,5, L/S:1/2)	WP3, WP2, WP5	Re-calibrate sensors; Implement real-time data validation checks to flag outliers or anomalies; Implement real-time user feedback to improve data accuracy. Monitoring and control parameters will be identified in the initial stages of the project based on extensive know how of partners acting on raw materials sectors.
4	Physics-based ROMs do not have sufficient accuracy (WP6, L/S:2/1)	WP6	ROMs will be enriched with data-based correction terms to increase accuracy.

Critical risks & risk management strategy

Grant Preparation (Critical Risks screen) — Enter the info.

Risk number	Description	Work Package No(s)	Proposed Mitigation Measures
5	Technological mismatch or interoperability issues of smart systems (WP8, L/S:2/2)	WP8	Development and modification to be conducted in parallel with dynamic communication. Deliverables and milestones introduced to facilitate monitoring of technical challenges. Industrial partners and technology providers will adapt monitoring and control systems, supporting industrial protocols like ModBus. The coordinator has an extensive know how on data and hardware interoperability and will support all actions
6	Issues in establishing proper collaboration and technical alignment in the pilot ecosystem (WP7,8, L/S:1:3)	WP8, WP7	The proposed workplan enables the application of corrective actions for any emerging issue; Feedback from the interaction with the end users and the scientific experts is foreseen for early problem identification. A lot of partners have been already collaborated in the past for developing and testing prototypes, enabling efficient and comprehensive implementation actions. A portfolio of materials, sensing and control equipment have been already utilised in prototypes and tested. Technology providers will be supported for the industial partners for identifying the optimum solutions able to be applied on time.
7	Limited engagement of stakeholders in project's activities and results (WP10-12, L/S:2/2)	WP11, WP10, WP12	Prediction and early revision of DEC strategy based on initial user engagement, monthly retention, and other metrics. Key DEC actions such as workshops, and resources (e.g., videos) will be accelerated to be made available earlier. Full utilisation of EC resources, collaboration with other initiatives

8	Project outcomes not compatible with existing market procedures expressed through standards (WP10-12, L/S:1/2)	WP11, WP10, WP12	A detailed overview on the relevant standards will be prepared at early project stage and will be considered during the development of the tools and processes.
9	Hesitation or disagreement in exploitation of results (WP10-12, L/S:2/2)	WP11, WP10, WP12	Exploitation & IPR management workshops will be held throughout the project starting from M9.
10	Delays in the project timetable (WP13, L/S:2/3)	WP13	Increase the monitoring activities. The consortium agrees on: (i) re- allocation of resources; (ii) parallel execution of tasks; or (iii) re- scheduling of activities or a suitable combination of those.
11	Significant number of partners, may cause delays or person in key organisation departs the project (WP13, L/S:2/2)	WP13	The implementation structure and resources are designed to allow for full completion of the activities without delays. Broad partner network makes it possible to replace partner while remaining within project scope.

Critical risks & risk management strategy

Grant Preparation (Critical Risks screen) — Enter the info.

Risk number	Description	Work Package No(s)	Proposed Mitigation Measures
12	A critical partner from the manufacturing process leaves the consortium (WP1-13, L/S:1/3)	WP2, WP1, WP7, WP4, WP5, WP10, WP9, WP8, WP11, WP3, WP6, WP13, WP12	The consortium has the experience and the expertise to identify another partner to join the project and take over the activities.

PROJECT REVIEWS

Project Reviews

Grant Preparation (Reviews screen) — Enter the info.

Review No	**Timing** (month)	**Location**	**Comments**
RV1	18	Greece, Athens, NTUA premises	
RV2	36	Spain, Barcelona, Eurecat premises	
RV3	48	Belgium, Brussels	

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HISTORY OF CHANGES		
VERSION	PUBLICATION DATE	CHANGE
1.0	17.06.2024	Initial version
		p.29 and p.30 explanations are given in relation to Criterion 2 (impact) following evaluators comments, in relation to CO_2 and H_2 estimated values
		p.28, table 5: "Barrier analysis of the PRIM-ROCK project" revision following evaluators comments related to key challenges, possible barriers and obstacles
1.1	1/7/2024	p.43: Ethics self-assessment' included in the proposal Part A, is copied into ethics section
1.2	31/7/2024	p. 13, 18, 23-26, 30-32, 39-41 - Beneficiary GENIKI METALLEUTIKI KAI METALLOURGIKI ANONIMI is replaced by HELLAS GOLD S.A. due to low financial capacity assessment. Beneficiary 18 actions related to FeNi production will be undertaken by NTUA. HELLAS GOLD S.A. will exploit PRIM-ROCK technologies for mixed sulhpide deposits.
1.3	13/8/2024	TABLE 4, p.29 revised

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#@APP-FORM-HERIAIA@#

NOMENCLATURE

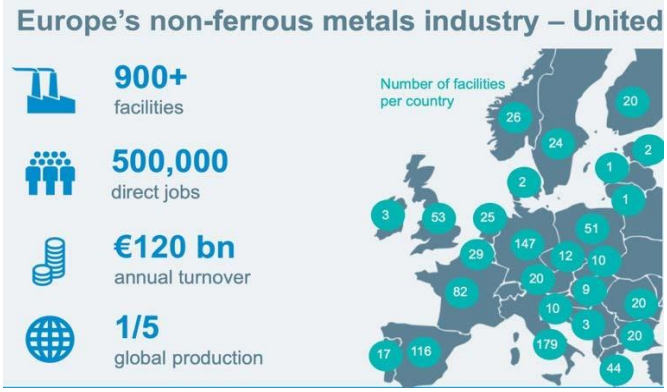
DFRK: Direct-fired rotary kilns, **IRFK**: indirect-fired rotary kiln, **MRK**: microwave rotary kiln, **MIP**: microwaveinduced plasma, **CKD**: cement kiln dust, **LKD**: Lime Kiln Dust, **IUC**: industrial use-case, **OPSR**: Optical Sorting Recognition, **XRT**: Chemical recognition.

1 Excellence

#@REL-EVA-RE@#

Introduction

Mineral resources are an important material basis for the entire economy. The International Mineralogical Association's (IMA) members annually produce 180 million tonnes of industrial minerals for the EU-28, Turkey, and Norway. These include calcium carbonate (precipitated and ground), dolomite, lime, bentonite, magnesite, and other valuable minerals[1]. Industrial minerals, either in a processed or natural state, are used to make building materials, ceramics, electronics, glass, medications and medical devices, paints, paper and plastics, and a myriad of other industrial and domestic products, contributing €14 billion to the EU economy[2]. The non-ferrous metals sector provides 500,000 direct jobs and 3 million indirect jobs in the EU, thousands more across wider Europe. These jobs are often highquality well-paid jobs, and trade unions have come together to fight to keep these jobs in Europe. European trade unions call for urgent action to level the global playing field to ensure Europe's competitiveness in producing high-quality green products, while preserving



[1] Processes4Planet SRIA 2022 [2] IMA-Europe

and creating jobs. Amongst the many processes employed in mineral processing, **calcination**, a crucial process in the minerals processing and cement industries,

involves heating materials to high temperatures. This process is integral to transforming raw materials into useful products and plays a vital role in the efficiency

Figure 1. European non-ferrous metals industry stats (Source: Eurometaux)

and sustainability of these industries. **Roasting** is a similar process for heating sulhpide ore to a high temperature in the presence of air. It is a step in the processing of certain ores. More specifically, roasting is often a metallurgical process involving gas–solid reactions at elevated temperatures with the goal of purifying the metal component(s). The gaseous product of sulhpide roasting, sulfur dioxide (SO_2) is often used to produce sulfuric acid. Many sulhpide minerals contain other components such as arsenic that are released into the environment. Rotary kilns have become essential in these processes, due to their diverse capabilities and high degree of control over process parameters.
While rotary kilns have become the backbone of many new industrial processes that make the world a more efficient and sustainable place, the technology cannot (or often should not) process material in just any shape or form. In addition to being inefficient, feeding unprepared material to a kiln can have several unintended consequences, ranging from an incomplete reaction or a failure to achieve the intended reaction, to unnecessarily long retention times and excessive energy costs. Direct-fired rotary kilns (DFRK) rely on direct contact between the process gas and the material in order to heat the material to the specified temperature, leading to fine and ultrafine particles becoming entrained in the process gas flow and carried out through the exhaust gas system. This results in a high rate of reprocessing and can significantly reduce the overall process efficiency, in addition to other issues.
In the cement industry, the amount of cement kiln dust (CKD) generated typically ranges 15–20%[2] of the produced cement, leading to an annual CKD production of approximately 12.9 metric tonsAssociated with document Ref. Ares(2024)6673367 - 20/09/2024. Similarly, the lime industry generates Lime Kiln Dust (LKD), with an annual production of approximately 1.8 to 3.6 million metric tons in the United States only[3]. CKD and LKD are usually stockpiled, and it is estimated that the total amount of kiln dust currently stockpiled throughout the country exceeds close to 90 million metric tons (100 million tons)[4], creating substantial issues in their management. In the critical metal industries, the wide use of carbon as a cheap reduction material as well as the lack of the relevant fundamental knowledge and expertise has limited the hydrogen use to a few cases of refractory metals. However, hydrogen could be used as an alternative to carbon reductant, provided that there is enough fundamental knowledge concerning the thermochemical feasibility and equilibrium, the kinetics, and the factors, i.e. metallic mineral mineralogical phase, affecting the hydrogen/metal oxide/sulphide reaction system. Moreover, it is necessary to deeply understand the technical problems arising in the case of treating complex metal ores and concentrates due to the various compounds and mineralogical phases present in the reaction system. In this context, the PRIM-ROCK project represents a new frontier in the metallurgical industry in view of developing and demonstrating the necessary fundamental knowledge that will pave the way to sustainable, more energy efficient mineral processes with low carbon footprint, in a significant number of critical metals. Moreover, it will develop entirely new, breakthrough technologies enabling the direct extraction of metals from the EU metal sulphide resources. The project will also demonstrate an integrated approach to treat critical metal minerals coupled with onsite renewable hydrogen production, powered by artificial intelligence. The knowledge developed in the PRIMROCK innovative processes directly targets the real scale application in the relevant industrial areas and will also provide the foundations for going beyond the specific cases studied in this project towards the development of a climate neutral critical metal hydrogen metallurgy on the long term. Both calcination and roasting processes present important optimisation potential dependent on the effective pre-treatment of raw material ores. In the ore sorting process, accurate, machine-assisted classification and sorting, as well as determining the optimal crushing size are key steps, greatly influence the efficiency and productivity of the production process, while also enabling also the utilisation of lower grade ores. For both sectors, process innovations are needed to improve resource efficiency, reduce waste and low-value by-product generation, and mitigate emissions and risk to worker health and safety.

1.1 Objectives and ambition

#@PRJ-OBJ-PO@#

[2] Al-Bakri AY, Ahmed HM, Hefni MA. Cement Kiln Dust (CKD): Potential Beneficial Applications and Eco-Sustainable Solutions. *Sustainability*. 2022; 14(12):7022. https://doi.org/10.3390/su14127022
[3] https://www.fhwa.dot.gov/publications/research/infrastructure/structures/97148/kd1.cfm and references therein

PRIM-ROCK aims to bring a step-change in the calcination and roasting processes, thanks to a holistic approach centred around ground-breaking process intensification (PI) and digital technologies, considering the following:

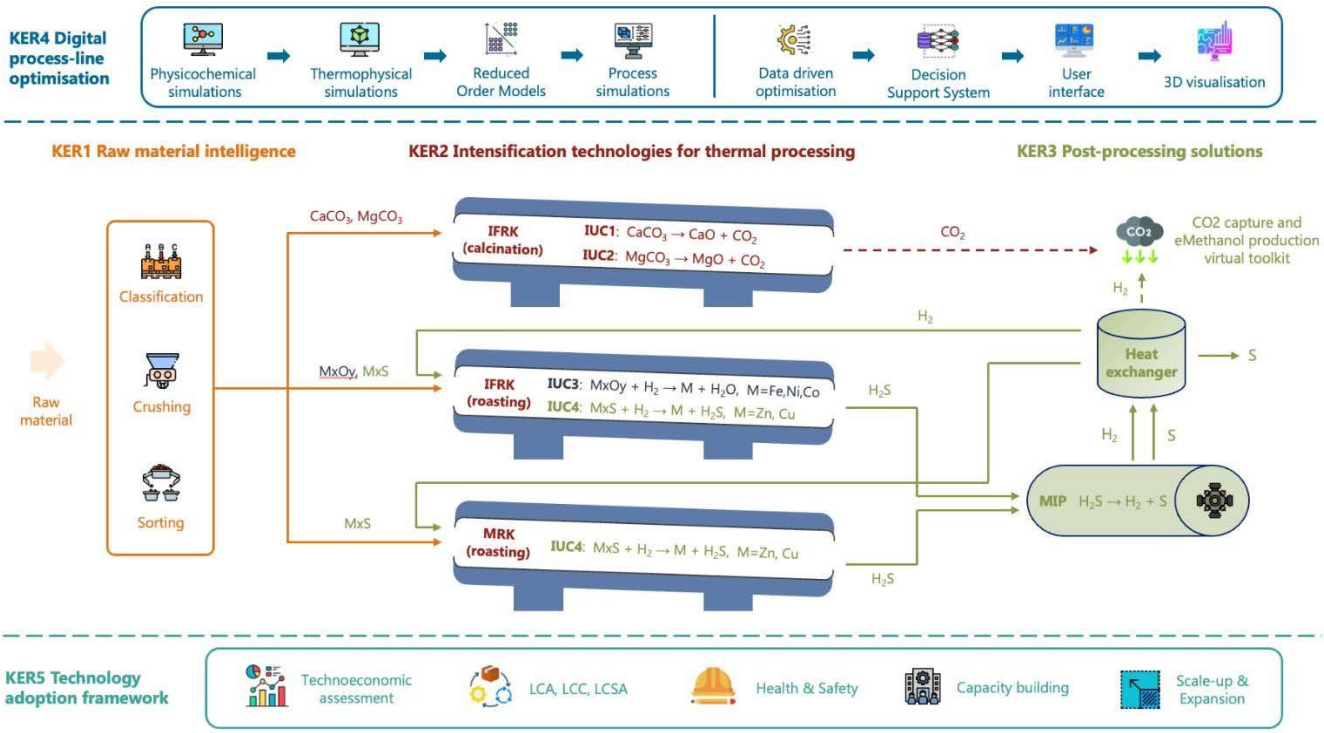


Figure 2. Overview of the process line addressing multiple types of raw materials.

(1) pre-processing stage, referred to in the project as the **Raw Material Intelligence (RMI) system (KER1)**, comprised of a process line featuring advanced sensorisation, AI-driven sorting, and decision-support, complete with real-time analysis capabilities and optimized crushing size determination for the raw materials. KER1 will contribute to process efficiency in a versatile manner, establishing an interface between pre-processing of raw materials and their thermal

treatment and **(a)** optimising pre-processing in accordance to process parameters, Associated with document Ref. Ares(2024)6673367 - 20/09/2024**(b)** enabling the adaptation of process parameters depending on raw material changes, even in real-time. Pre-processing will thus enable the resource efficiency of the corresponding processes to be increased by over 10%[4].

(2) replacement of Direct-fired rotary kilns (DFRKs) with Indirect-fired rotary kilns (IFRKs) and microwave rotary kilns (MRKs) as the process components **(KER2 – Intensification technologies for thermal processing)**. Replacing DFRKs with the electrified IFRKs and MRK removes the need for process gasses, solving the issue of fine and ultrafine material loss through drag, reducing emissions and improving the controllability and flexibility of the calcination and roasting processes. The capacity to process without losses of fine and ultrafine material is the major contributor to process efficiency, as losses can exceed 25% in many industries as previously described. Moreover, adoption of the technology removes the need for pelletisation and filters. Moreover, as slag or dam ring formation in the rotary kiln can be avoided, resource efficiency will increase by 6-12%[5]. This refers to a build-up around the circumference of the kiln's interior in such a way that material is either prevented from passing through or is significantly inhibited. Ring formation is extremely common, occurring in roughly 85% of commercial rotary kilns. It is typically the result of a change in slagging temperature. Thus, depending on the process, KER2 will contribute to resource efficiency in the 20-32%[5,6] range.

(3) improvements on post-process component level, enabling using the kiln for direct reduction through H_2 as a reducing agent, which will be subsequently recovered by decomposition through Microwave-Assisted Plasma (MIP) reactor (**KER3 – Post-processing solutions**), which requires a sustainable and low-cost H_2 supply that is

[4] Jan Spisak, Dusan Nascak, Eva Oravcova, "Increasing dust-chamber efficiency of rotary furnace for magnesite treatment", European Scientific Journal March 2015 edition vol.11, No.9 ISSN: 1857 – 7881 (Print) e - ISSN 1857- 7431

[5] Lime Kiln Principles and Operations. Terry N. Adams. Technical Consultant. Seattle WA

the goal of this activity. Replacement of C by H_2 and high levels of recovery will improve the resource efficiency of reduction processes of the selected lines by over 60% (FeNi case, from 45% to 90%[6] to pre-reduction stage). Additionally, an innovative heat exchanger prototype will be developed to at the post-process level to manage the thermal quenching effect of the plasma and cool the process gases in a controlled way, maximising recovery. Finally, due to the substantial enriched CO_2 produced by calcination, a virtual toolkit for decision support in capture and utilisation in e-fuels will be developed, enabling credible feasibility assessments.

(4) Enabling the integration of KER1-3 and maximizing the improvements in resource efficiency will be achieved through the digital infrastructure component (**KER4 - Digital process-line optimisation**). KER4 will deploy tailored, and exhaustive simulations of the pre-process, electrified process and post-process stages to obtain the optimal design of the prototypes, while process digital twins will be developed with data-driven functions for efficient monitoring (with 3D visualization capabilities) and further optimization. An innovative data platform will facilitate data fusion, bridging the models, the Process Control System (PCS) and the advanced sensor network of all stages. It is estimated that digital technologies will contribute to the maximization of the process-line resource efficiency by 3-5%.

(5) support the technological development process by addressing issues related to adoption of the technology through a comprehensive suite of transversal activities: Life Cycle Assessment (LCA), Life Cycle Costing (LCC), Life Cycle Sustainability Assessment (LCSA), Technoeconomic Analysis (TEA), GHG emissions avoidance, energy footprint analysis, technology scale-up roadmap, standardization, data management and security, business case and exploitation strategy development, and workplace assessment (including skills development). (**KER5 - Technology adoption framework**)

1.1.1 Project objectives

SO1. Contribute to resource efficiency gains through the Raw Material Intelligence (RMI) system (KER1) The characteristics of raw materials exert significant influence on processing efficiency and product quality.
PRIMROCK will leverage this interface by introducing a holistic system for raw material pre-processing through:

1. Design of the RMI component, creating a detailed blueprint for integrating advanced sensors, AI algorithms, and control systems into ore processing workflows, ensuring effective classification, sorting, crushing and optimization of processing parameters (T2.2–ICCS).
2. Introducing advanced sensorisation and control systems for the RMI system to accurately detect and analyse the composition of mineral ore streams, enabling dynamic adjustment of processing parameters for maximum efficiency and accuracy. Delivering an integrated sensor and control system prototype tailored for high-precision ore processing applications (T4.1–ICCS).
3. Deploying two AI-assisted sorting systems for tuneable segregation of mineral ore streams, utilizing nondestructive vision sensory devices and deep learning models. (T4.2–ICCS).
4. Developing an intelligent Decision Support System for raw material valorisation (RMI-DSS) that analyses physical and chemical properties of mined samples, along with sorting data and valorisation chain requirements, to determine optimal sorting thresholds and enable dynamic decision-making for sorting criteria (T4.3–CIBOS).

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5. Exploring non-destructive, real-time analysis methodologies using optical sensors, microscopy imaging sensors, and hyperspectral cameras to characterize ore and identify optimal crushing sizes with respect to the valorisation process. Conducting trials with lab-scale sorting systems to validate mineral separation and crushing size identification. (T4.4–ICCS).

KPIs: *sorting accuracy of metal oxides and metal sulphides >95%; Improvement in the overall utilization of mined ores by 40%; Increase classification models computational efficiency by 10%; 10% reduced waste in mined minerals by identifying optimal crushing size; Reduction of electrical energy consumption by 5%; CO_2 emissions reduction by 5%; Reduction of area/volume needed for pre-sorting by >35%; Laser-based detection 80-90% over the project duration for size.*

SO2. Achieve game-changing resource efficiency improvements through new process technologies (KER2) Addressing the issue of dust generation and losses through process gasses, PRIM-ROCK will develop a new,

[6] Artemova, A.V.; Maklakov, S.S.; Shiryaev, A.O.; Osipov, A.V.; Petrov, D.A.; Rozanov, K.N.; Lagarkov, A.N. Influence of Hydrogen Reduction Stage Conditions on the Microwave Properties of Fine Iron Powders Obtained via a Spray-Pyrolysis Technique. Magnetism 2023, 3, 90-101. https://doi.org/10.3390/magnetism3020008

indirectfired rotary kiln (IFRK) as well as a microwave rotary kiln (MRK), and optimise their design to vastly expand the spectrum of raw material size that can be processed via both the calcination and roasting processes, through:

1. Producing a lab-scale IFRK prototype at TRL4 for operational data generation and model training (T6.1–NTUA).
2. Performing advanced physicochemical (T2.3–NTUA, T2.4–ITA) and thermophysical (T2.3–CIRCE, T2.4–BUL) simulations for modelling the microscopic and macroscopic behaviour of the calcination and roasting processes in the IFRKs. Advanced modelling and design of the MRK (T2.5–CEIN). Developing physics-based Reduced Order Models based on the outputs of the thermophysical simulations, so that more efficient surrogated models can be obtained from off-line full order simulations to be embeded in higher-level models (larger scales) (T6.2–ITA, CIRCE). Simulating the whole process behaviour to obtain the optimal design architecture (T6.3–NTUA).
3. Delivering 2 IFRK prototypes, tailored for calcination and roasting (T6.4–NTUA) and 1 MRK tailored for sulphide roasting (T6.5–CEIN).

KPIs: *Design, development and delivery of 1 lab-scale IFRK, 2 pilot-scale IFRKs and 1 pilot-scale MRK; IFRKs: 2m, 4-10kg/h, 6 heating zones up to 1200º; MRK: tube reactors of 5-12cm diameter, 915MHz, 5-30kg/h; Technology capable of coupling of process steps (roasting+H_2 reduction); Eliminating reductive smelting and increasing the efficiency of reduction to over 80% for ferronickel; ROMs that can be executed in real time in simple devices and that can reproduce the process KPIs under steady-state conditions with > 90% accuracy; >90% conversion through raw material calcination.*

SO3. Leverage innovative intensified processes to further improve material efficiency through post-thermal processing treatment (KER3)

Adopting a circular and holistic approach the material efficiency is further leveraged utilising breakthrough, technically and economically feasible, and environmentally sound post-processes, through:

1. Modelling of a Microwave Induced Plasma (MIP) reactor tailored to transform the of hydrogen sulphide, generated (H_2S) during metal sulphide reduction, to Hydrogen (H_2) and Sulphur (S), towards the reutilisation of H_2 in the kilns (T2.6–EUT).
2. Modelling of a heat exchanger for controlled cooling of the gases coming from the MIP reactor (T2.6–BUL).
3. Developing a virtual toolkit for the exploitation of byproduct (enriched CO_2) valorisation in e-Methanol production, supported by technoeconomic analysis to demonstrate investment viability (T9.5–CORE).
4. Manufacturing and delivering of the MIP reactor at pilot-scale tailored to sulphide roasting (T6.6–EUT).
5. Manufacturing and delivering of the heat exchanger prototype (T6.7–BUL).

KPIs: *Hydrogen reduction of pure oxide and sulphide minerals at lab-scale at pilot scale (TRL6) with H_2 and S decomposition and recovery of both elements at >95% efficiency; Minimum mineral reduction degree 80%; The effect of the process parameters, including particle size, temperature, hydrogen partial pressure, reaction time, on the process recovery and kinetics, as well as the optimum operation conditions to achieve maximum metal reduction will be studied; Technical target: to achieve >97% reduction efficiency at optimum conditions), 35% reduced process energy footprint, process time.*

SO4. Development of digitalisation framework to optimize the resource efficiency of the new processes (KER4)

Development and deployment of a flexible, interoperable and scalable digitalisation framework comprised of:

1. A complete network of sensing techniques, such as SWIR (Shortwave Infrared), NIR (Near Infrared), VIS (Visible), Raman, Fluorescence and LIBS (Laser Induced Breakdown spectroscopy), coupled with HybridEnsemble Modelling for data fusion that can be readily tailored to pre-process, process, and post-process realtime monitoring requirements. (T3.2–IRIS)
2. A Process Control System (PCS) overviewing and managing the different tools in all stages of processing. The process control system embeds horizontal functionalities (e.g., logs, alerts, security and safety measures, and integration with legacy systems) for the smooth management of all stages of the processes. (T3.3, T5.1–MAGG).
3. Secure and interoperable multilayer data management platform, providing the resources needed for the planning and real-time monitoring of the process lifecycle, through a common framework of different spectroscopic process analytical technologies and implementing data fusion (T3.1, T5.3–IRIS).
4. AI-based data-driven models to identify the optimal parameter space for enhancing the resource efficiency of the IFRKs and MRK (T5.2–SCCH, CORE)
5. Decision Support System (DSS) and user-friendly interface for monitoring of the processes in (near) real-time and suggest optimal settings to the user (T5.4–SCCH, CORE)
6. 3D visualization of the processes leveraging advanced data interface technologies (T5.5–STAM)

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KPIs: *Increase the control and therefore the quality of the thermal processes; Improvement in the accuracy of predictive modelling (through the adoption of modern AI-driven models and transfer learning techniques) at 20% or higher, as compared to either existing predictive modelling results or expert-based predictions; Enhanced resource efficiency through process digital optimisation by ~5%.*

SO5. Deploy and consolidate a technology adoption framework

Establish the appropriate baselines and methodologies in the design phase of the project (T1.5–ADMR). Conduct the following types of assessments in the development and integration phase of the project (T6.8–ADMR). Consolidate the assessments in the deployment and finalisation phase of the project:

1. Assessment of energy efficiency and intensity, measuring and validating the energy consumption per unit of output across all processes and benchmarking energy use against industry standards and best practices, including a Life Cycle Sustainability Assessment (LCSA) and Evaluate Societal and Environmental Impacts: To quantify the broader societal and ecological benefits or risks associated with the project, ensuring alignment with environmental regulations and societal expectations (T9.2–PRINS)
2. Technoeconomic assessments for comparative evaluation against existing processes, including cost drivers and financial benefits of the proposed solutions and energy footprint calculation (T9.1–ADMR).
3. Assess and improve workplace effects and OHS performance: Develop strategies for workplace improvement and workforce development through capacity building (up/re skilling), addressing skills, organisational change, eliminating hazards and identifying risks (T9.3, T9.4–EUROC).
4. Business potential assessment and business plan development: To create a robust business plan that outlines project implementation, market entry, and scaling strategies, ensuring financial sustainability and market competitiveness of the project outcomes (T10.2,11.2,12.2–STAM).
5. Develop and implement a comprehensive strategy to identify relevant existing and emerging standards, and utilize insights from this analysis to inform the development process on the hand (e.g. ensuring interoperability for data sharing), as well as for engaging effectively with standardization bodies to actively contribute to ongoing standardization efforts, focusing on significant contributions to the development or modification of standards that align with the project's objectives and advancements (T10.3,11.3,12.3–UNE)
6. Foster Clustering and Networking, creating synergies with other projects, initiatives, funding programmes and stakeholders in the field, enhancing knowledge exchange, achieving multiplier effects, and fostering innovation ecosystems (T10.1,11.1,12.1–CORE).

KPIs: *LCA decrease global warm potential GWP100 by >80% (excluding H_2 production), assessments and OHSE carried out for each IUC, DEC KPIs (see Section 2.2).*

SO6. Establishment and demonstration of two pilot-scale units

1. Deployed in Greece (NTUA premises) comprising the RMI system and the IFRK, tailored to calcination.
2. Deployed in Spain (EUT premises) comprising an RMI system, IFRK, MRK, MIP and heat-exchanger, tailored to roasting.
3. Technological assessment of demonstrations (T8.5–NTUA)

IUC1,2 will be demonstrated in the Greek pilot unit, while IUC3,4-5 will be piloted in the Spanish one, with potential additional sample processing for comparative assessment according to initial results.

Table 1. Target process efficiency improvements in PRIM-ROCK

Industrial Use case	Pre-processing	Process	Process	Post-process
IUC#1 Limestone ($CaCO_3$)	**>50%** reduction of mass lost during handling (**4-6%** of total mass as fine/ultrafine) through detection (OPSR, XRT) and optimisation of particle size (crushing) based on process eliminating 95% of	**>90%** reduction in kiln dust losses due to drag (inc. ring formation) that comprise between **5%** (for concentrates) and **25%** of raw material lost or collected in filters (removal of pelletisation stage). to maximise recovery requirements losses	Virtual toolkit for feasibility assessment of Carbon Capture of enriched CO_2 produced by calcination and utilisation in e-fuels.	
IUC#2 Magnesite ($MgCO_3$)				
IUC#3 Laterite (Fe, Ni oxides)			Direct reduction with H_2 to increase efficiency from **45%** up to **90%** and eliminate pre- or post- reduction steps	Recovery of H_2 reducing agent through treatment in MIP and utilisation of Gas Cooling Heat Exchanger
IUC#4 Sphalerite concentrate (ZnS)				
IUC#5 Chalcopyrite concentrate (CuS)				**>90%.**

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Table 2. Pertinence of project objectives in relation to the call

HORIZON-CL4-2024-TWIN-TRANSITION-01-41	SO

To reach ambitious targets regarding resource efficiency, disruptive process technologies must be developed in addition to process efficiency options for existing technologies.	1, 2, 3
Develop disruptive process technologies to improve resource efficiency, such as those based on process intensification (e.g., 3D printed processes equipment, coupling of process steps, new processes that integrate multiple reaction steps, activation of molecules using renewable energy via alternative processes e.g. microwave, plasma);	2, 3
or to prevent and minimise waste generation by, e.g. processes that adjust in real time to feedstock changes or that have tighter processing control solutions to ensure higher yields from complex and fluctuating raw material feeds;	1
Where relevant advanced process technologies and their combinations need to be developed and supported by advanced materials innovation and the implementation of enabling digital technologies including advanced concepts on process control and data driven Artificial Intelligence.	1, 2, 3, 4
The proposals should include energy efficiency, techno-economic and life-cycle assessments considering the overall process. This should also include the assessment of possible societal and environmental impact and the effects on the workplaces (skills, organisational change, and others).	5
Proposals submitted under this topic should include a business case and exploitation strategy, as outlined in the introduction to this Destination.	5
Research must build on existing standards or contribute to standardisation. Where relevant interoperability for data sharing should be addressed.	4, 5
All proposals should build on or seek collaboration with existing projects and develop synergies with other relevant European, national or regional initiatives, funding programmes	5

1.1.2 Ambition

Innovations #1,2 Separation and sorting technology toolset as part of RMI (KER1)

Separation Technology for Optimal Mineral Crushing Size

State of the Art: In the ore sorting process, determining the optimal crushing size for mined minerals is a critical step. This factor greatly influences the efficiency of sensor-based sorting technologies. At the same time, numerous sensing technologies capable of providing insightful information and details regarding the materials' physical and chemical properties (i.e. optical, spectral reflectance/absorption, magnetic properties, etc.)[7] have emerged. In detail, colour-based characterization techniques have been designed and implemented, ample of identifying minerals of distinct colours (e.g. dark-coloured vs light-coloured) with high accuracy[8,9]. The systems' accuracy, however, drops drastically, circa 60%, when the samples under examination show colour resemblance (i.e. various tones of same colour). In addition, hyperspectral sensors are used to extract the ores' unique spectral signatures (up to 2500nm), which are closely related to their molecular vibrational properties, achieving an identification accuracy of up to 90%[10]. Nonetheless, despite the fast-paced development of the sensing hardware, no extensive analysis has been conducted on the correlation between the size of the input feedstock and the overall performance of the sorting system.

Breakthrough: Within PRIM-ROCK an exploratory analysis will be conducted to estimate the optimal, valorisation chain-specific crushing size of minerals, leveraging information acquired by spectroscopic (up to 5000nm) and microscopy sensors and taking into consideration the overall requirements (i.e. feedstock purity, grain size, particle size distribution, etc.) of the valorisation flow with the goal to process these materials further optimally. Certain process parameters, such as the size of crushing of mined minerals, will be correlated with the sorting effectivity.

Adaptive Sorting System for Ore Streams

State of the Art: Currently developed sorting systems delving into multi-sensory inference, utilizing an ensemble of sensors (e.g. optical RGB sensors, hyperspectral imaging, X-Ray technology, etc.) to enrich the acquired information for analysis[11]. Common ground to every sorting system is the need for a mechanism to segregate ores based on the output of the characterization process. To this end, various approaches have been proposed to automate the physical sorting of such materials that re-direct them to the respective bins by a) robotic arm[12], b) pneumatic means[8]. Depending on the sensing technology and the materials to be sorted, the separation accuracy of online methodologies

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[8] C. Bergmann, Developments in Ore Sorting Technologies, Council for Mineral Technology, Randburg, 2009, p. 75.

[9] C.Y. Xu, The application of photoelectric color separator and the optimization in mineral separation process of some mine, World Nonferrous Met., 2016, No. 17, p. 31.

[10] N. Okada, Y. Maekawa, N. Owada, K. Haga, A. Shibayama, and Y. Kawamura, 'Automated Identification of Mineral Types and Grain Size Using Hyperspectral Imaging and Deep Learning for Mineral Processing', Minerals, vol. 10, no. 9, 2020.

[11] D. Peukert, C. Xu, and P. Dowd, 'A Review of Sensor-Based Sorting in Mineral Processing: The Potential Benefits of Sensor Fusion', Minerals, vol. 12, no. 11, 2022.

[12] H.J. Wang, X.P. Liu, G. Wang, and S. Han, Optimization of mobile manipulator sorting path based on improved genetic algorithm, J. Beijing Univ. Posts Telecommun., 43(2020), No. 5, p. 34

ranges from 60% up to 90%[8]. The design and implementation of ore sorting systems, up to this day, however, focuses on the maximization of commonly used performance metrics, such as accuracy, precision, picks per minute, etc., hence constituting the sorting systems rigid and incapable of adapting dynamically to new sorting criteria. Although

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these metrics are crucial to quantify the system's overall capacity to execute fast and accurately the separation task, their maximization does not necessarily contribute towards the optimal valorisation path of ores. Therefore, at present, sorting systems combine powerful, high-resolution and information-dense sensors to marginally optimize their performance, without taking into consideration the valorisation requirements, the output stream's characteristics, and the available raw-materials.

Breakthrough: PRIM-ROCK 's goal is to upgrade the sorting process for mineral ore raw-materials by means of both hardware and software in order to constitute the system versatile and dynamically adaptive to a wider range of criteria, as they are defined by the overall valorisation process' needs. To this end, a lab scale RMI mechanism and a pilot scale RMI sorting mechanism will be designed and implemented capable of identifying and ultimately sort the raw materials from the mined ore stream, based on dynamic classification criteria (e.g. size, purity of sorted magnesite, etc.). An innovative intelligent Decision Support System (DSS) will be leveraged to estimate the optimal classification thresholds. Input to the DSS will be the combination of the optimal, valorisation flow-specific crushing size, the raw-materials' physical and chemical properties, as well as requirements of the upcoming valorisation flow, such as particle distribution, purity of input stream, etc. The classification of the ore streams will be supported by hyperspectral and microscopy imaging sensors, a pneumatic system will be responsible for the segregation of fine/ultrafine ores into their respective categories, while a delta robot equipped with an end-effector (e.g. gripper, vacuum) will handle coarser, bigger materials of the input stream, with accuracy more than 95%.

Innovation #3 New indirect-fired rotary kiln (IFRK) technology for the replacement of direct-fired rotary kilns (DFRK) (KER2)

State of the Art: Rotary kilns have become the backbone of many industrial processes, such as in roasting and calcination, particularly in recent years as the pressure to create a circular economy has continued to rise. Nowadays, in thermal processing Direct-fired rotary kilns (DFRKs) are mainly used, which rely on direct contact between the process gas and the material in order to heat the material to the specified temperature[14,15]. Direct-fired kilns can be either of the co-current design, or counter current design, referring to the direction that the process gas flows through the drum in relation to the material. While DFRKs are relatively simple in design, operation and maintenance and they are efficient in terms of heat transfer due to the direct contact between the material and the combustion gases, they also have some major drawbacks which turn out to be crucial in many applications. By adapting DFRK technology, off-gases flows with an inherent velocity are produced and particles that are too fine can become entrained in the process gas flow and carried out through the exhaust gas system. This results in a high rate of reprocessing and can significantly reduce the overall process efficiency. Dealing with this problem requires the design of the kiln to be centered around permissible gas velocities as opposed to heat transfer requirements. Additionally, DFRKs provide a limited atmosphere control as the atmosphere inside the kiln is essentially the same as the combustion atmosphere, where oxygen is also present. As a result, DFRKs cannot maintain an inert or non-reactive atmosphere and the inevitable oxidizing environment can affect the quality of materials that are sensitive to oxidation. Finally, the direct exposure of the material to the heat leads to a gradient in temperature, with parts of the material closer to the flame being hotter than those further away. This uneven temperature distribution can affect the consistency and quality of the process's material, leading to a non-uniform final product.

Breakthrough: In PRIM-ROCK a technology for electrified indirect-fired rotary kiln (IFRK) will be developed, where the kiln is heated from the outside, using a heat shroud, and the material is heated via contact with the hot kiln shell. Hence, IFRKs are able to treat all granulometry ranges, including finely divided solids as well, as they are not in intimate contact with hot combustion gases. As a result, the raw material losses are minimized. Furthermore, IFRKs can provide the necessary controlled environment for effective material processing, like the absence of oxygen at high temperatures and the utilisation of an oxygen free reductive environment. As the heat source is external, the indirect heating method allows for precise control of the temperature and provide more uniform heat distribution, minimizing the risk of localized overheating or thermal gradients within the material.

Innovation #4 Novel microwave rotary kiln for sulphide roasting (KER2)

State of the Art: Conventional rotary kiln installed in the industry typically rely on direct flames from fossil fuels and boast high capacity of production (around 200t/h) and require a residence time of over 2 hours for a proper heat transfer. Additionally, gases of combustion and reaction are intermixed, leading to uncontrolled atmospheres and non-extractable reaction gases. Microwave rotary kilns emerge as a promising solution for downsizing, energy efficiency

and fast processing time. They aim to bridge the industry gap, as exemplified by patent WO/2016/128596 from CEIN, and through various projects involving CEIN and NTUA, like DAPhNE, DESTINY and RemovAL, some of which have found industrial application. Carbothermal reduction with microwave heating has been

[14] S. Sarbishei, K.L. Tafaghodi, The effect of sulfur content of rotary kiln fuel on the composition of nickel laterite calcine, Fuel, 280 (2020), Article 118648, 10.1016/j.fuel.2020.118648

[15] Fu, X.; Chen, Z.; Xu, X.; He, L.; Song, Y. Deposits in Gas-fired Rotary Kiln for Limonite Magnetization-Reduction Roasting: Characteristics and Formation Mechanism. *Metals* **2019**, *9*, 764. https://doi.org/10.3390/met9070764

thoroughly studied[16] and successfully demonstrated[17] with the process carried out under controlled atmosphere in a Associated with document Ref. Ares(2024)6673367 - 20/09/2024 solid-phase flow microwave reactor, reaching TRL4/5 in microwave pilot plants developed during projects such as RemovAL, DESTINY and Sulfometae.

Breakthrough: PRIM-ROCK is set to revolutionize the roasting process through electrification, utilizing an ultrafast microwave process intensified by hydrogenation of sulphides. The innovative microwave rotary kiln will operate under a controlled atmosphere, capable of applying hydrogen as a reactant. It features a microwave applicator with electromagnetic modulation along the kiln, facilitating ultra-fast roasting processes and real-time tuning thanks to a unique monitoring and control system. High-power density modulation enables swift material transformation, typically in less than 2 minutes. Precise atmospheric conditions will be maintained using pressure differentials, ensuring adequate residence time for the hydrogen reaction. As the dry, granular material passes through the microwave rotary kiln, it is thermally transformed in the presence of reactive gases at high temperatures into separable, enriched intermediate metal products. Reaction gases will be continuously extracted and collected during the process.

Innovation #5 Hydrogen utilisation as a reducing agent for metal reduction from their oxide and sulphide ores

State of the Art: Historically, the metallurgical sector has predominantly utilized coal as a source for carbon, due to its abundance and cost-effectiveness. Carbon is used as a reducing agent to extract most metals by eliminating oxygen from oxide ores, releasing substantial carbon dioxide emissions during the process. As the process industries push towards decarbonisation, a decisive shift from carbon is enabled by the adoption of hydrogen as: **(a)** hydrogen serves as both a sustainable energy source and an eco-friendly reducing agent in metal production and **(b)** hydrogen can enhance performance due to many factors. These include the lack of contaminants found in coal and coke, that simplifies metal purification. Moreover, the reduction temperature is lower in H_2 based processes, while simpler, more efficient, and controllable continuous reactors can be utilised. Finally, consistent hydrogen supply through RES is possible, while simultaneously reducing reliance on coal mining and imports. Nonetheless, there are challenges. Hydrogen is only commercially utilized in a limited number of refractory metals (i.e., W, Mo) and partly utilized in Ni and Co metals production. Hydrogen reduction of metal oxides has been extensively studied at laboratory scale, particularly in regard to kinetics and reaction mechanism[18]. The challenge with sulphide ores, a significant source of critical metals, lies in managing the produced hydrogen sulphide gas, unlike in the case of oxide ores that produce water. Thermodynamic analyses indicate the potential for hydrogen to reduce oxides and sulphides of these critical metals. However, industrial-scale application faces hurdles such as: **(a)** underdeveloped thermodynamic models for solid-gas reactions and gas phase equilibria; **(b)** limited understanding of gas-solid reaction kinetics and mechanisms; **(c)** insufficient knowledge on the impact of ore pre-treatment on hydrogen reduction efficiency; **(d)** lack of comprehensive experimental data for scaling up to pilot and industrial levels; **(e)** uncertainties around the economic feasibility considering hydrogen costs; and **(f)** challenges in safely managing toxic gases like H_2S[19]. The utilization of hydrogen for reducing critical metals from their ores is still in early stages, necessitating further development of fundamental knowledge and resolution of key technical challenges.

Breakthrough: The main innovation proposed in this case is the hydrogen reduction of metals oxides and sulhpides in order to enhance metals recovery and eliminate the use of fossil fuels as all the processes are fully electrified. A breakthrough process for the treatment of Laterite (Fe, Ni oxides), Sphalerite concentrate (ZnS) and Chalcopyrite concentrate (CuS) with main advantages including: **(i)** simplification of the highly complex treatment process of ores concentrates and further utilization of low grade sources; **(ii)** recovery of the used hydrogen positively affecting the cost and sustainability of the overall sulhpide mineral reduction process; **(iii)** effectively addressing the conventional sulhpide ore processes environmental problem of SO_2(g) generation, through the efficient processing of H_2S and the consecutive production of H_2 and Sulphur. An efficient method to overcome the thermodynamic barrier and increase the copper yield is to shift the reaction equilibrium continuously to the right by introducing sufficiently large quantities of H_2 in the system and continuously removing the generated H_2S as soon as it is formed. The coupling of sulhpide reduction process with the continuous decomposition of the formed H_2S(g) to H2(g) and S(s) will allow the

continuous recycling and feed of the produced H_2(g) in sufficient quantities into the system and secure the economic performance of the process without adding excess cost.

[16] García-Baños, B.; Catalá-Civera, J.M.; Sánchez, J.R.; Navarrete, L.; López-Buendía, A.M.; Schmidt, L. High Temperature Dielectric Properties of Iron- and Zinc-Bearing Products during Carbothermic Reduction by Microwave Heating. Metals 2020, 10, 693. https://doi.org/10.3390/met10050693

[17] López-Buendía, AM, Brau, E, Sparis, D., Panias, D Davris, P., Balomenos, E. Iron source from bauxite residues by using microwave technology. Proceedings of the 41st international ICSOBA Conference, Dubai, 5 - 9 November 2023. https://icsoba.org/proceedings/41stconference-and-exhibition-icsoba-2023/?doc=88

[18] Rukini, A., Rhamdhani, M.A., Brooks, G.A. et al. Metals Production and Metal Oxides Reduction Using Hydrogen: A Review. J. Sustain. Metall. 8, 1–24 (2022). https://doi.org/10.1007/s40831-021-00486-5

[19] Physical Chemistry of Metallurgical Processes, First Edition. M. Shamsuddin. The Minerals, Metals & Materials Society. Published by John Wiley & Sons, Inc, 2016

Innovation #6 Hydrogen recovery from roasting by-product gasses through microwave-induced plasma (MIP) treatment

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State of the Art: The state-of-the-art production of critical metals from sulphide ores is highly complex involving many high temperature process stages and is directly associated with the production of sulfuric acid. The production of sulfuric acid adds significant cost to the CAPEX and OPEX of the overall metal extraction process and has low market value. In this respect the direct production of critical metals from sulphide minerals with the use of hydrogen is a very attractive process alternative as it significantly reduces the processing steps and considerably simplifies the overall metal production process. A major drawback of the direct reduction of metal sulphides with hydrogen is the formation of unwanted hydrogen sulphide gas (H2S) that is the main product of the reaction of the metal sulphides with hydrogen. Hydrogen sulphide is a toxic gas and cannot be emitted to the atmosphere as is. Therefore, a sustainable process for the direct hydrogen reduction of sulphide minerals should involve, as an integral part an efficient and economic process of the transformation of hydrogen sulphide to inert, easily manageable, harmless compounds. Some of the plasma systems used in the dissociation of H_2S using plasma include the use of a pulsed corona discharge reactor[20], a gliding arc discharge[14], and an ozoniser discharge[15], in single or coupled magnetrons[16]. Nonetheless, the challenge remains as H_2 and S recompose when cooling without rapid thermal quenching[17] [18].

Breakthrough: To address this challenge, a breakthrough, technically and economically feasible, and environmentally sound technology enabling the transformation of the generated hydrogen sulphide to Hydrogen (H_2) and Sulphur (S), is proposed by the PRIM-ROCK project. In the case of sulphide ores, the project proposes a breakthrough technology based on the use of hydrogen on an IFR$_K$ process to reduce the metal sulphides to metal and hydrogen sulphide which in turn is decomposed either with by a two-step thermochemical decomposition or with decomposition to H_2 and Sulphur in a Microwave Plasma reactor. The innovation will be supported by an innovative heat exchanger prototype that controls the cooling effect and inhibits the recombination, via thermal quenching. This enables the produced H_2 to be recycled in the sulphide reaction process and sell the unused H_2 and the produced Sulphur. The proposed process directly addresses all the inherent disadvantages of the conventional production of Ni, Co and Cu from sulphide ores like complexity and many high temperature production stages, unusually high CO_2 emissions, successfully addressing the environmental problems and rendering the production process from H_2 consumer to H_2 producer.

Innovations #7,8,9 Digitalisation

Multirange Spectroscopic Platform as part of Hybrid-Ensemble Modelling for data fusion

To date, there is no popular system that combines spectroscopy with other sensors to improve accuracy by data fusion. For this purpose, we will leverage an innovative **Multirange Spectroscopic Platform** developed by partner IRIS,

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[18] Microwave-Induced Plasma Torch for Thermal Decomposition of H2S into Hydrogen and Sulfur

whereby different spectroscopic Process Analytical Technologies (PATs) have been engineered into a common framework to open the possibility of tackling highly demanding monitoring applications by spectral data fusion in an effective and affordable way. This platform, called VISUM Fusion, is a disruptive framework that leverages all the common elements-power supply, embedded computer, dedicated software, user interface, communication means and regulatory compliance- plus a 'shared suite' of the specific sensors/spectrometers, the respective light sources, and the suitable probes, capable of delivering the following sensing techniques: SWIR, NIR, VIS spectroscopy; Raman spectroscopy; LIBS spectroscopy; Fluorescence spectroscopy. The disruption lies in the fact that we are activating these such techniques through shared spectrometers, as well as common illumination systems (VIS, NIR and SWIR can all use a common halogen lamp), power supplies, software drivers (if the spectrometers are purchased from the same manufacturer) and -in some cases- optical probes. LIBS, as a powerful analytical technique for the simultaneous acquisition of almost all the elements of the periodic table[19,27] is of relevance for monitoring applications in minerals processing, and as such will be added to the Fusion set-up by incorporating a pulsed laser (for generating plasma), an excitation probe and an inspection probe. The LIBS implementation also requires VIS high resolution spectrometers, each covering a narrow band to resolve fine peaks. The Fusion-Hybrid-Ensemble effectively addresses the requirement for multiple spectrometers, and just one or two will suffice. In this setup, LIBS will not operate in isolation; other technologies will accompany it, providing ample redundancy to compensate the lack of resolution. This approach results in a more economical LIBS configuration. This makes the platform ideally suited for experimental work related to the measurement or detection of compounds at low concentration, often in Associated with document Ref. Ares(2024)6673367 - 20/09/2024 complex matrices and with intrinsic barriers that do not match the ideal scenario of any of the traditional technologies. In the Machine Learning (ML) realm, ensemble modelling is a process where multiple diverse models are created to predict an outcome, either by using many different modelling algorithms or using different training data sets[20]. The ensemble model then aggregates the prediction of each base model and results in once final prediction for the unseen data. The motivation for using ensemble models is to reduce the generalization error of the prediction. In this project, we will elevate this concept to the next level by incorporating a new paradigm: Hybrid Ensemble, which extends the rather novel concept of ensemble from its original use in ML towards the sources of instrumental data when such sources involve a plurality of different physics. The concept is particularly suitable for implementing data fusion from a combination of spectroscopic analysers such as LBIS, NIR, Raman; it is intended for extracting more sensitive and accurate predictions on chemical composition and physical properties than could be retrieved by merely using these analysers alone or even by merging the multivariate data from different sensors to be further modelled under a single common predictive model.

Reduced Order Models for the design of the indirect-fired rotary kiln:

State of the Art: The physical-chemical processes involved in the process are complex and affected by many variables. Physics-based models can be used for simulation and analysis, but they will inevitably lead to computationally intensive simulations (CFD, FEM, DEM) that cannot be used in real-time, which hinders their usability in the Digital Twin used for process optimisation. Reduced Order Models (ROMs) can be used to achieve real-time ability. **Reduced Order Modelling** is a mathematical technique to transform complex and computationally costly models into real-time applications that can be integrated in production environments**,** transforming the complex mathematics running underneath the models into simple operations that can be solved on-the-fly with very low hardware requirements[21]. An example of state-of-the-art (SoA) implementation of a ROM approach is Twinkle[22], a library for building families of solvers to perform Canonical Polyadic Decomposition (CPD) of tensors, developed by ITA. This library is able to generate parametric ROMs as products of one-dimensional functions, applying tensor factorization techniques on data sets generated by running batches of virtual tests with the original model. It is especially focused on post-processing data from CAE simulation tools.

A complex system is usually described by the interaction of different parameters. An output variable of interest will depend on many other input variables, some of which may be intertwined. The approximation builds a modal representation of the multi-parametric model

$$F(v_1, \ldots, v_N) = \sum_{m=1}^{M} \alpha_m \prod_{n=1}^{N} f_{m,n}(v_n)$$

[19] Cécile Fabre, Nour Eddine Ourti, Christophe Ballouard, Julien Mercadier, Jean Cauzid, Handheld LIBS analysis for in situ quantification of Li and detection of the trace elements (Be, Rb and Cs), Journal of Geochemical Exploration, Volume 236, 2022, 106979, ISSN 0375-6742, [27] https://libs-info.com/understanding/11/

[20] Vijay Kotu, Bala Deshpande, Chapter 2 - Data Mining Process, Editor(s): Vijay Kotu, Bala Deshpande. Predictive Analytics and Data Mining, Morgan Kaufmann, 2015, Pages 17-36, ISBN 9780128014608, https://doi.org/10.1016/B978-0-12-801460-8.00002-1.

[21] V. Zambrano, R. Rodríguez-Barrachina, S. Calvo, S. Izquierdo , TWINKLE: A digital-twin-building kernel for real-time computer-aided engineering, SoftwareX 11 (2020) 100419

[22] V. Zambrano, R. Rodríguez-Barrachina, S. Calvo, S. Izquierdo. TWINKLE: A Digital-Twin-Building Kernel for Real-Time ComputerAided Engineering, under revision in Software X, (2019)

solution with separated variables and explicit dependency on model parameters. In other words, the output of the system is described by means of independent functions, one for each variable separately and related to the output variable. The approach is based on the
assumption that a problem of N, not necessarily independent, variables can be rewritten as the product of N onedimensional functions, one for each of the variables of the system, as shown in the equation.

Breakthrough: Real time, physics-based ROMs will be developed based on the results of high-accurate, but also computationally demanding simulations, of the detailed physics taking place inside the electrified indirect-fired rotary kiln for roasting (i.e. reacting, multiphase flow with heat transfer). These ROMs will be then embedded and executed as a subroutine of higher-level models to enhance its accuracy without increasing the computational effort. ROMs will be implemented under modelling standards as Modelica or as standalone FMUs, so they can easily link to other codes/libraries of higher-level models. The resulting hybrid models will be exploited for analysis of what-if scenarios, and for the optimization of the process parameters.

Data-based predictive modelling and model generalization:

State of the Art: Industrial processes involve physical phenomena that evolve over time and can be monitored through time-series sensor data. For example, in the lime calcination process[23], the processing time of the material may last over several hours, which increases the complexity of capturing the temporal relations of process parameters/control inputs with the product quality and efficiency. For this reason, calcination processes are often studied through fundamental principles of heat and mass transfer equations and CFD simulations[24]. However, given that the kilns under consideration have major technological differences from others, it is required that specific models are derived, specifically tailored to each kiln and each process. Due to such specialized modelling, a continuous validation and calibration of these models is necessary which cannot always be automated. In order to automatize the modelling and optimization of such complex processes, data-based modelling techniques and intelligent methods have been used instead, such as neural networks[33], artificial bee colony and multivariate regression splines for prediction[26], or machine learning methods such as linear regression, Lasso, support vector regression[27]. Although machine learning and data-based techniques provide a basis for automatizing the generation of predictive models, several challenges remain, namely **a)** the proper selection of relevant (or similar) and high-quality historical datasets and **b)** the proper selection of relevant parameters (features) that can better capture the temporal dynamic phenomena in the time horizon of interest (e.g., when generating predictions). This is because such processes involve nonlinear physical phenomena, thus machine learning models should be trained using data of similar operating conditions (e.g., similar temperature distribution in lime kiln processes[31]). However, even under conditions of properly selected datasets and feature parameters, training such AI-based machine learning models may still encounter several challenges, such as insufficient data, and changes in the characteristics (e.g., distributions) of the features used. Note that in many cases the digitization of such industrial plants has often been completed within the last few years, and there may not be sufficiently rich datasets for successfully modelling such processes through AI-based techniques, e.g., neural networks. This often leads to poorly performing models or models that cannot be generalized to new operating conditions (e.g., to small differences in the material composition, fuel properties, and configuration settings). In such cases, transfer learning techniques, such as domain generalization[28] and adaptation[29] , or model fine

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tuning[30], can be used to exploit additional datasets from similar operating conditions or simulation data. However, there has been a limited effort to address these challenges in time-series data in industrial processes[36], and to the best of our knowledge such techniques have not been applied in calcination and roasting processes.

Breakthrough: Given the absence of sufficiently rich and relevant datasets for modelling calcination and roasting processes through AI-based or data-based techniques, we target the development and implementation of a data processing pipeline that extends current practices in these processes by integrating the following main elements **a)** *data pre-processing* (involving data quality improvements, filtering, and data aggregations), **b)** *feature engineering* (involving the utilization of correlation and temporal causal techniques), **c)** *standard and sophisticated AI-based predictive modelling techniques* (e.g., regression models, recurrent neural networks and LSTM, variational autoencoders, and attention networks), **d)** *domain generalization/adaptation* techniques for exploiting datasets of different sources, i.e., DT simulations and historical data of similar conditions, and **e)** *model fine tuning or ensemble modelling* by utilizing the most recent data or synthesizing hybrid combinations with physics-based models. More specifically, domain generalization, in (d), corrects priorly available datasets (e.g., through optimal parameterization of correction factors or projections) that can retain the validity of the models when the operating conditions change, which provides robustness in model performance through time. Model fine tuning, in (e), allows for continuous adaptation of the models (e.g., regression models or LSTM) to new operating conditions by incorporating new data in the training process as it becomes available, or by establishing hybrid model combinations (e.g., with the physicsbased simulation models). The proposal team has experience in predictive modelling of industrial and calcination processes[31], feature engineering[32], transfer learning in manufacturing and time series datasets[38,36].

1.1.3 Positioning

Technology	Start	End
Lab-scale RMI characterization mechanism: A high accuracy system that identifies and characterizes ore streams to determine the most efficient crushing size	4	6
RMI sorting mechanism: A high accuracy, high performance sorting system for the segregation of mineral ore streams, while capable of dynamically, on-line adjusting the sorting hyperparameters	4	6
Estimation technology for optimal, valorisation flow-specific crushing size Associated with document Ref. Ares(2024)6673367 - 20/09/2024: An ore pre-treatment module based on correlation analysis of crushing size and valorisation flow	4	6
Intelligent Decision Support System for raw material valorisation: An intelligent tool to estimate the optimal sorting thresholds, based on the calculated crushing size, the physical and chemical properties of minerals and the requirements of the overall valorisation flow	4	6
Indirect-fired rotary kiln (IFRK): An electrified rotary kiln for the efficient calcination and roasting of mineral ores	4	6
Microwave rotary kiln (MRK): A microwave kiln for the reduction of metal sulphides	4	6
Microwave-induced plasma reactor: Plasma treatment for the recovery of H_2 from hydrogen sulphide gasses	4	6
Process Control System (PCS): For the efficient and user-friendly control of the IFRKs	4	6
Multirange Spectroscopic Platform: An innovative data infrastructure tailored to the pilots' monitoring challenges	4	6
Reduced Order Models: Advanced models for reducing the order of the physicochemical and thermophysical simulation results for the IFRKs, while maintaining accuracy	4	6
Data-driven models: AI-based algorithms for the optimization of the IFRKs and MRK resource efficiency	4	6

#§PRJ-OBJ-PO§#

1.2 Methodology

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Concept

The PRIM-ROCK project aims to meet the pressing demand for resource efficient mineral processes by introducing innovative solutions with potential applications across various resource-intensive industrial sectors. The project is committed to the transition from fuel-based to electrified thermal processes, advancing both the decarbonisation of the industrial sectors and reduction of the material waste, increasing at the same time the competitiveness of the industries through higher productivity. At its core, PRIM-ROCK focuses on the calcination and roasting processes of mineral ore streams and aims to deliver three state-of-the-art rotary kiln prototypes to replace the highly inefficient conventional methods. These comprise two novel indirect-fired rotary kilns, one for the calcination of lime and magnesite and the other for the reduction of metal oxides and sulphides, in addition to a microwave rotary kiln specifically tailored for sulphide ores. To further drive decarbonization, hydrogen will be employed as the reducing agent for metal oxides and sulphides. To exploit the full potential of the kilns and maximise resource efficiency, the raw material will be suitably pre-processed through an advanced and automated system, encompassing classification modules, crushing and sorting capabilities. Following a holistic and circular approach, PRIM-ROCK addresses also the post-processing stage by developing a cutting-edge microwave-induced plasma reactor to dissociate hydrogen sulphide produced during the sulphides roasting into hydrogen and sulphur, reclaiming hydrogen for reuse. To prevent recombination of hydrogen sulphide a heat exchanger system will be utilized for controlled cooling before H2 is reintroduced in the kilns. Additionally, a virtual toolkit will be developed for assessing the viability of capturing the enriched CO_2 generated by the calcination process, while utilizing the recovered H2 to produce e-Methanol. The project's innovative approach includes a multifaceted digital framework addressing industrial processes in a highly integrated way. A network of sophisticated sensors will be designed and developed to enable the in-depth and accurate monitoring and understanding of the systems physical behaviour, while a process control system will provide the required access to the critical operational parameters. The project will consolidate an interoperable and multilayer data management platform, providing the resources needed for the planning and real-time monitoring of the process lifecycle, through a common framework of different spectroscopic process analytical technologies and implementing data fusion. From the design stage, a complete set of simulations will be conducted, including physicochemical, thermophysical, reduced order modelling of their results and process simulations, to obtain the optimal design of the prototypes. Hybrid digital replicas of the prototypes will be consolidated with the utilisation of AI-based data-driven models, along with decision support systems, user-interfaces and 3D visualisation features, to maximise the resource efficiency of processes. Finally, to ensure the industrial adoption and uptake of the PRIMROCK solutions, a complete analysis of the techno-economic feasibility, sustainability and its overall societal and environmental impact will be conducted. The breakthrough technologies will be demonstrated in 3 different ASPIRE sections, namely Minerals (Magnesite - TERNAMAG, Laterite – NTUA for exploiting Greek FeNi deposits), Cement (Limestone - FORN) and None-ferrous metals (Sphalerite, Chalcopyrite – KGHM and mixed sulhpide oresHEGOLD).

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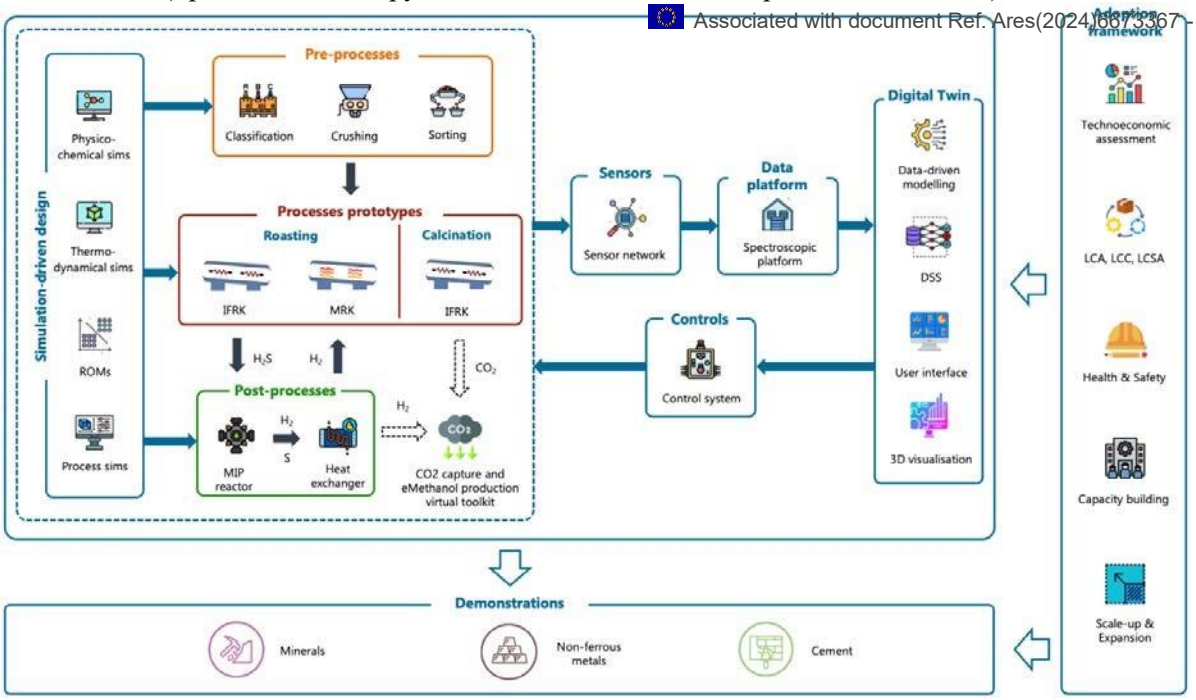

Figure 3. Overall concept of the PRIM-ROCK project

KER1 – Raw material intelligence system (ICCS, CIBOS)

ICCS is set to create an advanced automated system incorporating multiple sensing modules and deep-learning (DL) algorithms. The goal is to efficiently and in real-time identify and separate materials, of various grain sizes, detected in ore streams. Building on the knowledge gained from the Horizon Europe MASTERMINE and W2W projects, enhanced DL approaches will be developed to specifically identify and analyze complex non-linear spectral patterns in ore streams. In detail, hyperspectral cameras, ample of densely sampling the EM spectrum up to 5000nm, will be utilized along with an industrial high-speed RGB camera to optically recognize the particle size distribution and chemical composition of each sample, in relation to processing parameters, and hence perform the localization of the sample along the conveyor belt.

Moreover, information regarding the colour of the samples, i.e. levels of white, will be extracted in to order to assist the identification process, as well as to characterize it based on that feature. In tandem with the cameras, microscopy imaging sensors will be employed to gain further intel on the microscopic structure. In this manner, the samples' unique fingerprint will be defined through the combination of spectral, colour (RGB) and micro-spatial features. For classifying minerals within ore streams, a deep convolutional architecture, such as ViT or GAN, will be employed to extract joint features from both the hyperspectral and the microscopy imaging sources. Specifically, the raw input streams will firstly be treated by mechanical means, such as vibrational separation, cleaning, crushing, etc., ensuring even and uniform distribution along the belt that conveys the samples through the multi-sensory module. Once samples have passed through the multi-sensing module the joint spectro spatial information will be processed by the deep convolutional model to binary classify the samples. Based on the sample's size, the most prominent actuation sub-system (i.e. delta-type robotic manipulator fitted with either electromechanical or suction-based grasping mechanisms to handle larger samples, or pneumatic separation through pressurized air discharge for finer materials with median particle size below 3mm) will be selected, and guided by the previously extracted geometrical features, to perform the physical separation of the samples by placing them in the corresponding container according to the classification results. At the end of the conveyor's belt, an additional bucket is placed to gather unidentified/unsorted objects, that will be re-introduced to the system in a second iteration of sorting. In case they fail to be sorted again, an offline analysis of those samples will be conducted to identify their specific spectro-spatial properties and provide feedback to the sorting system. In this manner, the DL models will be updated to better fit the ore streams distribution, and hence increase the systems sorting efficiency.

A pilot case will be designed, focused on testing and validating two advanced and adaptive automated sorting prototypes, tailored for the calcination and roasting processes, respectively. These will facilitate the pre-processing of the mineral ore streams to **a)** identify and characterize and **b)** sort the apparent raw material. Initially an RMI characterization mechanism, will be designed and implemented to take mineral ore streams as input and characterize their content. The pre-treatment unit, responsible for processes such as mechanical pre-sorting, cleaning and crushing, will be installed prior the characterization system. The treated samples will, afterwards, be analysed using a combination of RGB, hyperspectral and microscopy imaging sensors, while DL models based on SoA architectures, e.g. CNN, Visual Transformers, GANs, etc., will perform the classification task and ultimately the characterization of raw material, based on its optical, i.e. sample's white levels, and spectral properties. In this manner, an extensive analysis of the system's capacity to analyse and characterize both coarse and fine grains will be conducted. The pilotscale RMI sorting prototypes will be developed, in parallel, employing the RMI characterization mechanism as the basis of its design. Specifically, the proposed sorting systems will utilize and build upon the same multi-sensor module, while incorporating a combination of robotic mechanisms, i.e. delta robot with a gripper or a vacuum endeffector for handling bigger and coarser samples, and a pneumatic system for the separation of fine/ultrafine grains. The sorted output stream will be then forwarded to the IFRKs. The integration of the lab-scale RMI characterization mechanism, post the IFRKs will, also, be investigated. In this manner, an automated characterization method for the products of the processes is proposed, thus a) allowing for better evaluation of the pipeline's overall efficiency and b) providing further feedback to optimize the crushing and sorting of the ore streams.

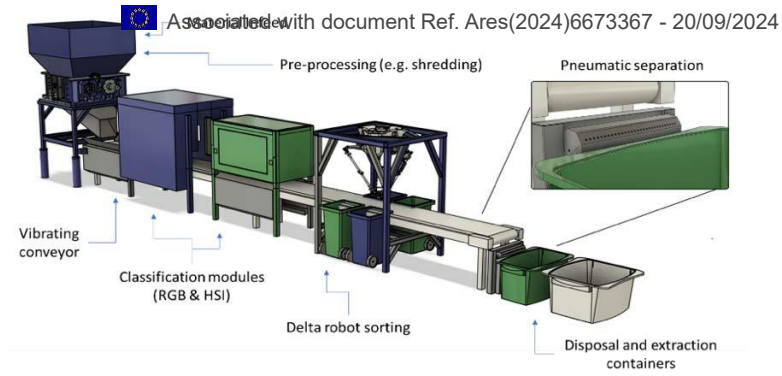

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Figure 4. Concept of the Raw Material Intelligence system

To increase the system's versatility an intelligent agent will be trained on data extracted during the sorter's operation time, to develop a Decision Support System (DSS) that dynamically estimates the optimal separation criteria and adapts on-line the sorted output stream. In detail, the intelligent DSS will leverage information acquired through the

operational phase of the sorter to select the optimal separation settings based on the properties of the input stream. Different combinations of pre-treatment approaches (e.g. crushing, washing, etc.) and classification schemes will be tested to identify the number of sorting scenarios the DSS may suggest and hence increase the overall performance of the system. Moreover, by utilizing the data collected from both RMI mechanisms, a continuous analysis of the composition of the input and output stream will be conducted, thus identifying changes in the streams' composition in terms of minerals and other substances concentration. In this manner, the deep learning classifier model will be regularly updated so that its output will dynamically adapt to circularity and sustainability criteria, such as aspects related to offer vs demand, logistics optimizations, supply chain management, etc. The systems' performance will be evaluated, based on absolute metrics (e.g. accuracy, picks per minute, etc.), as well as according to multidisciplinary, sustainability criteria, aiming to align the system's output characteristics with the rest of valorisation chain.

KER2 – Intensification technologies for thermal processing (NTUA, CEIN)

Indirect-fired rotary kilns for calcination and roasting of mineral ores

The process intensification component of the project will design, develop, and validate innovative and efficient technologies. Specifically, NTUA will construct two novel Indirect-fired rotary kilns (IFRKs) to replace the conventional Direct-fired rotary kilns (DFRKs) used in the calcination and roasting processes of mineral ore streams. In order to efficiently control transfer of gases and solids, hoods at feed and discharge ends of the kiln shell will be designed. Special seals between hoods and rotating shell will minimize ingress air and maintain required process atmosphere. As kiln shell will be heated externally, the selected material must withstand temperature, corrosion, and/or abrasion. Special attention will be given to enshrouding furnace for optimum heating up of the kiln shell with the electric emitters while isolating process gases from environment. Heating zones with electric emitters, insulation, wiring, and junction boxed for fixed kiln length increments will be designed to be easily serviceable, replaceable, and exchangeable. Sensing devices like thermocouples, H_2 mass flow, power meters, etc. will provide dynamic data for system monitoring and control.

The ores will be charged from the material bins to the kiln. The hydrogen needed for reduction will be pre-heated in a heat exchanger, before entering the reactor. To ensure a uniform gas distribution inside the reactor, a propriety gas distributor design will be applied. After reduction and final material sampling, the whole material is collected again in the material bins and discharged afterwards into a quenching vessel (cooler). Gas recycling stream will be passed to a heat exchanger and a gas cooler, for removing the formed water vapor during reduction. Unreacted hydrogen will be recycled and re-used in the process again after passing a recycle gas compressor, the heat exchanger, and the reducing gas furnace. The theoretical hydrogen consumption by the reduction will be added to the process before the heat exchanger.

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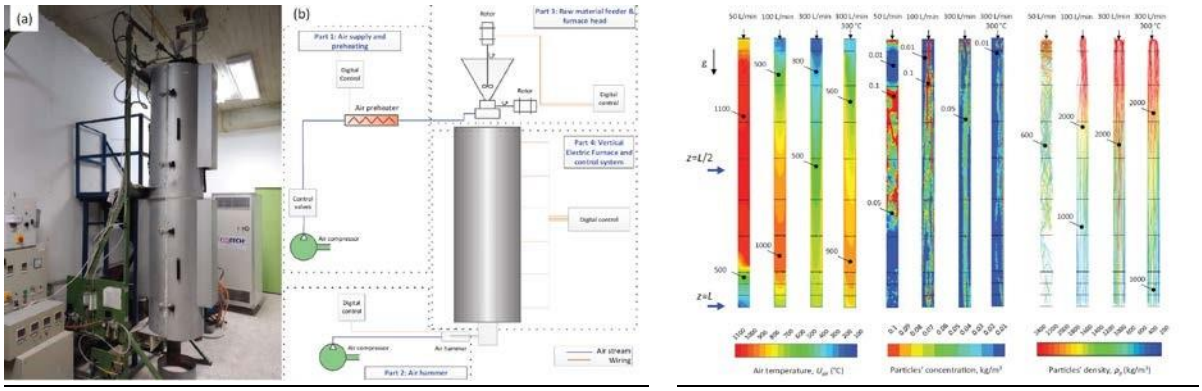

Figure 5. Top: Electrified vertical kiln for microspheres production, Bottom: validated CFD modelling (Contours of air temperature (left) and particles' concentration) for the electrified furnace[33]

NTUA team has an extensive knowledge, on research and technology development in the field of extractive industries, covering actions for designing, simulation and prototyping sustainable metallurgical routes by utilising H2 and energy

[33] Panagiotis M. Angelopoulos, Antonis Peppas & Maria Taxiarchou (2023) Modelling the Thermal Treatment and Expansion of Mineral Microspheres (Perlite) in Electric Furnace Through Computational Fluid Dynamics (CFD): Effect of Process Conditions and Feed Characteristics, Mineral Processing and Extractive Metallurgy Review, DOI: 10.1080/08827508.2022.2161536

efficient processes. The team is also equipped with equipment for physicochemical characterisation like ICP-MS, AAS, XRF, XRD, laser particle analysers, BET, TG, SEM, TEM and others.

Microwave rotary kiln for the reduction of metal sulphides

The development of the PRIM-ROCK project includes the construction of a microwave kiln for the reduction process of metal sulphides, designed with adjustable electromagnetic fields within the tube and featuring a novel system capable of modulating the signal to accommodate the dynamic changes occurring during the roasting reaction. This innovation is crucial for managing the transition of materials from moderate low dielectric properties to conductive states, necessitating precise control of the electrical field in a reactor designed for metallic materials. An advanced control system will collect thermal and microwave property data, which will be instrumental in developing the algorithms and software necessary for real-time tuning of the process. This level of control is essential for optimizing the roasting reaction and ensuring efficient treatment of materials.

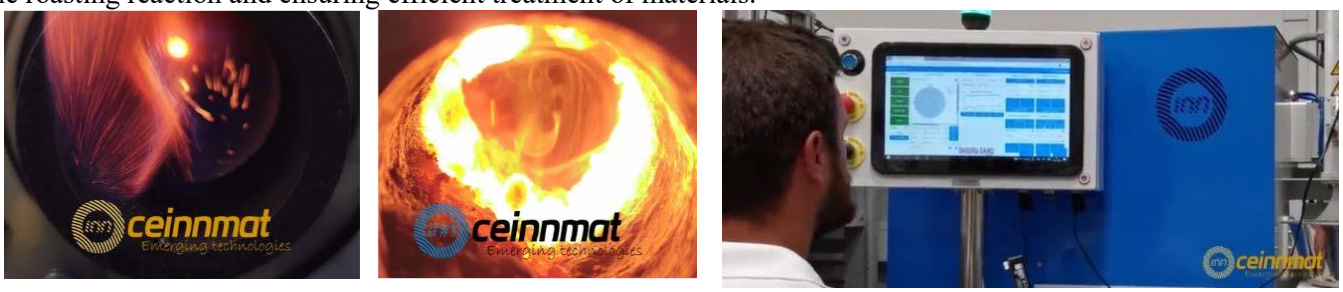

Figure 6. Tests with metallic component (left) sulphides (centre). Control system with microwave parameters, with polar Smith chart as base to be used in PRIM-ROCK (right).

Direct reduction of metal ores using hydrogen as a reducing agent

A breakthrough process for the treatment of metal sulphide concentrates is proposed involving hydrogen reduction of the concentrate in the IFRKs to produce metals and hydrogen sulphide as a result of the reaction between hydrogen with sulphur. The process is intended to be used in tandem with KER3 to achieve the sustainable use of hydrogen. The main advantages of the proposed process include: **(i)** simplification of the high complexity of metal sulphide concentrates, **(ii)** recovery of the used hydrogen affecting the cost and sustainability of the overall process, **(iii)** solving the environmental problem of SO_2(g) generation in the processing of sulphide ores through the production of sulphur. The process is simpler and less energy intensive. Nonetheless, as a new process, there are gaps in fundamental knowledge needed: where available, the results of lab experiments carried out by the project partners or available in the literature will be considered in the experimental design, but a significant number of lab tests will still be needed carried out to fill the gaps in the existing knowledge in terms of refining process kinetics and optimising the parameters of the reduction process. It is important to mention that although there is initial knowledge in the study of pure materials, this knowledge is not considered in many cases complete and needs significant work effort for additional testing. Concerning metal ores/concentrates and tailings very few references are known for the exploitation cases proposed in this project. The main outcomes from lab scale testing include development of proper mathematical

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models to predict the technical behaviour of pure oxide and sulphide materials as well as ores, concentrates and tailings under variable process conditions; identification of the optimum operation conditions in each of the studied systems; and determination of the quality of each product in terms of metal extraction and product purity. Based on the acquired knowledge from the lab scale testing, continuous operation bench scale tests will be carried out at the optimised conditions to produce adequate quantities (3 – 6 kg) to evaluate the process efficiency, hydrogen and energy consumption and the quality of the targeted metals. Tests will be performed in the modified IFRK. In terms of processing in the IFRK, the process is straightforward: the kiln will be loaded with sulphide ore, where the preprocessing (particle size, grinding, pelletisation) will be investigated. The optimal heat distribution will be defined across the three heating zones, in addition to other parameters (e.g., staying time, rotation, tilt, H_2 pressure). The main outcome of the bench scale tests includes the determination of the optimum conditions in each case studied in terms of process conditions, H_2 consumption and product quality to enable the engineering design of the pilot and demonstration tests. Depending on each case, the produced material can be a marketable ferroalloy, as in the case of nickel laterite treatment, or an intermediate product that needs to be further processed to extract and purify the contained critical metal. Considerable research effort will be allocated to the decomposition of the H_2S produced during the hydrogen reduction of sulphides, to H_2 gas and solid sulphur, as this process provides significant prospect to considerably enhance the technical and economic viability of the under-development solutions.

KER3 – Post-processing solutions (EUT, BUL, NTUA)

Microwave-induced plasma reactor for H_2 recovery

PRIM-ROCK will demonstrate at TRL6 an integrated and sustainable MIP reactor for the decomposition of H_2S gasses and recovery of H_2(g) and S(s), with subsequent reutilisation of H_2 for direct reduction in the process stage. As described, PRIM-ROCK will employ hydrogen as reducing agent to produce critical (Fe, Ni, Co) and important metals (Zn, Cu) in an environmentally friendly and economically sustainable way. The capacity of processing several materials with the same technology showcases the flexibility and adaptability of the solution. To realise the potential of the technology, it is imperative to generate fundamental knowledge, enriching existing models of kinetics[34] and especially on the plasma thermal quenching effect of the reverse reactions[35]. The process to be employed is comprised of the following elements:

(1) Reactor supply: To supply the reactor, a mixture of H_2S and an inert gas (e.g., argon) is used. The feed gas composition and supply rate will be extensively studied (projected 70-100% H_2S concentration, at flow rates of 0.53l/sec, preheating temperature).

(2) Plasma generation: The feed gas is then introduced into a plasma reactor, where it is subjected to a microwave field. The microwave field ionizes the gas, creating a plasma that dissociates the H_2S molecules into H_2 and S. The parameters in this stage are the power input that also influence the reactor temperature. **(3)** Product separation: The H_2 and S produced in the plasma reactor are separated from each other using a condenser. The separation process is a critical step in the process, as it determines the purity of the final products. **(4)** Product recovery: The H_2 and S are then collected in separate containers. This stage is crucial due to the cooling effect, as outside the optimal cooling temperature the original reactant H_2S gas is reformed. For this stage, a plasma thermal quenching analysis is very important to help design the optimal heat exchanger that will retain the hydrogen and sulphur products, inhibiting recombination. Following the simulations, experimental work first at bench and then arelevant scale is foreseen. Where available, the results of lab experiments carried out by the project partners or available in the literature will be considered in the experimental design.

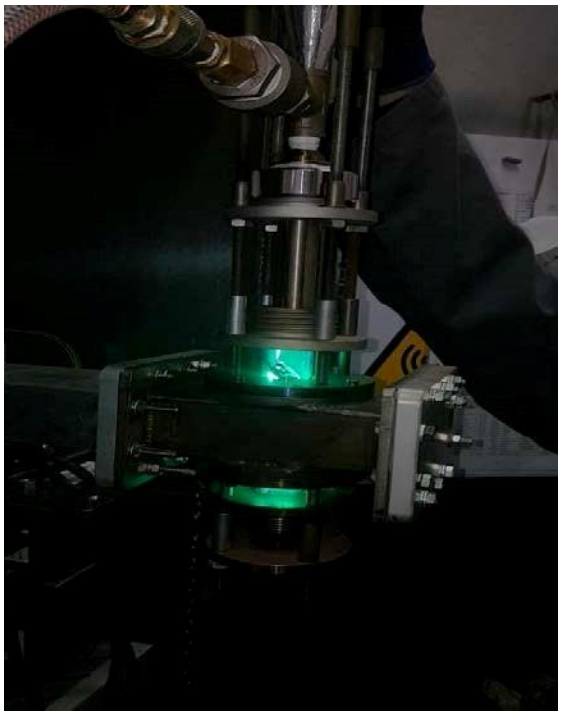

Figure 7. Microwave Induced Plasma reactor (MIP) at EUT premises

Design of a heat exchanger system to fit the existing reactor

[34] Alexander Friedman, Plasma chemistry. Cambridge University Press, 2009, pp. 744–745

[35] Microwave-Induced Plasma Torch for Thermal Decomposition of H2S into Hydrogen and Sulfur, Mohamed Sassi and Naji Amira, Masdar Institute of Science and Technology

Cooling of a high temperature surfaces (over 2000°C) presents an extremely challenging heat flux values that makes the use of conventional heat exchangers designs unviable. This is mainly due to the nature of suitable coolants for this temperature range that takes the heat exchanger solution to the use of liquid metals as coolants, which brings an extremely demanding safety process that is usually cost-prohibitive in most applications. However, with the advances in materials science and the adaptation of the heat pipe technology in many high temperature applications, an innovative heat pipe heat exchanger for such high heat flux applications can be a viable, cost effective and safe heat management systems for the cooling application of such walls. It is also due to the ability to control the heat transfer process withing the heat pipe system (boiling – condensation of the working fluid) through an innovative design for the heat pipe condenser system that twill keep the heat pipe system operating at a lower temperature than the allowed value within its structure to keep the heat transfer process sustained. An example system was developed by BUL that was tested in a representative environment to cool hot surfaces (1400°C - 2500°C) using a water-cooled heat pipe system with water as its working fluid. During the operation of the system, the internal working temperature of the heat pipe was 250°C thanks to the innovative condenser design for this heat pipe that facilitated the required active heat removal form eth heat pipe system that was absorbed form eth high temperature wall through radiation. An example of the system can be seen in the photos.

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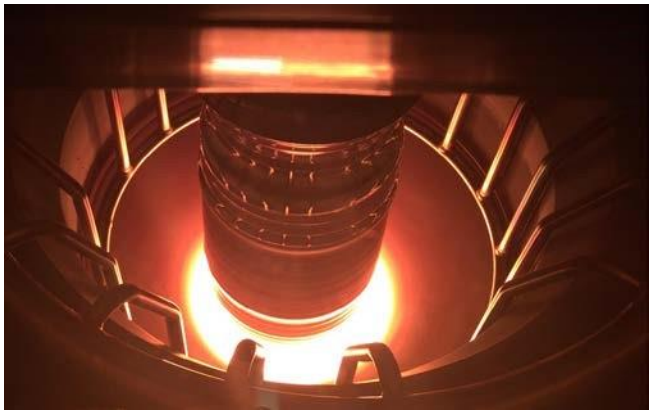

Figure 8. Heat exchanger used in the silicon industry (>1500ºC) for controlled cooling (BUL)

CO_2 capture and eMethanol production virtual toolkit

The Virtual Toolkit will simulate in a dynamic way a promising configuration for the exploitation of enriched CO2, produced via the calcination process of IUC1 (Lime) and 2 (Magnesite). Within the energy system's boundary conditions, based on the user's inputs that will set the objective function for each scenario, the recovery of CO2 through a virtual CCUS unit will feed a virtual e-Methanol production unit, and the system will be modelled for technoeconomic and environmental analysis. A Mixed-Integer Linear Programming (MILP) will be employed for the operational multi-objective optimization algorithm, while Reinforcement Learning (RL) will also be considered. Black-box models will be employed for the CCUS and e-fuel production unit, meaning that a data-driven approach will be preferred over physical modelling. This will require the collaboration of domain experts. The toolkit will provide reliable energy balance information, including emissions and costs and deliver long-term scenarios for investments according to end-user priorities and constraints.

KER4 – Digital process-line optimisation (IRIS, MAGG, ITA, NTUA, CIRCE, BUL, CORE, SCCH, STAM)

Scadalytics Industrial IoT data platform

The data from VISUM Fusion platform, and from any other relevant data source is collected in the Scadalytics Data Management Platform that consists of a series of layers. The Communication Interfaces and APIs layer is designed for data integration and interoperability not only with the VISUM Fusion, but can also receive data from SCADA/ERP, PLC, from other analysers, Industrial IoT or field devices and enriched by external data (economic, societal). A data processing layer performs data conditioning prior to its use by the data fusion layer. As discussed, a ground-breaking advancement is the combination the Multirange Spectroscopic Platform with hybrid-ensemble modelling for implementing data fusion. The most straightforward approach for implementing data fusion through hybrid ensembling involves identifying optimal pairs of instruments and machine learning methods within a specific ensemble architecture, to achieve both the most robust and accurate predictive algorithm. An instant evident advantage is a reduction in the number of observations (samples) required for developing models that are not prone to undergo overfitting. In fact, the lack of sufficient numbers of different samples (as many as required for building useful models) is the most frequent barrier to the wide use of machine-learning-based spectroscopic analysers. The hybrid ensembling concept addresses the limitation of two extreme approaches: on the one hand, modelling the merged data by means of just one weakly optimal algorithm and, on the other hand, applying the machine learning ensemble approach to a dataset that involves limited information provided by just one type of spectroscopic analyser. The outputs from this layer can be either sent directly to the Scada, can be interfaced through a self-contained User Interface, or can be sent up to an upper layer of intelligence for use by predictive models. Indeed, the Scadalytics platform, in combination with the VISUM Fusion, has the potential to move minerals processing towards process automation through multivariate process control (MVPC) and software sensors trained on historic process data. Scadalytics can also connect with a higher-level Digital Twin that may be in place or can act itself as the Digital Twin of the productive process that is being monitored. Modelling any novel intensified process or, in general, complex systems such as a plasma reactor entails not only managing a plurality of data sources but also describing processes that cannot be precisely emulated by deterministic approaches. That is the main reason for taking advantage of the combination of both Visum Fusion and Scadalytics for seamlessly acquiring highly informative data and building and hosting data-driven or hybrid digital twins of the involved processes in a common framework.

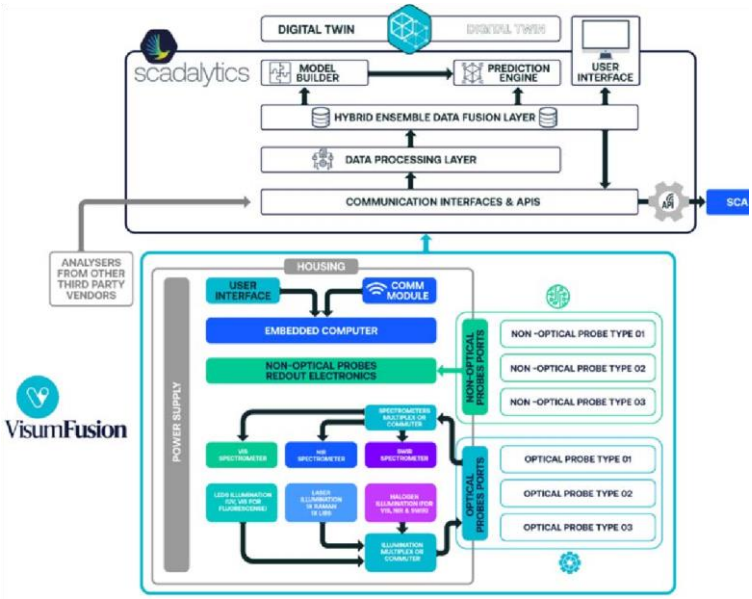

Figure 9. High level schematic of based on Multirange Spectroscopic- Hybrid Ensemble Fusion Platform and Scadalytics data management software

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Advanced sensor network for the process and post-process Figure 9 provides a schematic of the complete sensing network. It is based on the VISUM Fusion Multirange Spectroscopic Platform that incorporates several detection techniques (SWIR, NIR, VIS, Raman, Fluorescence and LIBS) in a common hardware platform, which not only allows us to use the optimum technique for each measurement requirement, but it also enables seamless data fusion, to obtain dramatic improvements in precision, accuracy, and limit of detection. This self-contained portable multirange spectroscopic platform is comprehensive enough to be configured for successfully tackling a high plurality of demanding applications, such as the in-line monitoring of the processing and refining of minerals and critical raw materials, while at the same time being highly modular to enable easy simplification as required for targeted applications and for the sake of affordability. To further extend the framework

Figure 10. VISUM PalmTM: Portable (handheld) NIR analyser for at-line control at different points of the process, from raw material to final quality control.

capabilities, VISUM Fusion can also fuse data provided by off-the-shelf compatible temperature and electrochemical sensors installed in each of the case study pilot plants.

Multiscale physics-based simulations towards the optimal prototypes design

To obtain the optimal design of the prototypes towards enhancing their resource efficiency PRIM-ROCK introduces a complete set of simulations to vastly analyse their physical behavior on different scales of interest. Special emphasis will be placed at the simulations of the IFRKs to generate knowledge that is currently scarce or underdeveloped in the literature, regarding the reaction kinetics, mechanisms and thermodynamics of the calcination and roasting processes involving ores of different physicochemical nature. First, the kinetics of the reactions will be assessed from molecular dynamics simulations (i.e. atomistic modelling). The simulations will rely on the reactive force field (ReaxFF) interatomic potentials, which allow to model large-scale systems where chemical reactions are occurring. This novelty approach links the quantum mechanics accuracy, which itself is computationally intense and limited to small systems, with the classical Molecular Dynamics (MD) algorithms and scheme to allow for exploring larger simulation scales and times than DFT methods. The ores will be modelled in the atomic level, considering the crystallographic structure of the selected mineral variety. In the case of roasting, hydrogen will be added to the model considering different concentrations and temperature conditions and allow it to react with the ore to monitor the evolution of the reduction reaction under the ReaxFF force field. Such simulations will allow for monitoring the different chemical species that are formed, or disappeared, during the chemical reactions, which will help in establishing a detailed reaction mechanism pathway. Besides, the precise monitorization of the concentration of the chemical species involved in each reaction step will allow for determining the kinetic parameters associated (i.e., activation energy, kinetic constant, order of reaction). This will not only help getting a better understanding of the reduction and carburization processes, but also give indications of the limiting physical and chemical processes. As a second step, the simulations will scale up, implementing a novel methodology to obtain the volumetric reaction rate at the particle level. To this end, the reaction kinetics obtained with ReaxFF will be considered at the gas-particle interface (surface reaction) to predict the overall volumetric reduction rate, as well as the materials properties (particle size, specific surface, porosity) and process parameters (such as residence time). This methodology will be developed using open-source tools such as LIGGGHTS (Discrete Element Method, DEM), OpenFOAM (Finite Volume Method FVM – Computational Fluid Dynamics CFD) and Python. Once the workflow is fully implemented, then it will be applied to study the calcination and roasting process at the particle scale under a wide range of scenarios (Design of Experiments, DoE), considering the range of the above-mentioned parameters of the process. The simulations are expected to be highly computationally demanding because of the complexity of the physics involved (multiphase, reacting, heat transfer flow) and thus the further direct application to the reactor scale will be unfeasible. Hence, the results of this physicochemical simulations at the particle level will be postprocessed using reduced order techniques. The resulting ROMs will be able to predict in real-time the volumetric reaction rate of the ore particle depending on the ore properties and operating conditions, and so, could be later used as a source term of the full thermodynamics IFRK simulations. Regarding the latter, a comprehensive CFD-DEM model will be developed, in which steady-state value of different relevant variables will be evaluated all along the dimension of the equipment. These include temperature, pressure, heat losses, emissions, calcination reaction rate and conversion, among others. The model will help to evaluate any change in operating conditions, geometry, raw material specifications or boundary conditions, providing a strong support to design phase by allowing virtual testing. The results will, finally, feed the full-scale process simulations, performed through simulators such as ASPEN, to model the whole process behavior and obtain the optimal design of the kilns.

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Digital Twinning of the process towards optimization of the resource efficiency (data-driven models, DSS, UI)

In PRIM-ROCK an AI-/data-driven modelling framework will be developed for supporting the monitoring and optimization of the calcination and roasting processes. Given that such techniques may require sufficiently large and rich datasets and may fail to generalize to operating conditions or scenarios that are not representative in the historical data, we will develop data-driven models that can be generalized to possibly unseen scenarios/operating conditions.

This would be achieved by utilizing a larger range of data sources, particularly data generated through the physics-based simulation framework developed in the project, or data generated by similar industrial use-cases. Such integration of alternative data sources, often referred to as *transfer learning* and *domain generalization/adaptation*, is graphically depicted in Figure 11. The upper (*process/data pre-processing layer*) layer captures a standard data-based modelling framework, also applied here, comprising data quality/pre-processing and feature engineering/selection techniques. These steps would be sufficient for most AI-/data-driven modelling approaches, such as different types of linear/nonlinear regression models or neural networks. However, as discussed earlier, such models may be sensitive to variations in the configuration of the processes. To this end, we augment this framework with the *knowledge-transfer layer* that aims at transferring additional knowledge to the model design through **a)** alternative data sources (e.g., simulations, similar processes), and/or **b)** model fine-tuning. In the first case, a feature/parameter alignment (or normalization) technique is implemented where features of different sources are transformed to become invariant to the details of the industrial process (e.g., dimensions of the kiln, material properties, etc.). This allows for building up significantly larger datasets for training a model that is robust to such variations in the process details. In the second case of model fine tuning, the trained model can further be adjusted through either **i)** additional training in the scenario of interest, or **ii)** hybrid/ensemble modelling, e.g., via combinations with the physics-based models.

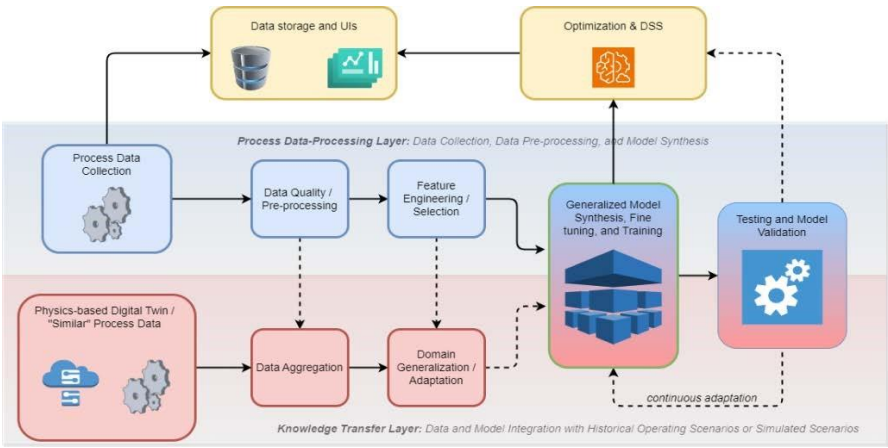

Figure 11. Concept of data-driven digital twinning towards DSS.

Process Control System

The process control system builds on MIRA, a Digital Twin Enabler platform developed by MAGG, which embeds the functionalities: **1)** to model the digital representation of the industrial process, specifying all relevant involved assets, resources, and data, **2)** to execute and monitor the operational aspects, **3)** to receive data to measure user defined metrics. The features specific to the process control will be implemented on top of MIRA's native capabilities to implement the process Digital Twin, following emerging trends for the monitoring of industrial systems over their life cycle in the context of estimation and control[36]. Besides modelling the real-life system, the underlying digital twin has the ability to operate in parallel with the real-life system and in an offline setting. The latter enables what-if analyses or investigations relating to different scenarios despite their occurrences. The figures below exemplify some of the possible data customizations.

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1.2.1 PRIM-ROCK methodology

The project's implementation will be carried out in three phases from a technological perspective, while DEC activities are distributed in WPs11-13 and Coordination in WP14.

P1 Preparatory & design phase (WP1-3, M1-18): The starting point of PRIM-ROCK is the consolidation of the project's technological specifications (T1.1), as well as the exact identification of the end-users' requirements and needs (T1.2) to establish a rigid framework for the development of the solutions until M6. This will be achieved through a detailed analysis of the use-cases involved and the characteristics of each novel process to properly align

[36] W. Birk, R. Hostettler, M. Razi, K.Atta, R. Tammia; Automatic generation and updating of process industrial digital twins for estimation and control - A review; Front. Control Eng., 25 August 2022, Sec. Control and Automation Systems, Volume 3 - 2022 | https://doi.org/10.3389/fcteg.2022.954858

the technological focus and investigate the potential for improvements. The critical parameters will also be identified for the smooth implementation of the monitoring and control systems (T1.3). Safety and security issues will be considered in this phase (T1.4) and the baseline for later efficiency calculations and comparisons will be established by M12 (T1.5). Based on these outputs, detailed multi-physics simulations (both on microscopic and macroscopic levels) will be conducted for processes and components to obtain the optimal design configurations towards increasing the overall resource efficiency (WP2). The project's digitalisation framework will also be designed during this phase, starting with the overall IoT reference architecture (T3.1) and the design of the sensor network (T3.2), as well as the process control and interfaces (T3.3) until M18.

Figure 12. Project phases and associated objectives

P2 Intelligence development and prototyping phase (WP4-7, M18-36): The second phase marks the progression from conceptual designs to tangible solutions. Key activities include the development and integration of intelligent systems (WP4) that harness the power of advanced sensor networks, AI algorithms, and data processing capabilities. These systems will undergo rigorous prototyping, aiming to refine the accuracy of real-time material sorting and processing. At the same time the Industry 4.0 framework will be deployed in WP5, featuring integration of automation and control systems (T5.1), data integration and orchestration through the project's IoT platform (T5.3), data-driven model development aligning with the project's overarching goal of enhancing resource efficiency (T5.2) and, finally, the integration of process digital twins with the associated decision support systems, interfaces (T5.4) and 3D visualisation attributes (T5.5). The IFRKs will be prototyped in WP6, first at lab scale to produce data (T6.1), and then following the simulation-driven optimisation of the heating mechanisms (T6.2, T6.3) the pilot-scale prototypes will be delivered (T6.4), as well as the microwave rotary kiln for sulphide roasting (T6.5), the microwaveinduced plasma reactor for H2 reduction (T6.6) and the heat exchanger for controlled gas cooling (T6.7). A preliminary assessment of the process feasibility from the technoeconomic perspective (T6.7). Once all the components are ready by M28, integration and validation will be carried out by M36 (WP7) in two pilot units: one in Greece (NTUA) and one in Spain (EUT).

P3 Demonstration phase (WP8-9, M37-48): During the Demonstration Phase, the focus shifts to demonstrating the developed solutions in relevant environment (T8.1, T8.2), accompanied by the various assessments aimed at validating the potential of the technology (T8.3). Simultaneously, a complete and comprehensive assessment of the impact of the PRIM-ROCK solutions will be carried out, including technoeconomic analyses, energy and intensity evaluations and societal and environmental studies (WP9).

Greek Pilot unit: H_2 Living Lab

Pilot activities at TRL 6 will be implemented in a H_2 Living Lab, where an ATEX IIC classified area is supervised by a Safety and Protection System (SPS). SPS manages signals are collected from different types of detectors like H_2 concentration, infrared flame detectors, pneumatic valves, compression status etc., which automatically activating pre-configured safety procedures in case of alarm. H_2 equipment is powered by a smart grid (2 PV plants and 6 wind generators, battery banks) operated through an intelligent energy flow monitoring and control system (54 smart meters and 1.800 monitoring nodes). A massive amount of data fromRES and H_2 facilities performance data, will be fed on models to be trained through AI.

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The Living Lab will create from the beginning of the project an environment, for a methodical approach focused on PRIM-ROCK technologies. An electrified 20kW rotary kiln will be fed with ore samples and H_2 produced from PEM electrolyser, where sulphates and oxides reduction rate will be evaluated towards processing parameters. Pilot trial will provide a data flow for evaluating physicochemical and simulation trained models accuracy. From the operational perspective, real-time (online) capacity which ensures maximum uptime of green-H_2 production via electrolysis or other means, which is a function of RES availability, energy pricing and storage SoC will be also validated. Representative reduced ores samples will be smelted for producing alloys to be characterised as intermediate products.

Figure 13. H_2 living lab infrastructure

Spanish Pilot unit: EUT premises

The Spanish pilot will be implemented in the EUT laboratory located in Barcelona. In the dedicated pilot area the MIP torch system is currently installed, which consists of a microwave head, protection and waveguides to deliver energy at the treatment point. The treatment point comprises modified waveguides to accommodate a tubular plasma capsule made of ceramic material, through which the material to be treated circulates. The magnetron and shields are water-cooled, while the plasma capsule is cooled by compressed air. There is the possibility of modifying both the waveguides and the plasma capsule to adapt them to different processes. That requires a brand-new design or modifying the current assembly. The ignition and material feeding are currently manual, but there is ongoing research into the possibility of automation. There are currently two 2 kW, 2.45 GHz water-cooled generators and a third on with the same characteristics and power, but air-cooled. A fourth generator, with 1.3 kW power, is currently dedicated to emergency support. Additional equipment is available, such as instrumentation, sensors, and data loggers, as well as equipment for material treatment before or after the use in the gas flow (sieve, mill, dust handlers, etc.). The entire system has laboratory-level processing capacity, making industrial-level production not possible without scaling for such purposes. The processing of solid materials (powder) is limited to less than g/min, and the gas processing level is at 1-3 l/min. However, multiple generators can be combined in series or parallel for specific tests. The system is versatile and can be adapted as required. It allows the generation of plasma from various types of gases (air, CO_2, oxygen, helium, argon, nitrogen, hydrogen, etc.) and can work with dusty plasmas.

Figure 14. Two microwaves working for hydrogen decomposition from water.

1.2.2 Industrial use-cases

IUC#1 – Lime production (FORNACI)

Organisation description: Fornaci Calce Grigolin has been operating for over 55 years in the production of high-quality lime with the aim of guaranteeing its customers a wide and complete range of products for the use of lime in various sectors with different purposes. The company boasts modern plants with high productivity and the continuous production cycle which, thanks to the use of wood sawdust as fuel, is considered zero impact as regards the Kyoto protocol. Another strategic factor guaranteeing continuity of supply is the high availability

of quarries belonging to the Grigolin group and own deposits of limestone raw material of high quality and homogeneity.

Figure 15. FORN lime production plant.

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Process description: The raw material to obtain lime is limestone: a sedimentary rock mostly composed of calcium carbonate that is extracted from special quarries. The limestone, coarsely crushed with a size of a few centimetres, is introduced into special furnaces where it is fired at high temperatures between 800 and 900°C. During this phase, a reaction called "calcination" takes place, which leads to the formation of calcium oxide with the release of carbon dioxide. The product thus obtained, also know as quicklime, takes the form of lumps of various grain sizes, which are then subjected to grinding and sifting. For cement industry an indicative material efficiency values for a raw mill, pyro-processing tower, rotary kiln, is 36%, 37% and 34[45]. Hydrated lime (or slaked lime) is obtained through the process of hydration of quicklime. The extinguishing of calcium oxide can take place by stoichiometrically dosing the water, with the formation of powdered hydrated lime, which in turn can be refined with specific wind separators to obtain micro-lime (or lime flower). When quenching takes place with excess water, Lime Putty is obtained, which undergoes suitable seasoning in special tanks.

Objectives: (a) improve the calcination process, (b) RMI to optimize kiln performance by removing unwanted materials in the raw material stream, (c) digitalisation framework through dynamic simulation of calcination process based on feedstock changes, data-driven models for the optimization of operational parameters, Reduced Order Models development and Hybrid Digital Twin of the process.

Implementation: Activities at FORN will start by deploying a gateway and all relevant infrastructure for data communication. All data relevant to raw materials will be shared, emphasizing in limestone particles size, as well as those relevant to the calcination process. FORN will also provide specifications on the UI and the expectations for results visualisation. Samples of raw materials will be shipped to partners for characterisation and to develop the digital models. The digital twin of the process will allow FORN to simulate the calcination performance, including kiln health state (e.g. temperature and pressure distribution), anticipating dangerous situations and removing noncompliant feedstock. Non-compliance can mainly consist of very fine limestone, that creates clogs in the kiln and reduces the air flow, and contamination by other materials, which can generate unwanted situations during calcination. Switching from batch to real-time analysis of fuel may also improve the calcination performance, preventing the plant from feeding contaminants in the kiln.

IUC #2 – Magnesite production (TERNAMAG)

Organisation description: TMAG is a magnesite and magnesia producer, member of GEK TERNAMAG GROUP, one of the leading business groups in Greece, with €1.2 billion turnover, over 5,000 employees, presence in 16 countries and half a century of market activity. It exploits the world-class magnesite deposits in the northern part of Euboea Island (Greece). Its large volume reserves consist of high-purity, boron-free, low-iron, cryptocrystalline magnesite. TERNAMAG currently boasts a high purity and consistency Caustic Calcined Magnesia (CCM) and Dead Burned Magnesia (DMB) products. Through a state-of-the-art beneficiation, calcination and sintering equipment, it can produce and supply a wide range of products, with MgO content up to 97% and customized grades. Its new stateof-the-art rotary kiln has an annual capacity of 60,000 MT sintered/calcined products. **Process description:** Beneficiation takes place in situ, including crushing, washing, screening/classification and use of magnetic and optical sorters. Hand picking is also used for the selection of high purity, special material. For the processing, calcination uses LPG fuel. Tailings are being processed with optical and magnetic separators and equal to 1.3 mMT. The low-grade magnesite (most of them fines and ultra-fines), is about 45% of the total exploiting magnesite and is treated as mine-wastes, generating considerable resource wastes and causing severe environmental consequences[46].

Objectives: (a) improve the calcination process, **(b)** production of very clean CO_2, **(c)** Colour detection for optical sorting, **(d)** digitalisation framework through dynamic simulation of calcination process based on feedstock changes, data-driven models for the optimization of operational parameters and Digital Twin of the process.

Implementation: IUC#2 will adopt the same implementation methodology as in IUC#1. Key activity in this case is the pre-processing of the raw material through the RMI system, utilizing deep convolutional models to characterize the content and an actuation system for sorting the apparent magnesite.

Figure 16. Top: TERNAMAG plant for magnesite production, Bottom: Rotary kiln for magnesite calcination.

[45] Tianming Gao, Litao Liu, Lei Shen, Fengnan Chen, "Analysis of material flow and consumption in cement production process", Journal of Cleaner Production 112 (2016) 553-565, Elsevier, http://dx.doi.org/10.1016/j.jclepro.2015.08.054

[46] Hao-Yu Zhu, Shu-Wen Shao, Ming-Zhi Guo, Shi-Yi Zhang, Yan Zhang, "Engineering properties and sustainability evaluation of crushed low grade magnesite mortars", Journal of Cleaner Production, Volume 425, 2023, 138979, ISSN 0959-6526, https://doi.org/10.1016/j.jclepro.2023.138979.

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IUC#3: Ferronickel production (NTUA)

Organisation description: Today, one of the biggest EU nickel producers is operating in Greece, producing approximately 20,000 tones of Ni contained in Fe-Ni per year. The process involves roasting reduction up to metallic iron in a Rotary Kiln (R/K) and smelting in an Electric Furnace. Drying, preheating and prereduction of the metallurgical mixture in rotary kilns, from which one is partially used for calcine or dolomitic lime production (needed for the process).

Figure 17. DFRK used for Ferronickel roasting

Process description: The development of the LM process emerged as an alternative to the Krupp-Renn method for processing Greek nickeliferous laterites, which proved ineffective. This novel approach included reducing ore to metallic iron through roasting in a Rotary Kiln (R/K), followed by smelting in an Electric Furnace (E/F). The initial ferronickel output from the E/F, containing about 15% nickel was further refined to 90-95% nickel in an oxygen converter, before undergoing electrolysis to produce pure nickel. However, the LM process encountered various challenges and was eventually replaced by a new method involved a preliminary reduction in R/Ks, with the subsequent smelting of calcine in E/Fs to produce ferronickel (15% Ni), which was then enriched to 20% Ni using an oxygen converter. Over time, numerous technological advancements have been integrated or tailored to enhance the process, establishing its reliability and competitiveness[47]. However, it's worth to mention that in a new FeNi industry (2017) at Republic of Kosovo, during the stages of crushing and calcination only 28% of Nickel contained is lost[48]. For FeNi production in Greece, 6,5% of raw material fed in rotary kiln is lost in flue gases[49].

Objectives: (1) Improve sorting of ores based on elements, concentration and particle size, **(2)** carry out reduction through roasting in H_2 atmosphere, replacing C.

Implementation: IUC#3 will adopt the overall methodology with the following changes: The main innovation proposed in this case is the hydrogen reduction of nickeliferous laterite in the IFRK to reduce iron, nickel and cobalt oxides, followed by the concentration of the metallic iron, nickel and cobalt by magnetic separation of the formed metal phase. Then, on industrial process level (beyond the project) production of refined ferroalloy with high nickel content in SAF. This process will eliminate the use of coal in the reduction process and will produce high nickel content ferroalloy, while simultaneously reducing the energy intensity of intermediate and final products. PRIMROCK exploitation strategy for FeNi production will be supervised by NTUA.

IUC#4&5 – ZnS and CuS (KGHM, HEGOLD)

Organisation description: KGHM is the main producer of Copper in Poland and a significant player among those controlling the world's copper reserves. The company's share is estimated at 40 million tonnes of copper in situ.

HEGOLD produces three concentrates, namely, lead/silver concentrate, zinc concentrate, and gold/pyrite/arsenopyrite concentrate which are sold through Greek ports

Process description: The deposit KGHM operates includes three main rock types: sandstone, shale and dolomite. Concentrates at Lubin (23,000t/d), Polkowice (22,000t/d) and Rudna (43,500t/d) use a conventional flotation process to produce 25–26% copper concentrates. The next step includes mixing, averaging and drying of copper concentrates mixture in order to be smelted in flash furnace to produce blister copper (99% Cu). The production of copper in the flash smelting process is accompanied with a large amount of copper-rich slag, which has to be treated with the aim of copper recovery. Copper flash smelting consists of blowing fine, dried copper sulphide concentrate and silica flux with air, oxygen-enriched air or oxygen blast into a hot (≈1200ºC) hearth-type furnace. The furnace is equipped with hydrocarbon fuel burners located concentrically in a reaction shaft. Burners are used to adjust temperature, so that the reaction in the shaft is performed properly, that is, a concentrate smelts at a specified rate. The products of the process are: (a) a molten copper-rich Cu-Fe-S matte, 45-65% Cu, which contains almost all of the copper from the concentrate plus unoxidized Fe and S; (b) a molten slag which contains iron oxide from iron oxidation plus gangue and flux oxides; (c) an off-gas which contains SO_2 from sulphur oxidation and N_2 from the input blast plus CO_2 and H_2O if supplementary fossil fuel is burnt in the furnace. In general flash smelting a significant amount of low-grade matte is produced which is an intermediate product of metal sulphides where valuable metals like Ni, Cu, and Co have been enriched. In the traditional smelt route, the low metal mattes are further converter blowing to reduce the iron content and other impurities forming the high-grade mattes like nickel–copper where almost 70 wt % Co and most part Ni, Cu are transferred to the converter slag, leading to the massive loss of valuable metals. Furthermore, the environmental issues of the emissions of oxy sulhpide gases in the pyrometallurgical process are inevitably. Consequently, developing an efficient and environmentally friendly method to synchronous extraction of nickel, copper, and cobalt from low–nickel matte is a highlighted issue[37]. During the process a massive amount of residue containing critical metals giving a promising alternative as a secondary raw material. HEGOLD built and commissioned in May 2017 a new selective flotation plant to process fresh ore from underground mineS at a throughput of around 400 ktpa. The company is seeking innovative technologies for direct reduction of sulphide concentrates.

Figure 18. Copper facilities of KGHM

[47] Zevgolis, "The evolution of the Greek Ferronickel production process"

[48] Zarife Bajraktari-Gashi, Behram Halilaj, "Material balance of the technological process in the new foundry of new ferronikel in Drenas 2017", Journal of Technology and Exploitation in Mechanical Engineering, Vol. 4, no. 1, pp. 29-35, 2018, ISSN 2451-148X

[49] Skartados, Konstantinos, "Detailed analysis of laterite pre-reduction in rotary kilns, correlated with overall efficiency at smelting. The LARCO case", PhD dissertation, National Technical University of Athens, 2000.

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Objectives: **(a)** ZnS and CuS roasting through IFRK and MRK, **(b)** metal reduction in H_2 atmosphere, **(c)** H_2 recovery through the MIP reactor, **(d)** controlled gas cooling via the heat exchanger, **(e)** reintroduction of recovered H_2 to the roasting process. **Implementation**: IUC#4&5 will adopt the full methodology of PRIM-ROCK, testing the roasting process through both the IFRK and MRK. Samples will be first pre-processed with the RMI system and subsequently sent to the Spanish pilot unit for roasting. The complete pilot-line will consist of: IFRK/MRK, MIP reactor and the heat exchanger. The utilization of the recovered H_2 in the roasting process, as well as for e-Methanol generation will be investigated.

1.2.3 Do No Significant Harm

The EU Taxonomy Compass was used to perform a preliminary analysis of the activities with the following assessed to their relevance to the 6 areas of the Taxonomy, education, data processing, hosting and related activities,

[37] Sun, Q., Cheng, H., Mei, X. et al. Efficient Synchronous Extraction of Nickel, Copper, and Cobalt from Low–Nickel Matte by Sulfation Roasting–Water Leaching Process. Sci Rep 10, 9916 (2020). https://doi.org/10.1038/s41598-020-66894-x

programming & consultancy. Substantial contribution and DNSH criteria are adhered to. PRIM-ROCK will contribute to 5 of the 6 areas, through the deployment of green processing technologies. In short, climate change adaptation and mitigation will be improved through the PRIM-ROCK. Likewise, water, pollution and circular economy transition will be addressed through the project.

1.2.4 Technical robustness of AI systems

In recent years AI projects and applications have increased exponentially, becoming one of the foremost strategic priorities. PRIM-ROCK will ensure the reliability, trustworthiness, technical and social robustness, transparency, as well as resilience of the AI systems (including classical machine learning, deep learning and distributed learning systems) that will be used based on the following measures: **(a) Training, validation, and evaluation of alternative and unbiased AI models** to identify and select technically robust models and exhibit the required performance of the tasks at hand. To achieve this, the CRISP-DM (Cross Industry Standard Process for Data Mining) methodology will be implemented, which involves the development and evaluation of multiple models, using an agile, iterative approach. Moreover, Exploratory Data Analysis (EDA) and model analysis will be applied in PRIM-ROCK to identify and mitigate potential biases. **(b) Development and use of transparent "white glass" models wherever applicable/acceptable and explanation of "black-box" models:** In cases where "white glass" models achieve satisfactory performance, they will be preferred over "black-box" models like deep learning. Nevertheless, when the use of "black-box" models is necessary the project will employ XAI (Explainable Artificial Intelligence) techniques in order to interpret the models and enhance their reliability. **(c) Use of Data Efficient AI methods applications where there is a lack of adequate quality data:** PRIM-ROCK will leverage data assets from its partners for the development of AI/ML systems. In case whereby the available dataset volumes are insufficient to create high performing AI systems, PRIM-ROCK will employ data efficient techniques such as few shot learning, model pruning, ensuring acceptable performance with smaller training datasets. **(d) Compliance with mandatory regulations (e.g., GDPR), as well as the requirements of the AI act, and the proposal of the European Parliament and Council Proposal for and AI Regulation:** PRIM-ROCK's AI systems adhere to mandatory frameworks and regulations, such as GDPR, and with emerging AI regulations at the European level. Hence, the AI systems will be thoroughly tested in terms of technical and social performance, particularly in high-risk scenarios.

1.2.5 Related R&D&I activities

In this section we present some of the research and innovation actions that are relevant to PRIM-ROCK and whose results will feed into the project. The partners involved will build upon the expertise gained in these actions and transfer the knowledge obtained. **KER1 Raw Material Intelligence system (ICCS, CIBOS)** ICCS draws experience from Wood2wood (GA: 101138789), PLASTICE (GA: 101058540), DiginTraCE (GA: 101091801) and REDOL (GA: 101091668) where they develop intelligent systems for valorisation of various waste streams.

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KER2 Intensification technologies for thermal processing (NTUA, CEIN) The KER2 team will use knowledge generated in previous projects on thermal processes research and prototyping, such as DESTINY (GA:820783), RemovAL (GA:776469) and ENEXAL (GA: 249710).

KER3 Post-processing solutions (EUT, BUL, NTUA) The KER3 team has expertise in the proposed post-process solutions and will leverage results from VIVALDI (CSIC), ETEKINA (GA: 768772), DREAM (GA: 723641) and H2SUSBUILD(GA: 214395). **KER4 Digital process line optimisation (IRIS, MAGG, ITA, NTUA, CIRCE, BUL, CORE, SCCH, STAM)** The KER4 team has expertise in digitalisation of process industries, with involved partners sharing common past and ongoing activities in AMBIANCE (GA:101058406), METAWAVE (GA:101138476), TRINEFLEX (GA:101058174), COGNIPLANT (GA:869931), CAPRI (GA:870062) and TEAMING.AI (GA: 957402).

KER5 Technology adoption framework (ADMR, PRINS, EUROC, STAM, UNE, VDU) The KER5 team will leverage experience from DiG_IT (GA:869529), INPERSO (GA:101069820), INN-PRESSME (GA:952972), STREAM-0D (GA:723082), FLEXIndustries (GA:101058453), ECOFACT (GA: 958373), SALEMA (GA: 101003785), BEATLES (GA: 101060645), VOLATILE (GA:720777) and AccelWater (GA: 958266).

KER6 PRIM-ROCK demonstrator portfolio (FORN, HEGOLD, TERNAMAG, KGHM, NTUA) FORN has participated in COGNIPLANT, HEGOLD in MultiMiner (GA: 101091374) and NETHELIX (GA: 101092365), TERNAMAG in MASTERMINE (GA:101091895) and TERRAVISION (GA: 101138643), KGHM in illuMINEation (GA:869379) and NetHelix (GA: 101092365). As part of the extrovert R&I strategy (WP10,11&12), PRIM-ROCK will also interface with SPIRE-SAIS51 to exchange expertise and lessons learned with regards to up/reskilling the European

workforce and engaged target groups. Likewise, through interfacing with H4C-ECOP52 and H4C EUROPE the Consortium will gain valuable experience in establishing H4Cs and their precursors.

1.2.6 Interdisciplinarity

PRIM-ROCK is an ambitious project aiming for the rapid acceleration of the adoption of breakthrough and smart solutions in the process industry ecosystem. This is an effort that by itself requires the harmonious collaboration of disciplines: applied sciences such as Process Engineering (**PE**), Computer Sciences (**CS**) with an emphasis on Data Science, Social Sciences & Humanities (**SSH**), Economics (**Eco**), Environmental Sciences (**ES**) and Physics, Chemistry & Mathematics (**PCM**).

WP	PE	CS	SSH	Eco	ES	PCM	Multi- and inter- disciplinary advances in PRIM-ROCK
1	✓	✓	✓	✓	✓	✓	Technological specs, process requirements, safety and baseline for assessments
2,3,4	✓	✓				✓	Simulations and designs of the prototypes, data reference architecture, monitoring & control design, RMI sensors & control systems, sorting prototype, classification, DSS
5,7	✓	✓	✓				Data infrastructure, control network, data-driven modelling, DSS, UI, 3D visualisation, Integrated and validated pilots
6	✓	✓		✓	✓	✓	ROMs, process simulations, prototyping, feasibility assessment
8	✓	✓					Demonstrations and technological assessments
9			✓	✓	✓		TEA, LCA, LCC, LCSA, health, scaleup, regulations

1.2.7 SSH dimension

In the modern mission-oriented innovation paradigm, a stronger recognition of non-technological aspects and of the added value of an open innovation processes are key[53]. This is due to the multiplicative effect of technological innovation when societal, economic and environmental contexts are considered and ultimately aligned. The project will address these issues on multiple levels: (a) on human resource, skills and labour market level, the project will carry out targeted training activities to up/reskill European workforce in order to improve the acceptance and safety of new technologies (WP8). (b) Moreover, through cooperation with local and regional education and training providers on the regional/local level (within companies and H4C), bridging with schools and universities, and developing new teaching materials, part of the project's extrovert R&I strategy (T8.1). (c) Finally, the project will deliver tools and resources (LCA models, novel business models, governance approaches, mapping reports and other open access results) to contribute to the development of a common tool database of good examples creating social value, ultimately stimulating both inter- and cross-sectoral transfer of innovations.

1.2.8 Gender equality strategy, abiding by the principles of diversity and inclusion

To fulfil project objectives, as well as to maximise its impact, it is necessary to specifically consider expectations and needs (incl. skills) of women with regards to the PRIM-ROCK results, since women in industries report a male dominated landscape. In this line, PRIM-ROCK will take care of offering a gender-neutral environment, for both contents and framework. User studies will be carried out on gender-balanced user panels. PRIM-ROCK will develop

[51] https://www.aspire2050.eu/sais

[52] https://www.h4c-community.eu/the-h4c-ecop-project/

[53] https://www.cbe.europa.eu/strategic-research-and-innovation-agenda-sria

a Gender Equality Plan supported by the results of a co-designed digital survey, engaging the industrial communities

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involved with its DC and addressing intersectional and context specific challenges. Rethinking the work environment, endorsing successful past approaches and inventing new practices when necessary and combining dedicated training sessions PRIM-ROCK aspires to set out the overall objectives for integrating an equality and inclusivity culture. Complementarily to that, actions to involve women in results exploitation will be sought, namely promoting women entrepreneurship reaching them through relevant Women in STEM initiatives. Moreover, women Web entrepreneurship is supported and encouraged with close access to the WeHubs, which connects business ecosystems and provides them with mentoring tools to support women entrepreneurs in the digital sector. Integrating an equality culture on training, rethinking the decision-making process and interventions on the built environment and inventing new practices, will be placed in the core of our activities with established methodologies that adhere to the European Institute for Gender Equality (EIGE) guidelines for gender mainstreaming. Furthermore, PRIM-ROCK will stay informed and get inspired by the newly established EC equality platform for the energy sector. The long-term impact

evoked by the project's equality strategy and actions can contribute to addressing inequalities in the clean energy transition, a need put forward by the Gender Equality Strategy 2020-2025[38].

1.2.9 Open science management

Open science practices (OSPs) are intrinsic to the project methodology. Also, the methodology of PRIM-ROCK aligns with the open access policy of the EU to the greatest extent possible, as well as ensures the protection of the financial interests of all participating parties in the consortium, in compliance with the principle "as open as possible, as closed as necessary". Moreover, the OSPs will contribute to the quality of research within the project by improving its transparency and traceability taking into consideration the project aspects like gender dimension or workers needs and concerns as part of a responsible and inclusive R&I effort. Within PRIM-ROCK the following set of OSPs will be implemented: **Early and open sharing of research:** To attain PRIM-ROCK objectives, it is crucial to validate research outputs with stakeholders engaged in the mineral processing industry, key early adopters of the breakthrough innovations. PRIM-ROCK will be careful to engage stakeholders from multiple TGs (see Section 2.1) to co-design and co-create the process innovations for roasting and calcination kiln technologies, and the related pre- and postprocessing. PRIM-ROCK will mobilise an inclusive and participatory approach to effectively involve them in the decision-making processes and other activities over the duration of the project. Moreover, in addition to the OSPs contributing to advance research by enhancing its transparency and traceability, the participatory approach can be critical for fostering opportunities for the necessary long-term commitment of key actors. **Providing open access to research outputs:** PRIM-ROCK partners are fully committed to ensure open access to peer-reviewed publications that adhere to the terms of the Horizon Europe Grant Agreement, utilizing specific platforms in compliance with OpenAIRE and community's preferences and requirements. Publications in Open Research Europe (ORE) will be conducted as well as in several relevant open journals and peer-reviewed publications with a CC-BY license or equivalent licenses.

1.2.10 Research data management

PRIM-ROCK will handle heterogeneous data along the project lifecycle, aiming to provide open access to the data excluding data for commercial purposes or other legitimate interests of beneficiaries. During the PRIM-ROCK project, several data are utilized and are exploited which are related to the specifications of the end-users of the project, the critical process parameters relating to the roasting and calcination kiln technologies and processes, and the related pre- and post-processing, as well as the breakthrough process innovations that are developed and promoted by the project. Different openness levels are established in the PRIM-ROCK project: **Open (O), Restricted (RE)**, and **Confidential (CO).** Crucial technical data remains confidential within the project duration and accessible exclusively for project partners to protect commercial interests. Additionally, the **Data Management Plan (DMP)** will be established to define the specifications of data procedures and will be accessible by stakeholders and the community in the form of database records or spreadsheets. The responsibility for data management lies with all project partners, overseen by the Project Coordinator, and reasonable expenses for data management are provisioned in the budget.

PRIM-ROCK is deeply committed to complying with the FAIR principles, according to below: **Findability**: Research data and outputs will be provided with metadata that can be harvested and identified with Digital Object Identifiers (DOIs). Moreover, generic documentation (publications, deliverables) will be easily findable through accessible URLs. Open data sets will be shared and linked on the official website of the project or other websites such as ResearchGate, as well as published in repositories like Zenodo, ensuring compliance with OpenAIRE standards and requirements. **Accessibility**: Open access of data is ensured using Zenodo open-data repository or similar platform. In PRIM-ROCK, accessible formats are widely utilized and have open specifications. Metadata is aimed to persist even when the data are no longer sustainable, as well as promote long-term digital preservation. Regarding confidential data and IPR, these will be stored in a centralized server repository under the latest available version of the Creative Commons Attribution International Public License (CC BY) or Creative Commons Public Domain Dedication (CC 0), where partners will share technical documents and data obtained from the research and testing throughout the project. **Interoperability**: Interoperability is an important consideration and highly relevant for PRIM-ROCK and will be adhered to through standards and ontologies. Furthermore, open files formats and MIME media types will be prioritized, and instructions will be included to define the ways for opening and

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[38] https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX%3A52020DC0152

managing the datasets. **Reusability**: PRIM-ROCK uses datasets that will be linked to relevant publications to streamline their understanding and reusability. Data and metadata follow standard terms. Moreover, the documentation includes a readme file and will be provided to avoid misuse or misinterpretation and keep integrity of data.

#@CON-MET-CM@# #@COM-PLE-CP@#

2 Impact

#@IMP-ACT-IA@#

The key outcome of PRIM-ROCK are two pilot-lines for the resource efficient processing of ores for the cement, mineral and non-ferrous industries. The pathway to impact in the project is divided in two periods. The first concerns the time during and shortly after the project, where the uptake of results from TGs creates short and mid-term effects, i.e., outcomes (described in §2.1.2) – and is projected until 2030, enabled by the DEC strategy (§2.2.1). The second period concerns the timeframe beyond 2030, where meaningful impact is expected thanks to an increased adoption of the PRIM-ROCK ecosystem and individual results in a significant number of replication and transferability cases (§2.1.2). The Key Exploitable Results (KERs) and individual results (R) are shown in Table 5, while market related information (e.g., market size, commercialisation cost) is described in §2.2.2.

Stakeholder Target Groups (TGs)

Five Stakeholder Target Groups (TGs) have been identified as the main beneficiaries of the results:

Table 3 Stakeholder Target Groups involved in the project

Stakeholder TGs	Main message and strategy
TG1 – Industries: Industrial end-users from the P4P sectors. Industrial associations	*Process flexibility is not only about energy and emissions, but also productivity, efficiency, and resilience.* **D:** Advertising, presentation and highlighting the projects innovations in forums and information channels. **E:** Agreements for tailored system commissioning, patenting, licensing, sales, SaaS, PaaS. Focus on ROI, payback, GHG savings and supportive tools (finance, planning) **C:** Industrial association and cluster involvement.
TG2 - Technology developers: EU OEMs, process sectors that are SME-centric.	*Resource efficiency can be a smart investment, both for providers and end-users.* **D:** Publication of pilot information, methodologies, results and best practices focused on the available results, commercialisation & manufacturing agreements. **E:** Licensing, PaaS, service and product models. **C:** Industrial & SME associations and clusters, promotion of the role of SMEs in process industries and decarbonisation.
TG3 – Research: Research performers such as universities, research organisations and associations.	*Research actors are integral in developing the knowledge to accelerate the adoption of new technologies.* **D:** Publication and presentation of the results, methodologies, datasets and other research outputs to relevant instruments and fora. Availability of open-access results. **E:** Opportunities for collaborative consultation and expert services for replication purposes. **C:** Clustering, shared activities with other national and EU supported initiatives.
TG4 - Financing & Governance: Local, regional, national and international authorities, public procurement.	*Ambitious policies are needed to address challenges posed by politico-economic conditions. PRIM-ROCK will provide evidence-driven conclusions to support decisions, investment, analyse risks and opportunities.* **D**: Publication of pilot information, methodologies, results, best practices and opportunities. **C**: Networking, policy briefs, interest representation.
TG5 - European society: General public, consumers, workforce, associations.	*Greening the industry and the economy leads to more and better employment opportunities and quality of life.* **D**: Advantages resulting from the project's implementation will be disseminated. Positive effects on quality of life, environmental and economic activity will be highlighted via online material, video, TV segments and interviews. **E**: Opportunities generated as part of the new business models will be highlighted. **C**: Engagement activities (e.g., workshops) to improve acceptance.

(OA: Open-Access Research Output and Other Results) 101178403 - 20/09/2024

Table 4. Results of the PRIM-ROCK project

KER1	Raw Material Intelligence system	Product / Service					
R1.1	RM monitoring and control systems (CIBOS)	Prototype	No	Publications	Patent	TG1 - Plant owners	Sales and support to plant owners (TG1), piloting service for SMEs, licensing agreements with OEMs
R1.2	Pre-processing line (ICCS)	Prototype			Patent	TG1 - Plant owners	
R1.3	RMI DSS (ICCS)	Software			License	TG1 - Plant owners	
KER2	Intensification technologies for thermal processing	Product					
R2.1	Molecular & Physicochemical models (ITA)	Models	Partial	Publications	Knowledge	TG3 - Research com.	Upscaled testing, manufacturing agreements for direct sales to TG1, licensing agreements with TG2, research agreements with TG3
R2.2	Thermophysical models (NTUA)	Models	Partial	Publications	Knowledge	TG3 - Research com.	
R2.3	Reduced order models (ITA)	Models	Partial	Publications	Methodology	TG3 - Research com.	
R2.4	Process models (NTUA)	Models	Partial	Publications	Knowledge	TG2 - EU Industry & SMEs	
R2.5	IRFK prototype (NTUA)	Prototype	No	Publications	Patent	TG1 - Plant owners	
R2.6	Microwave rotary kiln (CEIN)	Prototype	No	Publications	Patent	TG1 - Plant owners	
R2.6	Control system (MAGG)	Product	No	Publications	License	TG1 - Plant owners	
KER3	Post-processing solutions	Product					
R3.1	Multi-physics model of the plasma reactor (EUT)	Models	Partial	Publications	Knowledge	TG3 - Research com.	Manufacturing agreements with TG2, research agreements with TG3
R3.2	MI-Plasma reactor for H2 recovery (EUT)	Prototype	Partial	Publications	Patent	TG2-OEMs	
R3.3	Gas cooling heat exchanger (BUL)	Prototype	No	Publications	Patent	TG2-OEMs	
R3.4	Virtual toolkit (CORE)	Software	No	Publications	License	All	Service provision to all TGs
KER4	Digital process line optimisation	PaaS/Service					
R4.1	Monitoring configuration design (IRIS)	Know-how	No	No	Knowledge	TG1 - Plant owners	Individual or as KER provision of digitalisation services to TG1 (industries and SMEs) and to TG2 OEMs and service providers
R4.2	Data orchestration (CORE)	Platform	No	No	License	TG2 - EU Industry & SMEs	
R4.3	AI-based data driven optimisation (SCCH)	Software	No	Publications	License	TG2 - EU Industry & SMEs	
R4.4	Optimisation DSS & UI (SCCH)	Software	No	Publications	License	TG2 - EU Industry & SMEs	
R4.5	Digital Twin development methodology (CORE)	Methodology	Partial	Publications	Knowledge	TG3 - Research com.	
R4.6	3D Visualisation of process line (STAM)	Software	Yes	No	Knowledge	All	Service provision for training
KER5	Technology adoption framework	Know./Service					
R5.1	Technoeconomic & energy assessment (ADMR)	Service	Open	Publications	Open	TG1 - Plant owners	Service provision , licensing
R5.2	Societal & environmental impacts (PRINS)	Service	Open	Publications	Open	TG4 - Public & workforce	D&C to all TGs, services provided to clients, high level feedback through consolidated reporting, research agreements with TG3
R5.3	Workplace Impact Assessment (EUROC)	Service	Open	Publications	Open	TG1 - Plant owners	
R5.4	Business plan (STAM)	Service	No	Publications	Knowledge	TG1 - Plant owners	
R5.5	Standardisation reports (UNE)	Service	Open	Publications	Open	TG2 - EU Industry & SMEs	
R5.6	OHSE Reports and protocols (EUROC)	Service	Open	Publications	Open	TG1 - Plant owners	
R5.7	Policy and regulatory report (VMU)	Report	Open	Publications	Open	TG4-Policy makers	D&C, workshops, white papers
KER6	PRIM-ROCK demonstrator portfolio	Demo					
R6.1	IUC1 Limestone processing (FORNA)	Demo	Partial	Publications	Knowledge	TG1 - Plant owners	Pilot-as-a-service, direct commissioning agreements with plant owners, manufacturing agreements with OEMs
R6.3	IUC2 Laterite processing (NTUA)	Demo	Partial	Publications	Knowledge	TG1 - Plant owners	
R6.2	IUC3 Magnesite processing (TERNAMAG)	Demo	Partial	Publications	Knowledge	TG1 - Plant owners	
R6.4	IUC4 Sphalerite processing (KGHM)	Demo	Partial	Publications	Knowledge	TG1 - Plant owners	
R6.4	IUC4 Chalcopyrite processing (KGHM)	Demo	Partial	Publications	Knowledge	TG1 - Plant owners	

Barrier analysis

Each of the results will be exploited according to the exploitation plan (§2.2.1.4). To assess the adoption potential (via DEC) of the results, the collective experience and sector knowledge of the PRIM-ROCK Consortium is distilled into a barrier analysis, as shown below.

Table 5. Barrier analysis of the PRIM-ROCK project

Barriers most relevant to PRIM-ROCK	Means of overcoming barriers
High upfront costs required for investment in resource efficient, smart and electrified technologies	The Industrial Emissions Directive (IED) covers the largest installations producing raw materials in the EU. This directive requires facilities to adopt the so-called Best Available Techniques (BAT) to obtain the operation permit, which is managed by the competent authorities at a national or sub-national level. Only through an integrated value chain approach can real advances in low carbon technologies be understood, achieved and evaluated on a long term. Using a value chain approach would make it easier to boost the contribution of existing economic activities, but also to create and develop new more efficient low-carbon activities, through the introduction of new materials and investments. The project promotes a comprehensive capacity to emulate process lines and run pilots (KER6) producing credible and actionable technoeconomic information (R5.1)

Process industries are intrinsically conservative and hesitant regarding both data sharing and investment in new and unproven technologies.	The raw materials sector is sustainable in a low carbon economy. The potential for growth exists and is compatible with the targets of a low carbon economy. The concept of circularity should be part of process design and operation from the very beginning – big data analytics and cross-sector data integration can achieve this ambitious goal. In the future, innovation, increasing energy-cost and CO_2 prices, and investment support schemes might make these measures more financially attractive. Digitalisation approach will provide reliable data on configuration performance (KER4), by virtue of digital twins, validated through TEA (R5.1), but also through LCA (R5.3), safety (R5.7) and other aspects (KER5).
The accelerated technology development is not accompanied by sufficient capacity building for the EU workforce.	The consortium will utilise both research and industrial know how for delivering a comprehensive roadmap and tools for accelerating new technologies adaption. The project will support the adoption of new technologies through capacity building (R5.4), and visualisations (R4.6) used as training tools.
Lack of incentives and standards to drive the market towards resource efficiency	Policy and regulatory report (R5.8)

2.1 Project's pathway towards impact

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Contribution to the expected outcomes and impacts

EO1 Achieve a step change in the process industry's green transformation by improving by at least 30 % the industrial processes resource efficiency compared to the state of the art:

The PRIM-ROCK project is focused on achieving breakthrough resource efficiency primarily through improvements in the handling and processing of fine and ultrafine materials. The results contributing to this breakthrough are:

Process inefficiencies	Limestone ($CaCO_3$)	Magnesite ($MgCO_3$)	Laterite (Fe, Ni oxides)	Chalcopyrite (CuS)	Sphalerite (ZnS)
Pre-process losses	~5-6% of total mass lost during handling (fine / ultrafine), ultrafine fraction is collected and pelletized in the Mg and FeNi processing, which increases costs, material (bentonite) and energy footprint				
Thermal treatment	15-25% mass loss	15-25% mass loss	15-20% mass loss	5% mass loss	
Post-process	Significant process and energy CO_2 emissions		Intrinsic limitations of smelting & leaching, 10-15% losses in slag mattes		
Pre-process	At least 30% improvement in material handling and pre-processing through the RMI (KER1)				
Thermal treatment	90% elimination of drag losses, substantial improvements in energy intensity & productivity				
			Pre-reduction removal	Minimise losses (>90%)	
Post-process	Virtual model of CO2 capture and utilisation		Recovery of H2 and S		

Conventional processing: Total losses are estimated according to the calculations of S1 and summarized here: In the pre-processing stage losses in the area of 4-6% are reported. In some cases (e.g., laterite) pelletisation is employed, increasing the resource and energy footprint of the process significantly. In the thermal treatment stage, 5-25% mass loss due to drag is observed as a worst-case scenario. For instance, in the cement industry indicative material efficiency values for rotary kilns and similar units' range between 34%-37%. Within this share, 6-12% as slag or dam ring formation is accounted for. This brings total losses in the range of 21-30% of initial raw material mass. Moreover, in the case of low–nickel matte blowing (treatment of intermediate product of nickel sulphide ore), almost 70wt % Co and most part Ni, Cu are transferred to the converter slag, leading to the massive loss of valuable metals, whereas in the magnesite industry, the low grade magnesite (mostly fines and ultra-fines), is about 45% of the total exploiting magnesite and is treated as mine-wastes, generating considerable resource wastes and causing severe environmental consequences .

PRIM-ROCK improvements: Conservative estimations for the pre-processing stage indicate that losses can be reduced to 2-3% at pilot scale (TRL6, project end) and at below 2% at TRL9 (3 years post project). This is due to the integration of classification and sorting to the RMI (KER1) that are tailored to specific process lines, utilising high fidelity process models (KER4), and well trained DSS. Likewise, for the thermal processing (KER2), the capacity to effectively eliminate losses (below 3% at pilot scale and below 1.5% in commercial scale), is the main enabler for the breakthrough in process efficiency, but also comes with substantial improvements in productivity, intensity and emissions reduction. Moreover, in IUC3-5, where **(a)** C is used as a reducing agent and/or **(b)** multiple reduction stages are conventionally used, the removal of both elements will enable achieving breakthrough status.

Process inefficiencies	Limestone (FORN)	Magnesite (TERNAMAG)	Laterite (NTUA)	Chalcopyrite (KGHM)	Zinc (KGHM, HEGOLD)
Production (t/a)	120,000	100,000	20,000	341,000	25,000

Total losses (%)	28%	30%	21%	25%	26%
Improvement	**32%**	**36%**	**20%**	**30%**	**31%**

Considering the production metrics for the end-users of the project, it is observed that in each case, the benchmark of 30% is exceeded with the exception of FeNi. In the FeNi (IUC3) however, the focus is on the direct reduction through H_2, that will increase the reduction efficiency from 45% up to 90%, achieving a maximum of 100% in process efficiency, with a more conservative projection around the 60% mark until TRL8 where sufficient training and optimisation data will reach close to the theoretical maximum.

EO2 Reduce the CO2 intensity of the process industry and contribute to the climate neutrality goal; and EO3 Overall positive environmental and if relevant health and safety impact demonstrated;

In terms of emissions, these industries generate over 2.2MtCO_2/a, with up to 0.58MtCO_2/a associated with material losses. The following factors will contribute to the reduced emissions: **(a)** the main contributors are the electrified kilns (IFRK and MRK of KER2), where we project the CO_2 footprint to be reduced by 25-43%, depending on each country's energy mix, and without considering additional renewables. In the case of Zinc production and mainly for the ISF process, the direct emissions constitute the major part. In Europe, the ISF technology has suffered due to higher cost constraints since the early 2000's, which has led to the closure of almost all these plants. Taking into account the production level of the European zinc plants, the amount of electric energy consumed by those plants and the need for a secured continuous supply of this energy, in-house production from renewable resources such as wind or photovoltaic is not a realistic option[58]. ISF is a carbothermic reduction process which utilises carbon both as a heat source but also as reducing agent. 40% of roasting and calcination processes CO_2 emissions are related to thermal energy from fossil fuels burning[55], which will be minimised by IFRK technology utilising green electricity. **(b)** the replacement of C as a reducing agent with sustainable HAssociated with document Ref. Ares(2024)6673367 - 20/09/20242 will achieve significant reductions in laterite,

chalcopyrite and zinc processing, the case of sulphide ores processing, the amount of hydrogen produced from the generated hydrogen sulphide (H_2S) decomposition is about 60 kg per tonne of contained metal, For Zinc case the metal produced corresponds to 25,000 t/a and to 73,236 t/a of processed sulhpides, able to reproduce 4.4 t/a of H_2 through the microwave-induced plasma reactor. The annual global production of zinc exceeds 13.5 million tons, with the majority (85%) produced from sulhpide concentrates, which on theoretical level can correspond to over 22,000 tons of H_2 if sufficient infrastructure were to be installed.

Nonetheless, the sustainable H_2 loop will provide an excellent degree of flexibility to the end-user industries. The mass and energy balance of electrified calcination can reduce CO_2 emissions can reduce more than 20% in relation to a coal-fired calciner[57]. Moreover, a hybrid system integrating an electrified calciner with an amine-based CO_2 capture unit using only internal heat integration can completely decarbonize the cement industry. Thus, PRIM-ROCK will generate substantial savings as shown below, in total over 770ktCO_2 per year, assuming commercial scale operation in the 3 pilot lines. It should be noted that carbon credits (assumed at a conservative price of 55€/t) contribute to very substantial savings, totalling almost 80M€. State-of-the-art magnesia production processes [61] calculates the unavoidable geogenic CO_2 emissions amount equal to 1.09 tn CO_2 /tn MgO, proposing as the only sustainable perspective for this type of emissions the CCU.

Process parameters	Limestone (FORN)	Magnesite (TERNAMAG)	Laterite (NTUA)	Chalcopyrite (KGHM)	Zinc (KGHM, HEGOLD)	Total
CO_2eq (tCO_2/t)	1.50	4.20	13.40	4.10	3.00	
Process emissions	178,800 tCO_2	421,000 tCO_2	116,400 tCO_2	1,394,690 tCO_2	113,412 tCO_2	2,224,302 tCO_2
Energy consumption	150 GWh/y	311 GWh/y	3200 GWh/y	1125 GWh/y	147 GWh/y	4934 GWh/y
Material losses (t/a)	33,600 t/a	30,000 t/a	4,200 t/a	86,955 t/a	9,451 t/a	164,206
CO_2eq (tCO_2)	50064 t/a	126300 t/a	24444 t/a	355646 t/a	29487 t/a	585,941tCO_2
CO_2 intensity (final)	0.943 tCO_2	2.406 tCO_2	4.382 tCO_2	2.756 tCO_2	1.976 tCO_2	
CO_2 savings (t/y)	65,684	180,429	28,761	454,912	41,595	771,380
Carbon savings (€/a)	€ 6,221,371	€ 13,231,429	€ 4,820,139	€ 51,687,773	€ 3,949,956	€ 79,910,667
Emissions footprint	37%	43%	25%	33%	37%	

Data was collected from Eurostat, ASPIRE roadmap, Statista and sector specific reports for lime[39], ferronickel[40,41], zinc[42], copper[43] and magnesite[44,61], as well as data provided by end-users and domain experts in the Consortium. With regards to environmental, health and safety the main effects tiggered are shown below:

Relevant result	Human health / safety	Environmental	Uptake by 2030
Pre-processing line (R1.2)	Automation reducing accidents	>30% reduction of material discharge due to handling	**TG1**: Adopted by >3 EIIs in pilot scale
IRFK prototype (R2.5) Microwave rotary kiln (R2.6)	Reduction of dust discharge >90%, reducing risk to worker health	Process waste reduction by over 50% per unit of product	**TG1**: Uptake by >5 EIIs in TRL7 scale
Virtual toolkit (R3.4) and visualization (R4.6)	Utilisation for upskilling / reskilling purposes, reducing accidents	Projections of process footprint and environmental effects	**TG1 & TG3**: Uptake by >5 stakeholders
Societal & environmental impact assessment (R5.2) and Workplace IA (R5.3)	Definition of effects and improvement measures to facilitate the safe uptake of new technologies	Definition of process and product footprint and mitigation	**TG1&TG4**: Used in decision-making by >8 stakeholders
OHSE Reports and protocols (R5.6)	Definition of process risks and new mitigation measures	Mitigation to environmental damage due to mishandling	**TG1&TG4**: Decision making in>4 cases

Finally, it should be noted that the improvement in process efficiency will reduce by a corresponding amount the need for primary extraction of raw materials. Moreover, as dust loss during the pre-process and process stages is the main contributor to this effect, there are established protocols dealing with health and safety aspects of these materials, that will be directly impacted by the near elimination of this waste stream, as this dusting not only leads to environmental and health issues but also represents a loss of material, with significant economic implications.

EO4 Enable the techno-economic feasibility of novel technologies and processes, demonstrated and validated at suitable scale against current industrial processes to produce the same products and EO5 Enable the increase of the competitiveness and resilience of the European process industry;

From a technoeconomic perspective, it is important to consider: **(a)** the upfront costs (CAPEX), **(b)** the running costs (OPEX) and **(c)** the improvements on productivity level.

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Relevant result	CAPEX / OPEX / product	Uptake by 2030
RMI DSS (R1.3) Control system (R2.6) AI-based data driven optimisation (R4.3)	Increased CAPEX/OPEX (see investment below), but increased productivity by enabling the adaptation of pre-processing and process parameters, effectively leveraging flexibility, and improving process efficiency.	**TG1**: Optimisation adoption by at least 10 process plants, with 3 being SMEs **TG2**: Licensing to at least 15 service providers.
IRFK prototype (R2.5) Microwave rotary kiln (R2.6)	Increased CAPEX, reduced OPEX, significant productivity increase (also enabled by optimisation), short payback (<5y)	**TG1&4**: Investment of at least 20M in the upscaling and commercialisation
Technoeconomic & energy footprint assessment (R5.1)	Definition of specific TE metrics, interfacing with business plan (R5.4) to streamline the adoption.	**TG1&4**: At least 10 case-studies analysed with >7 committing.

The productivity (t of product/h) is listed for each of the end-users, along with investment projections for the adoption of IFRK in IUC1-5 and MFRK in IUC5 to achieve the same level of productivity. Initial designs show that a capacity of 800kg kiln is comfortably within design specifications for calcination, whereas for processes requiring H_2, the capacity will be half, at 400kg. Due to the efficient, controllable and optimised heating, staying time in the kiln will be reduced by 80% for the IFRK and by 90% for the MFRK, meaning that the IFRK will process 800kg in 7.5 mins compared to the 3t/h of conventional kilns and the MFRK will process 400kg in 6.5mins compared to the 1.5t/h for reduction processes, a productivity increase ranging between 9-16%. Nonetheless, this increase comes with substantial investment as the CAPEX for switching to IFRK/MFRK at the current production capacity of plants would range between 2.5M€ for a calcination IFRK and 5.5M€ for a roasting MFRK, plus an additional 1M€ for the plasma treatment. If the investment were to be made, however, the payback period would range between 0.3 to 1.84 years, rendering PRIM-ROCK an excellent investment opportunity.

[39] https://www.eula.eu/wp-content/uploads/2019/02/A-Competitive-and-Efficient-Lime-Industry-Summary_0.pdf
[40] Energy Consumption and Greenhouse Gas Emissions of Nickel Products
[41] Decarbonising the lime industry: State-of-the-art https://doi.org/10.1016/j.rser.2022.112765
[42] SECTORAL ROADMAP ZINC 2050 - Eurometaux
[43] Energy efficiency • Energy intensity in copper and gold mining
[44] Evaluation of GHG emissions from the production of magnesia https://doi.org/10.1016/j.jclepro.2016.06.118 [61] The European Magnesite/Magnesia Industry – Euromines 2020

Process inefficiencies	Limestone (FORN)	Magnesite (TERNAMAG)	Laterite (FeNi production)	Chalcopyrite (KGHM, HEGOLD)	Zinc	Total
Productivity (t/h)	17	14	3	47	5	
Investment (€)	€ 6,510,417	€ 5,341,880	€ 2,469,136	€ 46,694,053	€ 7,713,156	
Material losses (t/a)	33,600	30,000	4,200	86,955	9,451	164,206
Cost savings (€/a)	€ 414,000	€ 3,750,000	€ 716,800	€ 6,521,625	€ 459,828	12M€
Prod. Improv.	16%	12%	9%	11%	13%	
CAPEX	€ 2,500,000	€ 2,500,000	€ 3,200,000	€ 3,500,000	€ 5,500,000	
Carbon savings (€/a)	€ 6,221,371	€ 13,231,429	€ 4,820,139	€ 51,687,773	€ 3,949,956	€ 79,910,667
Total savings (€/a)	€ 6,635,371	€ 16,981,429	€ 5,536,939	€ 58,209,398	€ 4,409,784	€ 91,772,920
Payback (years)	1.03	0.33	0.47	0.84	1.84	

Regarding resilience, PRIM-ROCK will address this major challenge related to the processing of low-grade ores. Currently, there is no economic value (or the value is negative) in low grade ore bodies and industrial residues until new economic methods are provided to extract them. PRIM -ROCK will enable the utilisation of low-grade fines and ultrafine secondary resources for critical metals production. As already mentioned, Magnesite industry utilises mainly high-grade ores while in parallel low-grade ores are used for backfilling or treated as wastes. PRIM-ROCK will deliver innovative flexible and efficient production routes able to further enhance magnesium's strong reducibility for replacing rare metals including titanium, zirconium, beryllium, uranium, and hafnium. With the economy growth and ever-increasing demand for higher living quality, the thirst for magnesium products accelerates the widespread exploitation of magnesite ore. Furthermore, in the case of low–nickel matte blowing (treatment of intermediate product of nickel sulphide ore), almost 70 wt % Co and most part Ni, Cu are transferred to the converter slag, leading to the massive loss of valuable metals. Consequently, developing an efficient and environmentally friendly method like PRIM-ROCK to synchronous extraction of nickel, copper, and cobalt from low–nickel matte is a issue.

Contribution to expected impacts, scale and significance

To assess the scale and significance of the impacts, a brief replication study is carried out. The projection considers the direct replication (directly available cases such as owned plants, units contracted for management such as services) and indirect replication (client networks and collaborations). The replication potential is significant, due to the high added value of the project for stakeholders, and interest in the technologies developed. For the indirect replication, contacts will be engaged through the Advisory Board. As part of business planning and upscaled studies, the replication potential will be refined, and a more detailed analysis will be presented.

	Italy	Greece	Poland	Other	Total
Plants operated / owned	8	2 + 5 (TERNAMAG group)	7	1	22
Service provision	4	6	1	12	23
Indirect replication	6	8	3	15	32
Materials processed	Mg, FeNi, Co, Cu, Sb, and Sn				

The PRIM-ROCK Consortium is well placed to address challenges related to the supply of sustainable and costefficient critical raw materials, with end users from Greece, Italy and Poland. The potential critical metal proven and probable reserves in EU represent significant metal ore quantities, including: (i) 220 million tonnes of low-grade nickeliferous/cobaltiferous laterite resources containing 1.7 Mt Ni and 0.1 Mt Co found in Greece; 70 Mt and the reserves at 26.5 Mt of variable grade copper sulphide ores, also containing nickel, cobalt and PGMs, and representing about 3.5% and 3% of the global resources and reserves, respectively. These are mainly found in Poland, Finland,

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Bulgaria, Sweden, Spain and Greece and contain 41 Mt Cu.

	Adopters	Production capacity	Cost savings	Emissions averted	Investment leveraged
TRL8 (Q1 2030)	1 line	123,470 t/y	€ 18,354,584	154,276 tCO_2	€ 3,337,197
TRL9 (Q1 2031)	3 lines	370,410 t/y	€ 55,063,752	462,828 tCO_2	€ 10,011,591
TRL9+ (Q1 2032)	8 line	987,760 t/y	€ 146,836,672	1,234,209 tCO_2	€ 26,697,577
TRL10 (Q1 2033)	15 lines	1,852,050 t/y	€ 275,318,759	2,314,142 tCO_2	€ 50,057,956

To realise the scale and significance of the impacts, the successful conclusion of the project in 2028 is needed, where a TRL6 pilot line will be deployed. Following 1 year (2029), an upscaled (1/20) prototype will be tested, with a fullscale system commissioned in 2030. Market entry will follow on 2031, where at least 3 full-scale systems will be deployed. From there on, it is a matter of upscaling manufacturing capacity (2032) and achieving the status of an established system in the EU market by 2033, where cumulatively 15 process lines will host the full PRIM-ROCK system, with production capacity exceeding 1.8Mt/y, generating savings in excess of 275M€/y and associated emissions averted over 2.3MCO_2/y. Notably, these innovations will leverage investments over 50M€/y.

Contribution to destination impacts

***(a)** Global leadership in clean, climate-neutral and resilient industrial value chains, **(b)** circular economy and climate-neutral and human-centric digital systems and infrastructures, through **(c)** innovative production and manufacturing processes and their digitisation, **(d)** new business models, sustainable-by-design advanced materials and technologies enabling the switch to decarbonisation in all major emitting industrial sectors, including green digital technologies.*

PRIM-ROCK will contribute to all four aspects of the destination impact, as well as to the P4P KPIs as shown in Table 6 through: **(a)** the improvement of resource efficiency through electrification and decarbonisation of EU process industries as detailed in the previous sectors, accompanied by significant cost savings and improvements in productivity, addressed through KERs 1 and 2; **(b)** a comprehensive strategy and framework for digitalisation coupled with capacity building in KERs 4 and 5; ***(c)*** a holistic digitalisation pathway, adaptive to the needs of industrial ecosystems and focused on enabling actors to participate in the twin transition (KER4); ***(d)*** All of the above, demonstrating the promise of an integrated strategy to resource efficiency and the adoption of the technologies through KER5 and 6. Through these actions, PRIM-ROCK will contribute to the destination impact, as well as KSOs C,A and D.

Contribution to P4P KPIs

Table 6. PRIM-ROCK contribution to P4P KPIs

P4P KPIs[45] [63]	PRIM-ROCK Contribution
001.- Develop new electrified processes and Energy efficiency, ensuring process flexibility and capturing the full potential of renewable energies	Contribution to realizing the 100% total CO_2eq emission reduction potential demonstrated at TRL6 sustainable H_2 production (KER3), replacement of fossil fuels through electrification of thermal processes (R2.5 and R2.6), and the virtual toolkit for process scalability (R3.4) that will increase investment in this area, supported by the TEA (R5.1) and business plans (R5.4)
002.- Replace fossil fuels and feedstock by Renewable H2 and biomass in processes	
003.- Develop new efficient CO/CO_2 Capture and purification technologies	Virtual simulation toolkit for enriched CO2 capture and utilisation in e-fuels (R3.4)
005.- Design processes for maximum resource efficiency, including the development of materials for circularity	Contribution to 80% of Wastes reduction potential at TRL6, demonstrated through multiple sectors and processes (R6.1-4), especially through vastly improved processing technologies (KER2), replacement of C by H_2 for reduction (KER2), elimination of process steps through more efficient reduction (KER2), recovery of H2 for sustainable loops (KER3).
006.- Develop new processes for circularity of secondary materials from wastes/residues for all industrial processes	
0010.- Foster new framework conditions to generate a market for climate neutral solutions	Through the Technology Adoption Framework (KER5), covering technoeconomic aspects (R5.1), LCA, LCSA (R5.2) to OHSE protocols (R5.6) and standards (R5.7)
0011- Foster new skills and types of jobs and business development, including SMEs, through P4Planet programs and H4Cs	Over 1000 jobs created in industrial engineering and digital, new skills for equipment operation for operators and high-skilled profiles, with at least 1 integrated in specific academic curricula and 1 integrated company training programs.

Scientific, Technoeconomic and Societal Impacts

Scientific: By 2033, through open access results and the systemic approach to engagement of stakeholders across the value chain, important scientific knowledge will be generated, particularly regarding the affordable and sustainable improvement of building operation. The project will generate at least **32 (FWCI~2) publications and 2 in ORE**, a wealth of research data, and extensive capacity building in a pool of over **1000** individuals. These results will propagate into impactful research conducted and knowledge generated, projecting uptake by over **100 research organisations** using the open tools and data. The new directions in digital services and systems engineering, but also social sciences and humanities will contribute to **>100 jobs created in R&D**, mainly towards industrial engineering IoT, AI, ML, energy systems specialists, business/economist experts and social sciences. PRIM-ROCK will ensure Open Access and be efficient, transparent, and responsive to societal challenges aligned with EC´s goals on Open Science.

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Societal: By 2033, PRIM-ROCK will have contributed to EU policy priorities & global challenges by contributing to the areas highlighted in **Error! Reference source not found.**. This will be achieved through the extensive work on non-energy costs, market design and integration, digitalisation, policy, governance, and through continued interactions with local, municipal, national, and international stakeholders. Many of the topics will be related to implementations of the REPowerEU package.

[45] https://efce.info/3rd+European+Forum+on+New+Technologies/_/2%20-%20Ludo%20Diels%20-%20Process4Planet.pdf [63] APPENDIX F Processes4Planet Intervention Logic

Moreover, PRIM-ROCK will contribute to the SDGs by reducing process industry energy demand by **over 25%** and reducing CO_2 emissions by **over 30%, averting over 2200kt** of CO_2**.** The penetration of renewables will be increased by >30% Over

25,000 citizens will have been engaged, with **10%** participating in **8** H4C and U-IS with over 30 innovations deployed by SMEs.

Economical/Technological: By 2033, the cumulative improvements in resources efficiency, electrification and data-driven technologies will contribute to innovation-based growth by reducing energy costs for SMEs and industries, leading to significant savings but also confidence in the security of supply. Process industry productivity will increase by **10**% at least. Based on the PRIM-ROCK business plan (§2.2), we project that over 270**M€** will be generated as revenue for end users and over **1100** jobs will be the effects of PRIM-ROCK actions. Over 50**M€** will be leveraged for electrification and decarbonisation. Progress towards zero accidents will be significant.

2.2 Measures to maximize impact – Dissemination, exploitation, and communication (DEC)

#@COM-DIS-VIS-CDV@#

DEC strategy timeline

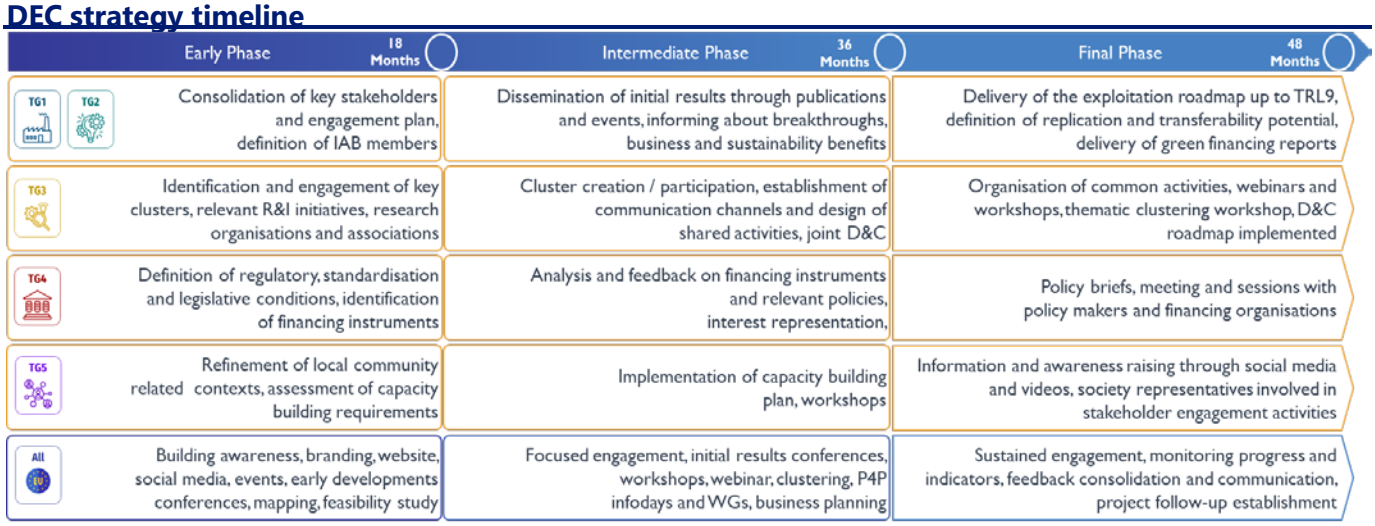

Figure 19 DEC Phases of PRIM-ROCK

PRIM-ROCK features a comprehensive strategy focused on maximising the uptake, diffusion, deployment and/or use of project's results by EU stakeholders, achieving the projected outcomes, and triggering the desired impacts. The strategy revolves around the DEC actions of WP9 and will be supported by comprehensive engagement activities (WP8). The strategy evolves to match the project's maturity, as the perception and needs of the Target Groups (TGs) become more apparent, as shown below. The Plan for Dissemination and Exploitation of Project Results (PEDR) is the main resource describing these activities, prepared by CORE as the Dissemination and Exploitation Manager (DEM). Activities will focus on 6 major stakeholder TGs to ensure the active uptake of results: TG1 Process plant owners, TG2 EU industrial ecosystem, TG3 Research community, TG4 Policy makers and financiers, TG5 Workforce and general public, encompassing 3 phases based on the progress of the project:

(a) Early Phase (M01-18, WP10): This phase is committed on building awareness for PRIM-ROCK, making the project visible and recognisable, sharing its objectives, values, and technological innovations. Visual identity and logotype, templates website and social media accounts are set to establish the PRIM-ROCK brand. Initial awareness is also focused on the mapping of key stakeholders and associated financial instruments that will facilitate uptake.

(b) Intermediate Phase (M19-36, WP11): This phase will concentrate on focused activities such as conference participation, workshop and webinar organisation, publications and other instruments added to the DEC toolset. By now, the design phase of the project has been concluded and its results are available, enhancing DEC efforts with tangible substance. Leveraging the momentum from the early phase, this phase is intended to further consolidate awareness and to establish meaningful dialogue on the project's projected value and continuation, especially with regards to the goals of the P4P. These efforts are bolstered by the targeted engagement of the TGs that will provide rapid feedback for the project's direction. An advantage of PRIM-ROCK is the high replicability potential (see §2.1) of the DC#s, that also involve a high number of influential stakeholders.

(c) Final Phase (M37-48): This phase will focus on maximizing target market and industry awareness about PRIMROCK's exploitable results. In this phase, with most results gradually available, stronger interfaces with the policy, governance, standardisation bodies and financiers are forged, in the form of meetings, sessions and policy briefs. This phase is committed to the project's continuation and longevity, but also to the realisation of projected impacts. During the final 6 months (WP13), with the participation of the TGs engaged, system upscale planning will be achieved with actual market-based feedback.

Dissemination and communication activities and instruments

The instruments employed for PRIM-ROCK's dissemination and communication activities are highlighted below.

Instruments and KPIs	Time	TGs
Publications: Peer-reviewed, open-access, journals (>20)	12-48	3

Communication materials: Effective communication with the P4P and wide intensive process industry sector and ecosystem: 2 brochures, 2 posters and 1 rollup (>1000 viewers)	6-48	1,2
Policy briefs: Policy briefs, reports, white papers, communiques (x2-3)	24-48	1,4
Newsletters: 1 every year informing regarding the progress of the project, results, opportunities for participation (>200 recipients each)	15-48	All
Conference participation: Participation in relevant conferenc, info days, and other events as Asso Ares(2024)6673367 - 20/09/2024 specified in paragraph below this table where PRIM-ROCK will be presented, especially events organized by P4P members and Innovation Clusters (est. >15 events)	12-48	All
Networking and joint events H2020/HEU projects/ initiatives: Networking activities, technical exchange and dissemination activities will be done with relevant R&I projects. 10-20 interactions, joint work on regulatory frameworks, joint D&C, cluster with P4P projects and link with H4Cs.	1-48	All
Focus groups / local regional stakeholder workshops: A series of focus groups in each pilot location will be done engaging with general public, addressing stakeholder and citizens perspectives and social acceptance. 1 focus groups/workshops per pilot location	12-48	5
Information, awareness, open-air activities 3 Information and awareness exhibitions: one per pilot and replication case, 100 recipients per information/awareness exhibition.	12-48	5
Digital/social media presence/networks Posts on social media (2-4 per month, >500 followers on LinkedIn, and Twitter)	30-48	3,5
Website A dedicated section of PRIM-ROCK website will be targeted to the research and innovation community, industry and public organisations, also including content targeting schools and universities > 75 visits per month and > 2 monthly updates	1-48	3,5
Videos: 3 videos to disseminate PRIM-ROCK, mainly through the pilot locations. >3000 viewers	3-48	5
Capacity building and skills development: These will be centred around the pilots to ensure that operators are trained on the correct and safe functioning of the PRIM-ROCK technologies.	12-48	1,5
Open Science KPIs: Early shared pre-prints/presentations/posters >4; open project repository: 1; Articles in ORE 2; Open research outputs: 3; Publications with Open Data: >2; Open data in the repository: >2; Articles in open peer review: 4; Researchers in open peer-review: 4.	18-48	All

2.2.1 Exploitation of project's results and business plan

Exploitation, market and IPR management activities will be executed and monitored by dedicated WPs (WP11-13) aiming to ensure future industrial implementation. PRIM-ROCK is directly in line with **(a)** the EC "Innovation Radar" which aims to identify and support high potential exploitable results, to represent a pathway to enacting, measuring and accelerating commercial and economic outputs and **(b)** with retaining competitive IP advantages within EU. All exploitation strategies will be described in the PEDR to be submitted as deliverables in M6, M18, M36 and M48 ensuring coherent business planning and sustainability of financing after the end of the project. Opportunities for further funding possibilities including EU, national and regional funding programmes (within Europe) will be mapped, and synergies will be investigated.

Business plan: Commercial exploitation of the PRIM-ROCK offering is projected to begin in 2031. Every result is primarily oriented to the wider industry engineering market, but also features significant potential in secondary markets, that are referred. PRIM-ROCK features 30 results, where all will feature different levels of commercial exploitation. For a more robust projection, the analysis is focused on the project's KERs.

KER1 - Raw Material Intelligence system: KER1 provides a complete system for the classification, analysis, preprocessing and segregation of raw materials according to process parameters. It has three main components: RM monitoring and control systems (R1.1), the Pre-processing line (R1.2) and the RMI DSS (R1.3). It will be commercialised by ICCS and CIBOS as the main beneficiaries. The relevant markets are sensor-based sorting machines for the mining market sector, standing at 110 M€ in 2022 and projected to reach 250M€ in 2031[46]. Exploitation will be product (hardware) and service-based through sales or leasing, at 1M€/unit at TRL9.

KER2 - Intensification technologies for thermal processing: KEP2 is comprised of R2.1 Molecular & Physicochemical models, R2.2 Thermophysical models, R2.3 Reduced order models, R2.4 Process models, R2.5

[46] Transparency markets, Yahoo Finance, GMI [65] Credence research

IRFK prototypes, R2.6 Microwave rotary kiln and the R2.7 Control system. NTUA and CEIN will exploit the prototypes and the other partners likewise their results. The global market stands at over 800M€, with 60% in Asia and 17% in the EU[65]. The exploitation model will again be product and service based, with licensing and manufacturing agreements with OEMs, at a price of 2,5M€ for a 800kg capacity unit (or 400kg with H2 at 3.5M€).

KER3 - Post-processing solutions: KER3 is comprised of different solutions, with R3.1 Multi-physics model of the plasma reactor (EUT) and R3.2 MI-Plasma reactor for H2 recovery (EUT) supported by R3.3 Gas cooling heat exchanger (BUL) and a R3.4 Virtual toolkit (CORE) dealing with the virtual post-process of CCUS and e-fuel generation. The main market all of these solutions are oriented towards EEIs, a €19.8b market according to the SRIA. Commercialisation will be both product and service based, with tailored systems commissioned at 1M€/unit. **KER4 - Digital process-line optimisation:** KER4 delivers a scalable and adaptable process for the digitalisation of process industries, and their optimisation. It is comprised of R4.1 Monitoring configuration design (IRIS), R4.2 Data orchestration (CORE), R4.3 AI-based data driven optimisation (SCCH), R4.4 Optimisation DSS & UI (SCCH) and R4.5 Digital Twin development methodology (CORE). The corresponding digital services market is surging at 1.5Tn€. Exploitation will be service or product as a service based with a 100,000€ sales price with subscription. **KER5 - Technology adoption framework:** KER5 will deliver a holistic service to support decision-making, investment and operation of project technologies. It is comprised of R5.1 Technoeconomic & energy footprint

assessment (ADMR), R5.2 Societal & environmental impacts (INS), R5.3 Workplace Impact Assessment

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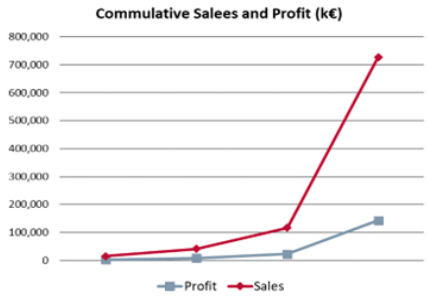
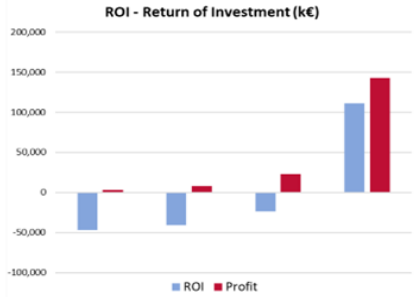
(EUROC), R5.4 Business plan (STAM), R5.5 Standardisation reports (UNE), R5.6 OHSE Reports and protocols (EUROC), R5.7 Policy and regulatory report (VDU). KER5 is relevant to the consultation market, that exceeds 100b€, with technical consulting at ~20%. Exploitation will be service based.

KER6 - PRIM-ROCK demonstrator portfolio: KER6 will be used as the whole system product offering, addressed to the process industries, a €19.8b market according to the SRIA.

Financial projections: The prospects for the commercialisation of PRIM-ROCK were projected until 2033. By 2031, we expect the project to achieve market readiness, as the project will conclude in Q4 2028. In brief, the opportunity is financially attractive and likely to deliver substantial returns on the investment over the forecasted period. By the end of the 4th year, the cumulative profit stands at €162M, which is a substantial sum, especially considering the initial investment of €10M and commercialization costs of additional €35M, implying that not only the initial investment is recovered within the four-year period, but also sizeable profit is generated. The table indicates aggressive sales growth year-over-year, with substantial increases in both sales and profit. The final market share percentages are very low, which suggests that even with the growth, PRIME-ROCK is operating in a large market with considerable expansion potential. Finally, with a new job created for every €120k in sales, over 1360 jobs will be created. The project is well-positioned to offer lucrative returns, while the cumulative profit indicates healthy growth potential. However, it is essential to note that these projections are contingent upon successful execution of StreamSTEP and the realization of the forecasted sales.

				Years after commercialisation-->		End of 1st Year		2030	End of 2nd Year*	2031	End of 3rd Year*	2032	End of 4th Year*	2033			
Exploitable project result	Partner	Item Price (€)	Profit/ Item (€)	Initial Market size (m€)	Intial Market Share (%)	Sales (initial)	Sales (k€)	Profit (k€)**	Sales (k€)***	Profit (k€)**	Sales (k€)***	Profit (k€)**	Sales (k€)***	Profit (k€)**	Final Market Size (m€)*	Final Market Share (%)	Market Share Increase (%)**
Raw Material Intelligence system	ICCS	1,000,000	193,550	58	3.45%	2.00	2,000.00	387.10	3,619.01	700.46	10,585.14	2,048.75	79,950.52	15,474.42	90	88.82%	2476%
Intensification technologies for thermal processing	NTUA, MAGG, ITA	2,500,000	576,920	136	3.68%	2.00	5,000.00	1,153.84	9,224.93	2,128.82	26,172.27	6,039.72	212,247.58	48,979.95	211	100.56%	2635%
Post-processing solutions	EUT, BUL, CORE	1,000,000	130,430	1,980	0.10%	2.00	2,000.00	260.86	3,654.49	476.66	10,688.91	1,394.15	80,734.35	10,530.18	3,073	2.63%	2501%
Digital process line optimisation	CORE, SCCH, ITA, STAM	110,000	28,520	1,500,000	0.00%	5.00	550.00	142.60	975.71	252.98	2,768.22	717.72	22,669.35	5,877.55	########	0.00%	2556%
Technology adoption framework	ADMR, PRINS, STAM, EUROC	50,000	22,070	20,000	0.00%	8.00	400.00	176.56	702.51	310.09	2,013.67	888.83	15,529.61	6,854.77	31,039	0.05%	2402%
PRIM-ROCK demonstrator portfolio	End Users	7,000,000	1,166,670	1,980	0.35%	1.00	7,000.00	1,166.67	12,790.71	2,131.79	37,785.30	6,297.57	291,404.24	48,567.51	3,073	9.48%	2582%
Total Sales/Annual Profit							16,950.00	3,287.63	30,967.36	6,000.79	90,013.50	17,386.75	702,535.65	136,284.38	Total sales/profit		
Cumulative Sales/Profit							16,950.00	3,287.63	47,917.36	9,288.42	137,930.87	26,676.17	840,466.52	162,959.55	Cumulative sales/profit		
Cumulative profit - initial investement								-46,712.37		- 40,711.58		- 23,324.83		112,959.55	ROI		
New Jobs (Cumulative) ****								28		79		224		1,360	Cumulative jobs		
General assumptions made:						Initial Investment (m€)		15.00	* Annual Market Size Increase			7.6%	** Annual Market Size Increase (average)		76%		
						Commercialisation Cost (m€)		35.00	*** Gross Profit (average)			23%	**** 1 new job every k€		120		





2.2.2 Project results and management of intellectual property

Project methodology and results will be published, but products and models developed will be evaluated by an IPR committee within the Project Consortium to evaluate any harm to property rights. Regarding Intellectual Property (IP) rules, at proposal stage the fundamental aspects in relation to Background IP, Foreground, Results Ownership, Exploitation strategies and Access Rights have been discussed to ensure the necessary IP framework is understood and robust. During the preparation and signing of Consortium Agreement the partners will update/confirm their background IP and create an IP baseline. PRIM-ROCK will follow the IPR Management Guidelines from the proposal stage right through to the end of the project and IPR issues will be further elaborated and agreed during the negotiation of the Grant Agreement (GA). Moreover, during project implementation, the Consortium will take address IPR issues and market uptake, as part of the GA sessions. **Consortium Agreement (CA):** IP issues will be covered in the CA that will be drawn up based on the DESCA model. All consortium partners will agree and sign acceptance of the CA prior to the signature of the GA. The maintenance of the status of result ownership and rules accessing background knowledge vital to the successful completion and exploitation of the project results by each beneficiary are addressed in WP10-12. **Background IP:** The Background knowledge of the partners is identified at proposal stage and updated prior to the writing of the CA. Background IPR will remain the property of the Originator Company or organisation. However, where needed, during the period of the project, all background knowledge necessary for carrying out the project will be available for use by any other partner of the consortium, with access rights being granted royalty-free by the owning partner. For post-project exploitation, needed access will be once again granted by the owning partner, either royalty-free or by charging a license fee under reasonable terms, thereby promoting collaboration and the advancements of the research and innovation employing NDAs as needed. **Patents and Knowledge Sharing:** The novel results of the project that are eligible for patenting will be actioned as soon as possible. In cases of joint inventions, the ownership will be shared in accordance with a notified agreement. A framework for access rights for jointly conducted research will also be reported in the DECP. Other IPR tools, such as trademarks, copyrights, utility models etc. will also be actively assessed for the different aspects of novel innovation, e.g. trademarks will be considered for individual components for branding and commercialization purposes in other industries by the DEM in consultation with the consortium partners. #§COM-DIS-VIS-CDV§#

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2.3 Summary

Target groups and needs	Results	DEC Measures	Outcomes (by 2031)	Impacts (by 2033)

TG1 – Industries: Industrial end-users from the P4P sectors. Industrial associations. They are the main beneficiaries of the project's results. They need to improve the resource efficiency, productivity and resilience of the EIIs. They also need access to new tools and resources to support technological innovation. **TG2 - Technology developers**: EU OEMs, process sectors that are SME-centric. They need to scale up their operations, expand networks, clients and business, improve resilience and innovate. **TG3 – Research**: Research performers such as universities, research organisations and associations. The research community needs access to new tools and knowledge, operational data and insights provided directly by the industry and society. **TG4 - Financing & Governance**: Local, regional, national and international. They need access to clear and actionable information on a high level, particularly with regards to technoeconomic areas, risk of investment, environmental and societal impacts, feedback on policies and regulations. **TG5 - European society**: General public, consumers, workforce, associations. They need to improve their quality of life through more and better paying jobs, reduced pollution and more resilient societies.	**KER1** - Raw Material Intelligence system: **R1.1** RM monitoring and control systems, **R1.2** Pre processing line, **R1.3** RMI DSS **KER2** - Intensification technologies for thermal processing: **R2.1** Molecular & Physicochemical models (ITA), **R2.2** Thermophysical models (NTUA), **R2.3** Reduced order models (ITA) **R2.4** Process models (NTUA), R2.5 IRFK prototype (NTUA), **R2.6** Microwave rotary kiln (CEIN), **R2.7** Control system (MAGG) **KER3** - Post-processing solutions: **R3.1** Multi physics model of the plasma reactor (EUT), **R3.2** MIPlasma reactor for H2 recovery (EUT), **R3.3** Gas cooling heat exchanger (BUL) R3.4 Virtual toolkit (CORE) **KER4** - Digital process line optimisation: **R4.1** Monitoring configuration design (IRIS), **R4.2** Data orchestration (CORE), **R4.3** AI-based data driven optimisation (SCCH), **R4.4** Optimisation DSS & UI (SCCH), **R4.5** Digital Twin development methodology (CORE), 3D Visualisation of process line **KER5** - Technology adoption framework: R5.1 Technoeconomic & energy footprint assessment (ADMR), **R5.2** Societal & environmental impacts (PRINS), **R5.3** Workplace Impact Assessment (EUROC), **R5.4** Business plan (STAM), **R5.5** Standardisation reports (UNE), **R5.6** OHSE Reports and protocols (EUROC), **R5.7** Policy and regulatory report (VDU) **KER6** - PRIM-ROCK demonstrator portfolio: **R6.1** IUC1 Limestone processing (FORN) **R6.2** IUC2 Laterite processing (NTUA for Greek FeNi production) **R6.3** IUC3 Magnesite processing (TERNAMAG) **R6.4** IUC4Zinc processing (KGHM) **R6.5** IUC5 Chalcopyrite processing (KGHM)	**Dissemination**: Product /service showcasing and exhibitions, wider energy sector related events, conferences, fairs, scientific publications, technical workshops, meetings with policy makers, technical factsheets, videos. Oriented towards all TGs. **Communication**: Workshops, interviews, surveys, physical-online communities, collaboration with EU R&I initiatives and energy communities, organisations and partnerships, information awareness, website, forums, networking. **Exploitation**: Know-how and IP protection, openaccess, licensing, product as a service, replication strategy, introduction into existing service portfolios, packaging, establishment of sales networks, franchising. Smart contracts, shared savings, CO2 tax valorisation. Measures apply to all results according to the tables in 2.1.	• Adoption of project's main result by at least 3 process industries in operational scale. • Digitalisation framework and optimisation applied in at least 10 plants (mineral, non-ferrous, cement, ferrous sectors) • Reduction of losses in raw material handling by over 50%, RMI system adopted by >3 • Reduction of losses in process stage by over 80%, Electrified Rotating Kiln (IFRK/MRK) uptake by >5 EIIs in pilot scale • Virtual toolkit used by >5 stakeholders to reduce the risk of CCUS, TEA applied in >5 industries for process upgrade • Investment of at least 20M in upscaling, KER5 used by at least 12 plant owners / operators	• 32 (FWCI~2) publications and 2 in ORE, • Extensive capacity building in a pool of over 1000 individuals. • Uptake by over 100 research organisations using the open tools and data. • The new directions in digital services and systems engineering, but also social sciences and humanities will contribute to >100 jobs created in R&D • Moreover, PRIM-ROCK will contribute to the SDGs by reducing process industry energy demand by over 25% • and reducing CO2 emissions by over 30%, averting over 2200kt of CO2. • The penetration of renewables will be increased by >30% • Over 25,000 citizens will have been engaged. • Process industry productivity will increase by 10% at least. • Over 270M€ will be generated as revenue for end users and over • 1100 jobs will be the effects of PRIMROCK actions. • Over 50M€ will be leveraged for electrification and decarbonisation. • Progress towards zero accidents will be significant.

3 Quality and efficiency of the implementations

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#@QUA-LIT-QL@# #@WRK-PLA-WP@#

3.1 Work plan and resources

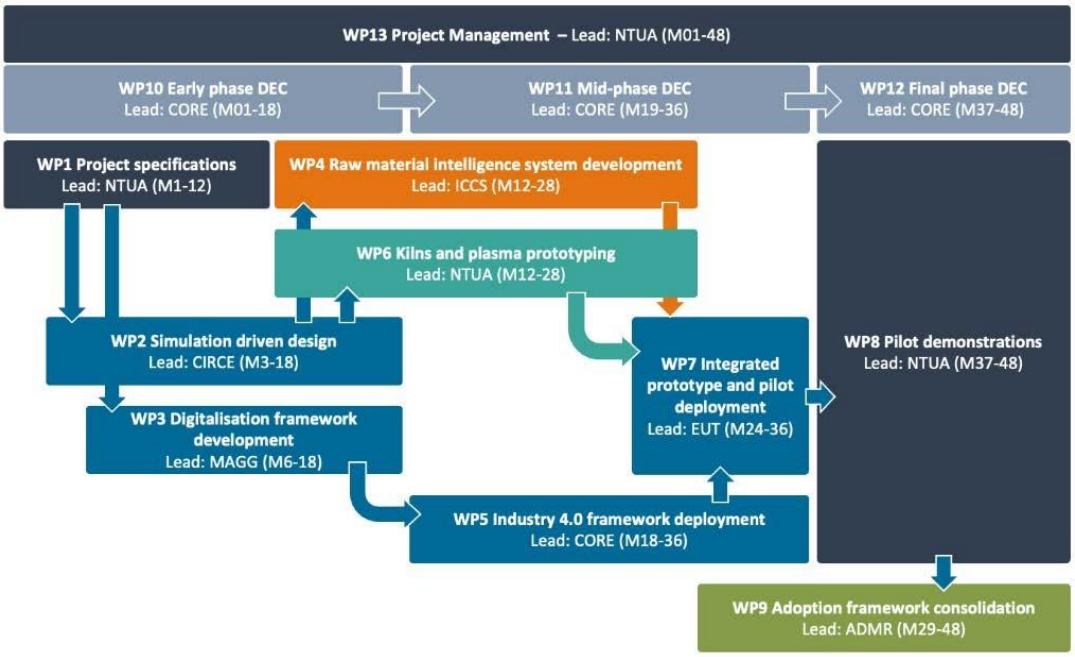

Figure 20. WP structure (pert-like)

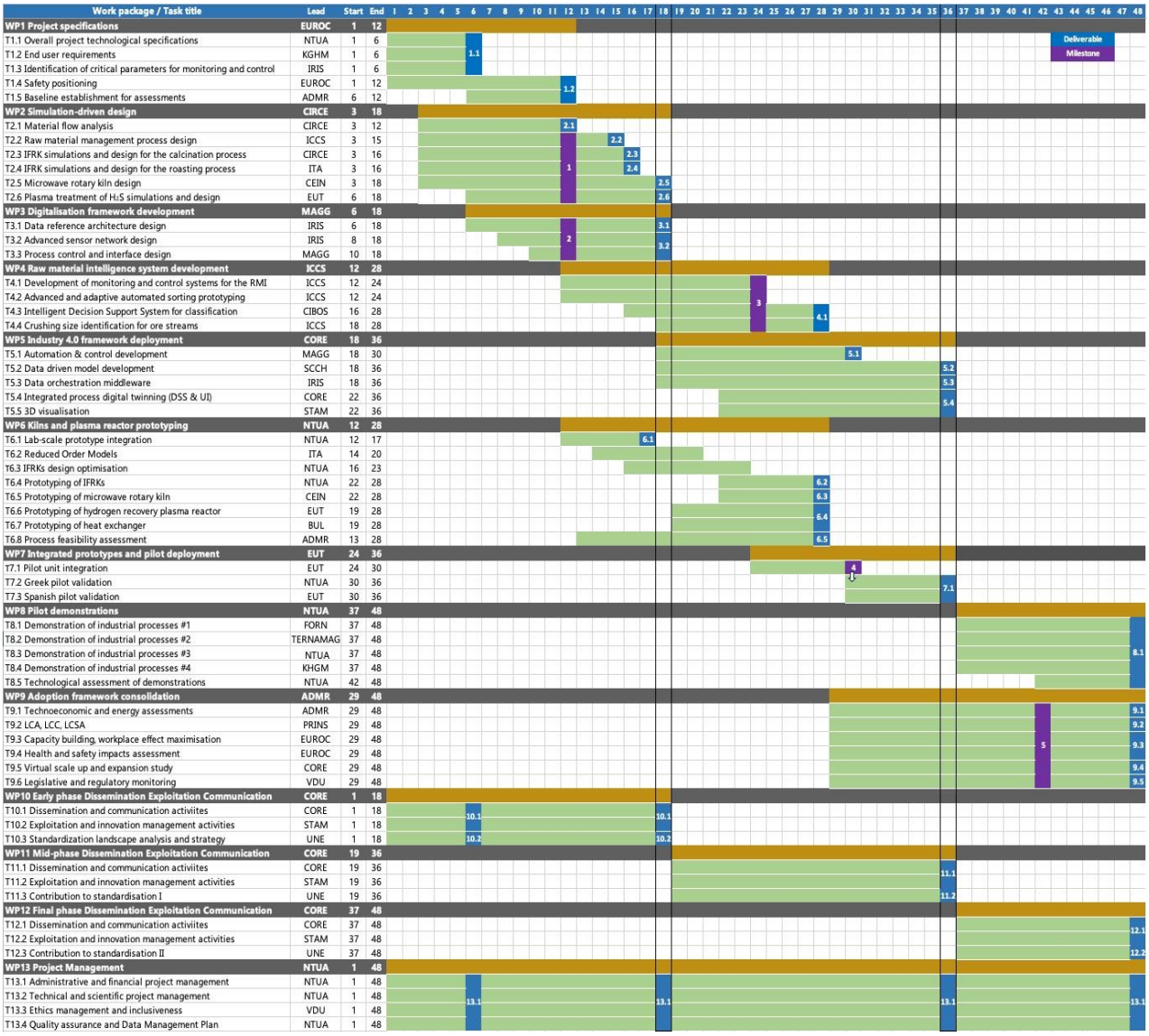

Figure 21. PRIM-ROCK's Gantt

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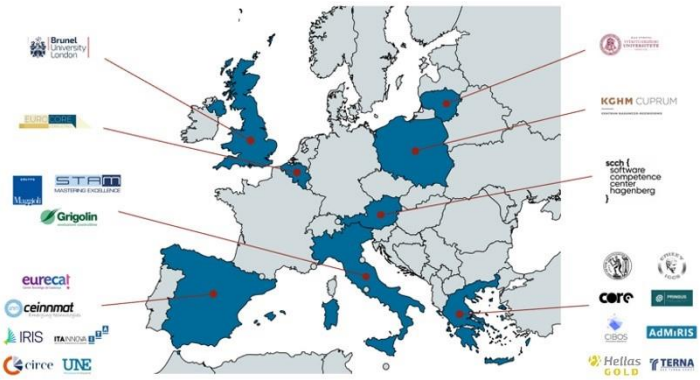
3.2 Capacity of participants and consortium as a whole

#@CON-SOR-CS@# #@PRJ-MGT-PM@# Matching of project objectives



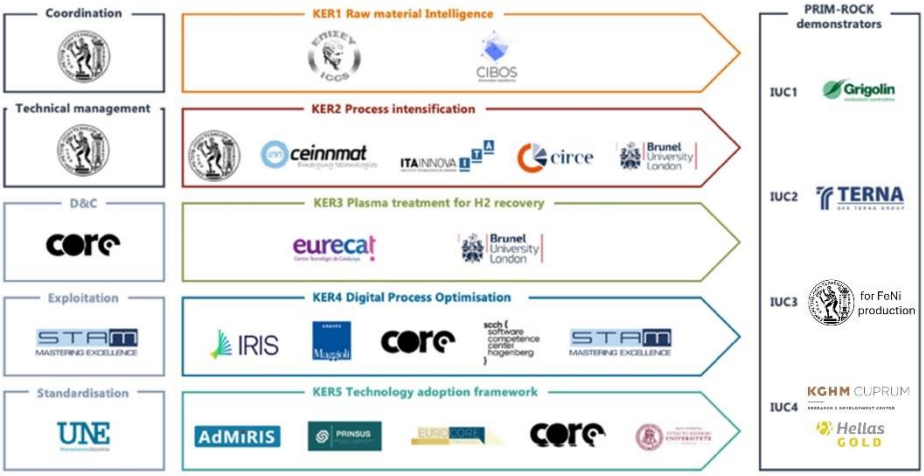
PRIM-ROCK gathers an inter- and trans-disciplinary Consortium composed of 22 partners from 8 countries, 7 Member States (Spain, Greece, Belgium, Italy, Austria, Poland, Lithuania), as well as United Kingdom. The Consortium features 7 SMEs (CORE, IRIS, STAM, ADMR, PRINS, EUROC, CIBOS), along with 6 RTDs (EUT, ICCS, CIRCE, ITA, SCCH, CEIN) and 3 universities (NTUA, BUL, VDU). Moreover, 1 Standardisation Body (UNE), and 5 Large Enterprises (MAGG, FORN, HEGOLD, TERNAMAG, KGHM) round out the partnership. The cohesive team of partners has been created considering: **a)** balanced complementary expertise and capacity, with clear roles and results; **b)** geographical differentiation adhering to the requirements of the call with regards to pilots but also replicability of results; **c)**

previous experience, proven expertise and motivation, for an effective and efficient management and execution of the project; **d)** good representation and involvement of stakeholder Target Groups; **e)** P4P partnership involvement and **f)** Newcomers to EU funded research.

3.2.1 Access to critical infrastructure

Have been described in 1.2.2. All the partners have the infrastructure to carry out development and piloting work.

3.2.2 Description of partners and corresponding roles



All partners will participate in WP1 and its activities, in WP8 and WP9 for demonstrations and technology adoption framework consolidation, respectively, as well as WP10-12 for DEC actions and WP13 with regards to reporting as part of project management.

#1 NTUA is the project's Coordinator. They consolidate technical specs in T1.1, conduct physicochemical modelling of the IFRK for calcination in T2.3, lead WP6 by optimising IFRKs design in T6.3 and construct the prototypes in T6.1 and T6.4, validate them in T7.2, and leads the demonstration WP8 by assessing the technologies in T8.5.

#2 CORE is an EL SME. They undertake data-driven optimization of the roasting process via IFRK and MRK in T5.2, lead T5.4 where the process DSS and UI will be developed and conduct the scale-up & expansion study in T9.5. They also lead the CDE WPs (WP10,11,12), overseeing the communication and dissemination activities.

#3 EUT is an ES RTD, responsible for conducting the simulations and design of the MIP reactor in T2.5, as well as its prototyping in T6.6 for the recovery of H2. They also lead WP7, undertaking the integration of the prototypes in the pilot units in T7.1 and the validation of the ES pilot in T7.3.

#4 ICCS is an EL RTD. They consolidate end-users' requirements in T1.2 and are responsible for the RMI system. Particularly, they design the system in T2.2 and lead WP4, where they develop the monitoring and control systems for the RMI (T4.1), develop the sorting prototype in T4.2, the classification DSS in T4.3 and the crushing size identification system in T4.4.

#5 CIRCE is an ES RTD. CIRCE leads the simulation and design WP2, where they lead the analysis of the material flow in T2.1, as well as T2.3 conducting the thermophysical simulations of the IFRK and develop the ROMs in T6.2.

#6 ITA is an ES technology center. ITA leads T2.5 by conducting the physicochemical modelling of the IFRK for roasting, as well as T6.2 where they develop the associated ROMs.

#7 SCCH is an AT RTD. They lead the efforts for the data-driven calcination process optimisation in T5.2 and develop the associated DSS and UI in T5.4.

#8 BUL is a UK university, leading the development of a heat exchanger prototype for the controlled cooling of H2 and S gasses in T6.7, based on the designs of WP2. Moreover, BUL carries out the thermophysical modelling of the

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IFRK for the roasting process in T2.4.

#9 CEIN is an ES technology provider SME. CEIN leads the design of the MRK for the sulphides roasting process in T2.5 and manufactures the prototype in T6.5.

#10 IRIS is an ES RTD, which leverages their expertise in monitoring and sensor networks, as well as on the associated data integration. IRIS leads T1.3, T3.1 and T3.2, developing the foundation of the digital framework. Then, in T5.3 IRIS integrates the middleware, that is refined in WP7 and demonstrated in WP8.

#11 STAM is an IT consulting SME. STAM leads the 3D visualisation of the digital twins in T5.5 and support the integration of industrial IoT sensors in the FORN case for operational data collection. They are also the project's Exploitation Manager (WP10,11,12).

#12 ADMR is an EL SME and the main responsible for the coordination of the technology adoption framework, supporting the individual assessments and other activities. ADMR leads T1.5 and T6.8, coordinating these activities, as well as WP9 and the associated TEA task (T9.1).

#13 PRINS is an EL SME, experts in LCA, LCC and LCSA. They leverage their knowledge leading T9.2.
#14 EUROC is a BE SME and the project's expert on health and safety. They lead T1.4, T9.3 and T9.4, as well as provide support in WPs6, 7 and 8.
#15 MAGG is an IT LE and experts in control systems and automation. MAGG leads WP3 and T3.3 designing the control system and interface, develops the system in T5.1, implement in WP7 and demonstrate in WP8.
#16 CIBOS is an EL SME. They lead the development and implementation of the RMI-DSS in T4.3.
#20 VDU is a LT university, responsible for the legislative monitoring (T9.6) and ethics management (T13.3).
#21 UNE is an ES Standardisation Body and will address the standardisation aspects of the project in WPs 10-12.
End users: **FORN**, **TERNAMAG**, **HEGOLD, FeNi Greek producers (through NTUA)** and **KGHM** will have a similar participation profile. Their role will be to **(a)** providing requirements, parameters and support for the design of process technologies that will emulate conditions relevant to industrial production. In the case of FORN, STAM will support data collection, in the case of TERNAMAG, sample analysis will also be carried out onsite; **(b)** Sending raw material samples to the pilot unit for comparative assessment of the process performance. *#§CON-SOR-CS§# #§PRJ-MGT-PM§#*

4 Ethics self-assessment

Ethical dimension of the objectives, methodology and likely impact
PRIM-ROCK is focused on improving the calcination and roasting processes, thanks to a holistic approach centred around groundbreaking process intensification (PI) and digital technologies. The Consortium commits to upholding the highest ethical standards for research, as delineated in the European Code of Conduct for Research Integrity of ALLEA and ESF, to observe and promote the principles of integrity in scientific and scholarly research. These principles include honesty in communication, reliability in performing research, objectivity, impartiality and independence, openness and accessibility, the duty of care, and fairness in providing references. For the Consortium, it is of the utmost importance that ethical issues are appropriately addressed from the outset of the project. To facilitate the practical application of ethical and legal principles the EC has identified key issues that could arise during project's development. PRIM-ROCK entails the collection of meaningful data, with a view to producing insights that will successfully inform the project's activities, enabling evidence-based results. The Consortium is aware and committed to the EU and national legislation regarding data protection and data ethics. A Data Management Plan (DMP) will be formulated from the early stages of the project and updated when deemed necessary. It will describe the complete data management life cycle of the data to be collected, processed and/or generated by the project and will lay out the approach for their sound and FAIR management. All data referring to user groups participating in the demo cases and other project activities (e.g., interviews, surveys, workshops, etc.) will be managed according to specific terms that will be detailed in the DMP, protecting the data subjects' rights and freedoms. All information related to project results and data exchanged in the project will be treated as confidential, thereby providing a first line of protection and security. Artificial Intelligence: The consortium places significant importance on ethics and AI and will take all the necessary steps to ensure that the development, training, and integration of AI is inclusive, explainable, has a positive purpose and use data responsibly. Task 13.4 will cover data lifecycle aspects as well as data governance through the delivery of the DMP and its updates during the project lifetime.
Compliance with ethical principles and relevant legislations
NTUA, as the project's Coordinator and in close collaboration with the project's Steering Committee will establish the required governance systems to constantly monitor the progress of the project from its legal, ethical, and

societal dimensions and will record risks identified, mitigation suggestions, and actions taken, regularly reporti
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to the PO and the EC via its multi-version deliverables as described the implementation plan (Section 3). PRIM-ROCK will comply with data protection acts, directives, and opinio both at European and at National level. These include:

- The Convention for the Protection of Human Rights and Fundamental Freedoms, especially its arts. 7 and 8.
- The Charter of Fundamental Rights of the EU, specifically the article concerning the protection of personal data.
- Directive 95/46/EC of the European Parliament and the Council of 24 October 1995 on the protection of individuals with regard to the processing of personal data and on the free movement within the EU/ EEA of such data to be replaced in May 2018 by Regulation 679/2016 (the General Data Protection Regulation). - Directive 2002/58/EC of 12 July 2002 concerning the processing of personal data and the protection of privacy in the electronic communications sector (Directive on privacy and electronic communications) [2002] OJ L201/37, as modified by the Citizens Rights Directive.
- The opinions of the European Group on Ethics in Science and New Technologies in their report Citizens Rights and New Technologies: A European Challenge on the Charter on Fundamental Rights related to technological innovation.
- In particular recommendations related to ICT concerning data protection and individuals freedom and autonomy applicable national data protection legislation.
- As regards processing personal electronic data, protecting privacy in the electronic communications sector and retaining data generated or processed in connection with the provision of publicly available electronic communications services or of public communications networks (e.g. cloud, big data, open data, cookies etc.),
- PRIM-ROCK will comply with the relevant legislation:
- EU Directive 2002/58/EC of the European Parliament and of the Council of 12 July 2002 concerning the processing of personal data and the protection of privacy in the electronic communications sector (Directive on privacy and electronic communications).
- EU Directive 2006/24/EC of 15 March 2006 on the retention of data generated or processed in connection with the provision of publicly available electronic communications services or of public communications networks.

Artificial Intelligence: PRIM-ROCK will pay attention to ethical issues related to the way the developed AI operates, in terms of Bias elements caused by model, system, or process inconsistencies. The system will be designed and assessed in a way that ensures compliance with the recent EU's draft AI Act, the OECD's recommendations on AI, and the Council of Europe's report on the human rights dimensions of Algorithms.